

Model-Agnostic Interpretation of Machine Learning via Average Marginal Effects

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May 2026

Abstract

We propose a framework for interpreting any supervised machine learning model through average marginal effects, a standard inferential tool in the econometrics of nonlinear models. The framework is agnostic to the choice of learner — the same estimation procedure applies whether the prediction function is a neural network, a gradient boosted model, a random forest, or any other supervised learner — and produces a coefficient table directly analogous to the output of a parametric regression, with estimates, standard errors, and hypothesis tests. Standard errors are obtained via a nonparametric bootstrap with full model refitting on each replicate, capturing uncertainty about both model specification and estimation. The framework nests the classical generalized linear model result as a special case.

The theoretical contribution is twofold. First, we establish gradient consistency for neural networks in Sobolev space $W^{1,p}$, for gradient boosted models and Mondrian forests in the space of functions of bounded variation, and identify the obstruction to extending these results to Breiman forests. Second, we prove bootstrap validity for all three estimator classes via the Hadamard differentiability of the average marginal effect functional and the functional delta method, providing the first rigorous inferential foundation for machine learning average marginal effects. These results together

imply that the interpretability–accuracy tradeoff is not fundamental: it is an artifact of requiring parametric assumptions to make interpretation rigorous, and it dissolves once those assumptions are dropped.

Empirical validation on three datasets spanning binary classification and continuous regression demonstrates that average marginal effects are stable across model classes, that bootstrap standard errors narrow as model fit improves, and that Breiman random forests produce less stable gradient estimates than the other estimator classes — consistent with the theoretical result that gradient consistency for Breiman forests is harder to establish. Practitioners requiring random forest average marginal effects are advised to consider Mondrian forests, for which both gradient consistency and bootstrap validity are proved. Existing interpretability methods including SHAP values and partial dependence plots lack the inferential apparatus the proposed framework provides: they produce point estimates with no associated sampling distributions and no basis for hypothesis testing.

1 Introduction

The rise of flexible machine learning methods has created a persistent tension in empirical research. Models such as random forests (Breiman, 2001a), gradient boosted trees (Chen and Guestrin, 2016), and deep neural networks (Hastie et al., 2009) routinely achieve predictive accuracy that parametric models cannot match. Yet the standard tools of statistical inference — coefficient estimates, standard errors, hypothesis tests — apply naturally only to parametric models whose functional form is specified in advance. Breiman (2001b) framed this as a conflict between two cultures of statistical modeling: one oriented toward interpretable data-generating models, the other toward accurate algorithmic prediction. The conventional resolution has been to treat accuracy and interpretability as opposing ends of a spectrum, accepting reduced interpretability as the price of improved prediction. This paper argues that the tradeoff is not fundamental. It is an artifact of requiring parametric assumptions

to make interpretation rigorous, and it disappears once that requirement is dropped.

The machine learning literature has produced a rich set of post-hoc interpretability methods that attempt to explain black-box model predictions without parametric assumptions. Partial dependence plots (Friedman, 2001) visualize the marginal relationship between a feature and the predicted outcome by averaging over the empirical distribution of other features. LIME (Ribeiro et al., 2016) fits a local linear approximation to the prediction function in the neighborhood of individual observations. SHAP (Lundberg and Lee, 2017) assigns each feature a contribution to each prediction based on Shapley values from cooperative game theory, providing a unified framework that nests several earlier methods. These tools are widely used in practice and are surveyed comprehensively in Molnar (2025) and Doshi-Velez and Kim (2017). However, they share a fundamental limitation: they produce point estimates with no associated sampling distributions. A SHAP value, a PDP slope, and a LIME coefficient are descriptive quantities — there is no standard error, no confidence interval, and no basis for a hypothesis test. A researcher cannot determine whether a SHAP value is statistically distinguishable from zero, or whether two features have significantly different effects, or whether the effect of a variable differs across model specifications. For applied researchers who need to draw defensible conclusions from their models, this is a serious constraint.

Two prior contributions are closest in spirit to the present paper. Scholbeck et al. (2024) propose forward marginal effects for nonlinear prediction functions as a model-agnostic interpretability method, establishing a general framework for computing feature effects for any supervised learner. Their framework is descriptive rather than inferential: it characterizes how feature effects vary across the covariate space but does not provide standard errors, confidence intervals, or hypothesis tests, and it explicitly argues against summarizing feature effects in a single metric such as the average marginal effect. The present paper takes the opposite position: we argue that the average marginal effect is precisely the right summary quantity, that it has a rigorous population-level interpretation inherited from the

econometric literature, and that valid inference is achievable via a nonparametric bootstrap with full model refitting. Löwe et al. (2024) provide the `fmeffects` R package implementing this descriptive framework, demonstrating the practical demand for marginal-effect-style interpretation of machine learning models, but without the theoretical justification — no gradient consistency, no bootstrap validity, no characterization of when the marginal effects are well-defined — that the present paper provides through the Hadamard differentiability of the AME functional and the functional delta method.

The econometric literature has solved exactly this inferential problem for parametric models through average marginal effects (Greene, 2012; Wooldridge, 2010). For a generalized linear model with inverse link function μ and linear predictor $\eta = \mathbf{x}^\top \boldsymbol{\beta}$, the average marginal effect of variable j is $\text{AME}_j = \beta_j \cdot \mathbb{E}[\mu'(\eta)]$ — the regression coefficient scaled by the average derivative of the inverse link, averaged over the empirical distribution of the linear predictor. For the identity link this reduces to the OLS coefficient. For the logistic link it averages the logistic density over the sample. The result is a quantity measured in the natural units of the outcome, interpretable as the average change in the predicted outcome associated with a one-unit change in x_j , with standard errors obtained by the delta method or bootstrap. This is the output of `margins` in R and Stata (Leeper, 2021), and it is the standard form in which marginal effects are reported in applied economics, political science, sociology, and public health. The limitation is that it has been restricted to parametric models: to compute $\partial f / \partial x_j$ analytically, one must know the functional form of f .

This paper removes that restriction. We propose a framework for computing average marginal effects for any supervised machine learning model, requiring only that the prediction function $\hat{f}$ can be evaluated at a point. The average marginal effect of variable j is defined as the expectation of the partial derivative of f with respect to x_j , averaged over the empirical distribution of the features. For models where the derivative is not available in closed form — which includes tree-based models, for which classical derivatives do not exist everywhere — we estimate it via a centered finite difference with an adaptive step size scaled to the spread

of each feature. Standard errors are obtained via a nonparametric bootstrap (Efron, 1979; Efron and Tibshirani, 1993) that refits the model, including hyperparameter selection, on each bootstrap sample, capturing uncertainty about model specification as well as estimation. The result is a coefficient table — estimate, standard error, and significance level — directly analogous to the output of a parametric regression but with no restrictions on the functional form of $\hat{f}$. The framework nests the classical GLM result as a special case: when $\hat{f}$ is a logistic regression, the estimator recovers the classical AME automatically. When $\hat{f}$ is XGBoost or a random forest or a neural network, the same formula applies and produces output on the same scale and with the same interpretation.

The theoretical contribution of the paper proceeds in two parts. The first establishes gradient consistency for three classes of supervised learners, providing the asymptotic foundation for the proposed estimator. For neural networks, which live in Sobolev space $W^{1,p}$, gradient consistency follows from uniform $W^{2,p}$ boundedness and L^p consistency via the Rellich–Kondrachov compact embedding theorem (Evans, 2010; Hornik et al., 1990). For gradient boosted models with tree base learners, the natural function space is the space of functions of bounded variation (Ambrosio et al., 2000), where gradients exist as Radon measures and consistency is established via the BV compactness theorem (Bühlmann and Yu, 2003; Mason et al., 2000). For random forests, gradient consistency is proved for the Mondrian forest variant (Lakshminarayanan et al., 2014; Mourtada et al., 2017) and conjectured for Breiman forests (Breiman, 2001a; Scornet et al., 2015), with the obstruction to a full proof identified precisely. The second part establishes bootstrap validity for each of the three estimator classes. The key result is the Hadamard differentiability of the AME functional (van der Vaart and Wellner, 1996, Chapter 3.9), which shows that the AME varies smoothly under perturbations of the underlying estimator in the appropriate function space sense. Combined with rate conditions for each estimator class and the functional delta method, this yields bootstrap validity as a corollary for neural networks, gradient boosted models, and Mondrian forests. Bootstrap validity for Breiman forests remains open for the

same reason that gradient consistency is harder to establish — the data-dependent splitting rule couples partition geometry to response variation in a way that resists clean analysis.

We validate the framework empirically on three datasets spanning binary classification in social science, binary classification in finance, and continuous regression: the UCI Adult Income dataset (Kohavi, 1996; Dua and Graff, 2019), the UCI Credit Card Default dataset (Yeh, 2009; Dua and Graff, 2019), and the Ames Housing dataset (De Cock, 2011). Across all three applications we compare AMEs from logistic regression, random forests (Breiman, 2001a), XGBoost (Chen and Guestrin, 2016), and neural networks against SHAP values (Lundberg and Lee, 2017) and partial dependence plot slopes (Friedman, 2001). The AME and PDP estimates are in close agreement throughout, while SHAP values are on a different scale and in several cases have the wrong sign — a systematic divergence that is most visible in the regression setting where units are dollars and the magnitudes are large enough to make the discrepancy concrete. Because AMEs come with standard errors and SHAP values do not, the AME framework supports conclusions that the SHAP framework cannot. An additional empirical finding is that Breiman random forests produce less stable AME estimates than the other estimator classes, consistent with the theoretical result that gradient consistency for Breiman forests is harder to establish than for Mondrian forests. Practitioners who require random forest AMEs are therefore advised to consider Mondrian forests, for which both gradient consistency and bootstrap validity are proved.

The remainder of the paper is organized as follows. Section 2 defines the estimand, presents the estimator, and establishes its relationship to classical average marginal effects for the GLM family. Section 3 establishes gradient consistency and bootstrap validity for neural networks, gradient boosted models, and Mondrian forests, and identifies the obstruction to extending these results to Breiman forests. Section 4 presents the three empirical applications. Section 5 presents simulation evidence on finite-sample performance. Section 6 concludes.

2 Method

We propose a framework for interpreting any supervised machine learning model through the lens of average marginal effects. The framework is agnostic to the choice of learner — the same estimation procedure applies whether $\hat{f}$ is a neural network, a gradient boosted model, a random forest, or any other supervised learner. Crucially, the framework produces standard errors via a nonparametric bootstrap, enabling hypothesis tests and confidence intervals that existing model-agnostic interpretability methods do not support. SHAP values and permutation-based feature importance measures provide point estimates of variable importance but no basis for inference. The average marginal effect estimator proposed here yields a coefficient table — estimate, standard error, and significance — directly analogous to the output of a parametric regression, with no restrictions on the functional form of $\hat{f}$.

2.1 Estimand

Let $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^\top \in \mathbb{R}^p$ denote the feature vector for observation i , and let $y_i \in \mathbb{R}$ denote the outcome. We observe n independent draws $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ from some population distribution P . Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ denote the conditional expectation function $f(\mathbf{x}) = E[Y \mid \mathbf{x}]$. For binary outcomes, $f(\mathbf{x}) = P(Y = 1 \mid \mathbf{x})$ is the conditional probability of the positive class; for continuous outcomes, $f(\mathbf{x}) = E[Y \mid \mathbf{x}]$ directly. In both cases f is the same object — the conditional expectation of Y given $\mathbf{x}$ — and the method requires no modification across outcome types.

The marginal effect of variable j at observation i is the partial derivative of f evaluated at $\mathbf{x}_i$:

$$m_{ij} = \left. \frac{\partial f}{\partial x_j} \right|_{\mathbf{x}_i} \quad (1)$$

The population average marginal effect of variable j is:

$$\text{AME}_j = \mathbb{E} \left[\frac{\partial f}{\partial x_j}(\mathbf{x}) \right] \quad (2)$$

where the expectation is taken over the marginal distribution of $\mathbf{x}$ under P . This is the estimand. The conditions under which $\partial f / \partial x_j$ exists and under which the sample estimator defined below converges to AME_j are established in Section 4.

2.2 Estimation

Given an estimator $\hat{f}$ of f , the sample analog is:

$$\widehat{\text{AME}}_j = \frac{1}{n} \sum_{i=1}^n \left. \frac{\partial \hat{f}}{\partial x_j} \right|_{\mathbf{x}_i} \quad (3)$$

For most supervised learners the partial derivative $\partial \hat{f} / \partial x_j$ is not available in closed form, and for tree-based models it does not exist everywhere in the classical sense. We therefore estimate it via a centered finite difference:

$$\left. \frac{\partial \hat{f}}{\partial x_j} \right|_{\mathbf{x}_i} \approx \frac{\hat{f}(\mathbf{x}_i^{(j, x_{ij} + h_j)}) - \hat{f}(\mathbf{x}_i^{(j, x_{ij} - h_j)})}{2h_j} \quad (4)$$

where $\mathbf{x}_i^{(j, v)}$ denotes the vector $\mathbf{x}_i$ with the j -th component replaced by v . The centered finite difference is preferred over a one-sided difference because its approximation error is $O(h_j^2)$ rather than $O(h_j)$, which becomes material when h_j is not negligibly small.

The step size h_j is chosen adaptively as:

$$h_j = \max(10^{-4}, 0.05 \cdot \hat{\sigma}_j) \quad (5)$$

where $\hat{\sigma}_j$ is the sample standard deviation of the j -th feature. Scaling h_j to the spread of x_j ensures the finite difference is neither so small as to introduce numerical instability nor so large as to obscure local behavior. The floor of 10^{-4} guards against degenerate cases where $\hat{\sigma}_j \approx 0$.

For binary or categorical variables the partial derivative does not exist and the finite

difference is replaced by a contrast. The marginal effect for observation i is:

$$m_{ij} = \hat{f}(\mathbf{x}_i^{(j,1)}) - \hat{f}(\mathbf{x}_i^{(j,0)}) \quad (6)$$

which measures the change in predicted outcome when x_j is switched from 0 to 1 holding all other variables at their observed values. The sample average marginal effect is then $\widehat{\text{AME}}_j = n^{-1} \sum_{i=1}^n m_{ij}$, with interpretation identical to the continuous case: a change in x_j from 0 to 1 is associated with a $\widehat{\text{AME}}_j$ change in the predicted outcome, on average across the sample.

For binary classification models, $\hat{f}$ must return the estimated conditional probability $\hat{P}(Y = 1 \mid \mathbf{x})$ rather than the predicted class label. Class labels are discrete and their finite differences are identically zero; the method requires a continuous output. This is the predicted probability of the positive class, available as `predict_proba(X)[: , 1]` in scikit-learn and equivalent interfaces in other frameworks. The AME is then interpreted as the change in predicted probability associated with a one-unit change in x_j , on average across the sample.

2.3 Relationship to Classical Average Marginal Effects

Average marginal effects are a standard tool in the econometrics of nonlinear models (Greene, 2012; Wooldridge, 2010). For parametric models the estimator $\widehat{\text{AME}}_j$ defined above reduces exactly to the classical result. Consider a generalized linear model with inverse link function μ and linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$, so that $f(\mathbf{x}_i) = \mu(\eta_i)$. The marginal effect of variable j at observation i is:

$$\left. \frac{\partial f}{\partial x_j} \right|_{\mathbf{x}_i} = \beta_j \cdot \mu'(\eta_i) \quad (7)$$

and the population average marginal effect is $\text{AME}_j = \beta_j \cdot E[\mu'(\eta)]$. For the identity link $\mu'(\eta) = 1$ and $\text{AME}_j = \beta_j$ — the marginal effect is constant across observations and equals the OLS coefficient. For the logistic link $\mu'(\eta) = \Lambda(\eta)(1 - \Lambda(\eta))$ and the AME averages

the logistic density over the sample (Greene, 2012). For the probit link $\mu'(\eta) = \phi(\eta)$, the standard normal density evaluated at the linear predictor (Wooldridge, 2010).

In each case $\widehat{\text{AME}}_j$ recovers the classical result automatically — the finite difference applied to the model output numerically approximates $\beta_j \cdot \mu'(\eta_i)$ without requiring knowledge of the link function. The framework therefore nests the entire GLM family as a special case and reduces to familiar output when the model is parametric. This backward compatibility is what the `marginalEffects` package in R and Python implements (Arel-Bundock et al., 2024); the present framework extends that implementation with formal asymptotic guarantees for machine learning models.

2.4 Inference

Standard errors are obtained via a nonparametric bootstrap (Efron, 1979; Efron and Tibshirani, 1993). Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ denote the observed sample and let $\hat{\lambda}$ denote the hyperparameters of $\hat{f}$ selected by cross-validation on $\mathcal{D}$. For $b = 1, \dots, B$:

- (1) Draw a bootstrap sample $\mathcal{D}^{(b)}$ by sampling n observations from $\mathcal{D}$ with replacement, sampling $(\mathbf{x}_i, y_i)$ pairs jointly to preserve the joint distribution of features and outcome.
- (2) Initialize the hyperparameter search at $\hat{\lambda}$ and refit $\hat{f}^{(b)}$ on $\mathcal{D}^{(b)}$ with hyperparameters $\hat{\lambda}^{(b)}$ selected by cross-validation.
- (3) Compute $\widehat{\text{AME}}_j^{(b)}$ using $\hat{f}^{(b)}$ evaluated at the original sample $\mathcal{D}$.

The bootstrap standard error is:

$$\widehat{\text{se}}_j = \sqrt{\frac{1}{B-1} \sum_{b=1}^B \left(\widehat{\text{AME}}_j^{(b)} - \bar{\text{AME}}_j \right)^2} \quad (8)$$

where $\bar{\text{AME}}_j = B^{-1} \sum_{b=1}^B \widehat{\text{AME}}_j^{(b)}$.

Several design choices warrant comment. Sampling $(\mathbf{x}_i, y_i)$ pairs jointly preserves the joint distribution of features and outcome that $\hat{f}$ is trained to approximate; resampling features and outcomes independently would destroy this dependence structure. Refitting $\hat{f}^{(b)}$ with hyperparameter selection on each bootstrap sample ensures that the bootstrap standard errors capture uncertainty about model specification, not only estimation uncertainty conditional on a fixed architecture. Initializing the hyperparameter search at $\hat{\lambda}$ rather than from scratch makes this computationally feasible — the original model serves as a warm start, substantially reducing the cost of cross-validation on each replicate. Finally, evaluating $\hat{f}^{(b)}$ at the original sample $\mathcal{D}$ rather than the bootstrap sample $\mathcal{D}^{(b)}$ follows standard practice for smooth functionals of the empirical distribution (Efron and Tibshirani, 1993) and ensures that $\widehat{\text{AME}}_j^{(b)}$ depends on the bootstrap draw only through $\hat{f}^{(b)}$, which simplifies the asymptotic analysis in Section 4. Analysis of the variant that evaluates at $\mathcal{D}^{(b)}$ is left to future work.

3 Asymptotic Theory

3.1 Overview

The method agnostic interpretation framework rests on a theoretical foundation that connects the gradients of machine learning estimators to the gradients of the true underlying function. This section establishes that foundation. We show that for three estimator classes—neural networks, gradient boosted models, and random forests—the gradient of the estimator $\hat{f}_n$ converges asymptotically to the gradient of the true function f , and that bootstrap standard errors for the resulting average marginal effects are asymptotically valid. The average marginal effects presented in Section 3 are therefore consistently estimating population quantities with well defined interpretations, not heuristic approximations, and inference on those quantities is justified.

The connection between the theory and the method is direct. The average marginal effect

of variable j is defined as:

$$\widehat{AME}_j = \frac{1}{n} \sum_{i=1}^n \frac{\partial \hat{f}_n}{\partial x_j}(x_i) \quad (9)$$

For this quantity to be meaningful as an estimate of a population parameter, two things must be true. First, the gradient $\frac{\partial \hat{f}_n}{\partial x_j}$ must exist and be well defined—it must be estimating something real rather than an artifact of the particular model fitted. Second, it must converge to the corresponding gradient of the true function f as the sample size grows. Without these guarantees, the average marginal effects are descriptive statistics about the fitted model rather than estimates of population quantities. The theorems in this section provide exactly these guarantees. The corollaries then establish that bootstrap standard errors correctly characterize the uncertainty in those estimates.

The key insight is that the appropriate notion of gradient consistency depends on the function space the estimator naturally inhabits, which in turn reflects the inductive bias of the estimator class. This is not merely a technical observation—it reflects something fundamental about what each estimator class is doing. Neural networks with smooth activation functions produce smooth estimators whose gradients are ordinary L^p functions, and the appropriate framework is Sobolev space theory (Evans, 2010). Random forests produce piecewise constant estimators whose gradients are concentrated on leaf boundaries rather than distributed across the domain, and the appropriate framework is the theory of functions of bounded variation (Ambrosio et al., 2000). Gradient boosted models occupy a middle ground depending on the base learner. The average marginal effects computed in Section 3 are the same operation—averaging the gradient over the sample—but the theoretical justification for that operation differs across estimator classes in a way that reflects their fundamentally different structures.

This observation also clarifies why existing approaches to model interpretation lack the theoretical foundation developed here. Methods such as SHAP values and partial dependence plots are not computing gradients of the estimator in any well defined function space sense, and consequently there is no analog of the results below that would establish what population

quantity they are estimating or at what rate. The average marginal effect framework, by contrast, is precisely defined as a gradient operation, and the results below establish that this operation is doing what practitioners intuitively expect it to do.

Full proofs of all results are provided in the Appendix.

3.2 Notation and Preliminaries

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $f : \Omega \rightarrow \mathbb{R}$ be the true underlying function relating covariates to outcomes. Let $\hat{f}_n : \Omega \rightarrow \mathbb{R}$ denote an estimator of f constructed from n observations $\{(x_i, y_i)\}_{i=1}^n$. We assume throughout that Ω is bounded and Lipschitz, which ensures the compactness and embedding results we rely on are available. These are mild regularity conditions satisfied by any compact covariate space with reasonable boundary behavior.

We work with two function spaces depending on the estimator class. The choice of function space is not arbitrary—each space is the natural habitat of the estimator class in question, in the sense that estimators of that class belong to the space with probability one under mild conditions.

The **Sobolev space** $W^{k,p}(\Omega)$ for $k \geq 1$ and $p \geq 1$ consists of functions whose weak derivatives up to order k are in $L^p(\Omega)$, with norm:

$$\|g\|_{W^{k,p}} = \sum_{|\alpha| \leq k} \|\partial^\alpha g\|_{L^p} \quad (10)$$

The gradient $\nabla g \in L^p(\Omega)^d$ is a vector valued L^p function for any $g \in W^{1,p}(\Omega)$. Sobolev spaces are the natural setting for smooth estimators because they quantify not just how well the function is approximated in L^p but how well its derivatives are approximated simultaneously. This is exactly the property needed to justify gradient based interpretation— L^p consistency of $\hat{f}_n$ alone is insufficient because differentiation is an unbounded operator on L^p , and convergence of a function does not in general imply convergence of its derivatives

without additional regularity (Evans, 2010, Chapter 5).

The **space of functions of bounded variation** $BV(\Omega)$ consists of functions $g \in L^1(\Omega)$ whose distributional gradient Dg is a vector valued Radon measure with finite total variation $|Dg|(\Omega) < \infty$. The BV norm is:

$$\|g\|_{BV} = \|g\|_{L^1} + |Dg|(\Omega) \quad (11)$$

The BV framework extends the notion of gradient to functions that are not differentiable in the classical or even weak sense—precisely the situation with tree based estimators, which have jump discontinuities at leaf boundaries. For piecewise constant functions the distributional gradient is a sum of surface measures concentrated on the boundaries between constant regions, capturing the intuition that the “derivative” of a step function is a spike at the jump location. This makes BV the natural space for random forests and boosting with tree base learners. The total variation $|Dg|(\Omega)$ measures the total size of all jumps, weighted by the boundary surface area—a quantity that is finite and well controlled for any piecewise constant function with finitely many pieces. The standard reference for the theory of BV functions is Ambrosio et al. (2000).

The relationship between the two spaces is important. If $f \in W^{1,p}(\Omega)$ for $p \geq 1$, then $f \in BV(\Omega)$ with $Df = \nabla f \cdot \mathcal{L}^d$ —that is, the distributional gradient of f is absolutely continuous with respect to Lebesgue measure with density ∇f . This embedding $W^{1,p} \hookrightarrow BV$ means that assuming $f \in W^{1,p}(\Omega)$ is sufficient for both the Sobolev and BV results below. We make this assumption throughout.

Definition 1 (Well-Behaved Estimator, Sobolev Sense). *We say that $\hat{f}_n$ is well-behaved in the Sobolev sense if:*

$$\sup_n \left\| \hat{f}_n \right\|_{W^{2,p}} < \infty \quad \text{and} \quad \left\| \hat{f}_n - f \right\|_{L^p} \xrightarrow{p} 0 \quad (12)$$

Definition 2 (Well-Behaved Estimator, BV Sense). *We say that $\hat{f}_n$ is well-behaved in the*

BV sense if:

$$\sup_n \mathbb{E} \left[|D\hat{f}_n|(\Omega) \right] < \infty \quad \text{and} \quad \left\| \hat{f}_n - f \right\|_{L^1} \xrightarrow{p} 0 \quad (13)$$

In both cases the well-behaved condition has the same structure: a uniform boundedness condition in a strong space combined with consistency in a weaker space. The uniform boundedness condition is the key regularity requirement—it says that the estimator sequence does not become increasingly irregular as n grows, even as it fits the data more closely. This condition is verifiable from properties of the estimator class and is established for each class in the theorems below. The consistency condition is the standard requirement that the estimator converges to the truth and is available from existing results in each case.

We also state the rate condition required for the bootstrap corollaries. Let P denote the marginal distribution of the covariates x .

Definition 3 (Rate Condition). *We say that $\hat{f}_n$ satisfies the rate condition if:*

$$\sqrt{n} \left(\hat{f}_n - f \right) = O_p(1) \quad (14)$$

in the appropriate norm — $W^{1,p}(\Omega)$ for estimators well-behaved in the Sobolev sense, and $BV(\Omega)$ for estimators well-behaved in the BV sense.

The rate condition requires that $\hat{f}_n$ converges to f at the parametric rate $n^{-1/2}$. This is stronger than consistency alone and is the condition that makes the functional delta method applicable. It is verified for each estimator class in the corollaries below.

3.3 Core Lemmas

The theoretical results rest on five lemmas. Lemmas 1 through 4 establish derivative existence and consistency in each function space. Lemma 5 establishes the Hadamard differentiability of the AME functional, which is the key condition for bootstrap validity. All five lemmas are proved in the Appendix and applied across all three estimator classes.

The parallel structure of Lemmas 1 through 4 reflects the parallel structure of the Sobolev and BV frameworks and is the formal expression of the analogy between smooth and piecewise constant estimators. Lemma 5 then sits above both frameworks as the unifying result that connects gradient consistency to bootstrap validity.

Lemma 1 (Derivative Existence in Sobolev Spaces). *Let $\hat{f}_n \in W^{2,p}(\Omega)$ for some $p > 1$. Then $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$ exists as a well defined L^p function and satisfies:*

$$\left\| \nabla \hat{f}_n \right\|_{L^p} \leq \left\| \hat{f}_n \right\|_{W^{1,p}} \leq \left\| \hat{f}_n \right\|_{W^{2,p}} \quad (15)$$

Lemma 1 is essentially definitional—membership in $W^{2,p}(\Omega)$ implies by definition that the first and second weak derivatives exist and are in L^p . The content of the lemma is the norm bound, which establishes that the gradient operator $\nabla : W^{2,p} \rightarrow W^{1,p}$ is bounded and linear. This boundedness is what makes the subsequent consistency argument work—a bounded linear operator preserves convergence, so once we have convergence of $\hat{f}_n$ in the right space, convergence of $\nabla \hat{f}_n$ follows automatically. The reason we require $\hat{f}_n \in W^{2,p}(\Omega)$ rather than just $W^{1,p}(\Omega)$ is that we need one additional degree of differentiability to apply the compactness argument in Lemma 2. The relevant theory of Sobolev spaces and their norms is developed in Evans (2010, Chapter 5).

Lemma 2 (Derivative Consistency in Sobolev Spaces). *Let $f \in W^{1,p}(\Omega)$ and suppose $\hat{f}_n$ is well-behaved in the Sobolev sense. Then:*

$$\left\| \nabla \hat{f}_n - \nabla f \right\|_{L^p} \xrightarrow{p} 0 \quad (16)$$

Lemma 2 is the harder result and the one that does the real work in the Sobolev setting. The difficulty is that L^p consistency of $\hat{f}_n$ alone does not imply L^p consistency of $\nabla \hat{f}_n$ —differentiation is an unbounded operator on L^p and limits and derivatives cannot be exchanged without additional structure. The uniform $W^{2,p}(\Omega)$ boundedness condition

provides that structure. The proof applies the Rellich–Kondrachov compact embedding theorem, which states that a uniformly bounded sequence in $W^{2,p}(\Omega)$ has a subsequence converging strongly in $W^{1,p}(\Omega)$ (Evans, 2010, Theorem 5, Section 5.7). Combined with the L^p consistency condition identifying the limit as f , this gives strong convergence of $\hat{f}_n$ to f in $W^{1,p}(\Omega)$, from which L^p convergence of the gradients follows immediately from the boundedness of ∇ established in Lemma 1. The key insight is that the uniform boundedness condition ensures the estimator sequence lives in a compact subset of $W^{1,p}(\Omega)$, preventing the gradients from escaping to infinity even as the estimator fits the data more closely.

Lemma 3 (Derivative Existence in BV). *Let $\hat{f}_n \in BV(\Omega)$. Then the distributional gradient $D\hat{f}_n$ exists as a well defined vector valued Radon measure in $\mathcal{M}(\Omega)^d$ with:*

$$|D\hat{f}_n|(\Omega) < \infty \tag{17}$$

For piecewise constant estimators with leaf partition $\{R_j\}$ and leaf means $\{\bar{y}_j\}$, the distributional gradient takes the explicit form:

$$D\hat{f}_n = \sum_{j \sim k} (\bar{y}_k - \bar{y}_j) \cdot \nu_{jk} \cdot \mathcal{H}^{d-1} \llcorner_{\partial R_j \cap \partial R_k} \tag{18}$$

where ν_{jk} is the unit normal between adjacent leaves and $\mathcal{H}^{d-1}$ is the $(d-1)$ -dimensional Hausdorff measure on leaf boundaries.

Lemma 3 is the BV analog of Lemma 1 and is similarly structural in nature. The first part—that BV membership implies the distributional gradient exists as a Radon measure—follows from the definition of BV as developed in Ambrosio et al. (2000, Chapter 3). The content of the lemma is the explicit formula for piecewise constant estimators, which makes concrete what the distributional gradient actually looks like for a tree based model. The formula says that the gradient is not a function at all but a measure concentrated on the boundaries between leaves, with mass proportional to the jump in response values across

the boundary weighted by the boundary surface area. This is the precise sense in which a random forest or boosted tree has a gradient—not a pointwise derivative, which does not exist, but a distributional object that captures where and how much the estimator changes. This explicit representation is also what makes the total variation $|D\hat{f}_n|(\Omega)$ computable in terms of observable quantities—the leaf means and the partition geometry—which is essential for verifying the well-behaved condition in the theorems below.

Lemma 4 (Derivative Consistency in BV). *Let $f \in W^{1,p}(\Omega)$ for some $p > 1$, so that $f \in BV(\Omega)$ with $Df = \nabla f \cdot \mathcal{L}^d$. Suppose $\hat{f}_n$ is well-behaved in the BV sense. Then:*

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad (19)$$

If additionally $\mathbb{E}[|D\hat{f}_n|(\Omega)] \rightarrow |Df|(\Omega)$, then convergence is strict.

Lemma 4 is the hardest of the four function space lemmas and the BV analog of Lemma 2. The convergence is necessarily weaker than in the Sobolev case—weak-* convergence of measures rather than strong L^p convergence—because the BV space is not reflexive and the stronger norm convergence is not available in general. Weak-* convergence means that for any continuous test function ϕ :

$$\int_{\Omega} \phi d(D\hat{f}_n) \rightarrow \int_{\Omega} \phi d(Df) \quad (20)$$

which is the appropriate notion of convergence for the distributional gradients of piecewise constant estimators. In the context of average marginal effects, this means that integrated gradient information—which is exactly what the AME computes—converges to the correct population quantity. The proof applies the BV compactness theorem of Ambrosio et al. (2000, Theorem 3.23), which is the analog of Rellich–Kondrachov in the BV setting: a uniformly bounded sequence in BV has a subsequence converging in L^1 , and the L^1 consistency condition identifies the limit as f . The strict convergence upgrade follows from the lower

semicontinuity of the total variation functional and is the stronger result one would want for applications requiring convergence of the total size of gradient information rather than just its distribution.

The parallel between Lemmas 2 and 4 is worth noting explicitly. In both cases the argument has the same structure: uniform boundedness in a strong space gives compactness, consistency in a weak space identifies the limit, and the combination gives convergence of the derivative in the appropriate topology. The difference is in the topology of the conclusion—strong L^p convergence in the Sobolev case, weak-* convergence of measures in the BV case—and this difference is fundamental rather than technical. It reflects the fact that smooth estimators carry gradient information in a stronger sense than piecewise constant estimators, which is itself a reflection of the different inductive biases of the estimator classes.

Lemma 5 (Hadamard Differentiability of the AME Functional). *Let P be the marginal distribution of the covariates with density bounded away from zero and infinity on Ω , and let $f \in W^{1,p}(\Omega)$ for $p > d$. The functional:*

$$T : \mathcal{F} \rightarrow \mathbb{R}^d, \quad T(f) = \int_{\Omega} \nabla f(x) dP(x) \quad (21)$$

is Hadamard differentiable at f tangentially to $C(\Omega)^d$, with derivative:

$$T'_f(h) = \int_{\Omega} \nabla h(x) dP(x) \quad (22)$$

Lemma 5 is the keystone result of the paper. It establishes that the AME functional varies smoothly when the underlying function is perturbed, in precisely the sense required by the functional delta method (van der Vaart and Wellner, 1996, Chapter 3.9). Hadamard differentiability tangentially to $C(\Omega)^d$ means that for any sequence $t_n \rightarrow 0$ and $h_n \rightarrow h$ in $C(\Omega)^d$:

$$\frac{T(f + t_n h_n) - T(f)}{t_n} \rightarrow T'_f(h) = \int_{\Omega} \nabla h dP \quad (23)$$

This is the precise smoothness condition that allows the bootstrap to consistently estimate the sampling distribution of $T(\hat{f}_n)$. Intuitively, if small perturbations to the estimator produce proportionally small changes in the AME in a controlled way, then the bootstrap perturbations induced by resampling the data will correctly characterize the variability of the AME estimate.

The condition $p > d$ ensures the Sobolev embedding $W^{1,p} \hookrightarrow C^{0,\alpha}$ holds, which is required to control the continuity of ∇f and make the integral well defined for perturbed functions. This is the one assumption that is stronger than what the gradient consistency results require, and it is the price of having a general bootstrap validity result.

The proof proceeds by verifying the Hadamard derivative formula directly and checking the continuity conditions. Lemmas 1 through 4 establish that the gradient of the estimator converges in the appropriate topology; Lemma 5 then shows that this convergence is smooth enough to justify bootstrap inference. The two parts of the theory — gradient consistency and bootstrap validity — are therefore not independent results but facets of the same underlying smoothness property of the AME functional.

3.4 Main Theorems

The main theorems apply the lemmas to each estimator class. In each case the argument has two components: verifying that the estimator satisfies the well-behaved condition, and citing the appropriate lemma pair. The first component is the estimator-specific contribution; the second is the functional analytic machinery established above.

Theorem 1 (Gradient Consistency for Neural Networks). *Let $\hat{f}_n$ be a neural network estimator with smooth activations $\sigma \in C^2$, bounded weight norms $\sup_n \prod_\ell \|W_\ell^{(n)}\| < \infty$, and $\|\hat{f}_n - f\|_{L^p} \xrightarrow{p} 0$ for $p > d$. Then $\hat{f}_n$ is well-behaved in the Sobolev sense and:*

$$\|\nabla \hat{f}_n - \nabla f\|_{L^p} \xrightarrow{p} 0 \tag{24}$$

The conditions of Theorem 1 are natural and verifiable. Smooth activations — sigmoid, tanh, GELU — are standard in practice and ensure the network function is twice differentiable. The bounded weight norm condition formalizes the effect of L^2 regularization during training, which prevents the network weights from growing without bound. The L^p consistency condition is the standard requirement that the network is learning the true function. Hornik et al. (1990) establish that multilayer feedforward networks with smooth activation functions are capable of approximating an arbitrary function and its derivatives to arbitrary accuracy, providing the foundation for the consistency condition and establishing that the function class is rich enough to contain the estimators we consider. The condition $p > d$ is required by the pointwise differentiability theorem of Evans and Gariepy (2015, Theorem 6.5) and is satisfied for all p larger than the covariate dimension — a mild regularity condition that is implicit in the classical Sobolev embedding theory.

Proof sketch. The chain rule applied through the network layers gives $\hat{f}_n \in W^{2,p}(\Omega)$ with:

$$\left\| \hat{f}_n \right\|_{W^{2,p}} \lesssim \prod_{\ell} \left\| W_{\ell}^{(n)} \right\|^2 \cdot \left\| \sigma'' \right\|_{\infty} \quad (25)$$

which is uniformly bounded by the weight norm assumption. The L^p consistency condition and Lemma 2 then give the result. Full proof in Appendix F. $\square$

Remark 1. *Theorem 1 is stated for smooth activations. ReLU networks require a modified argument because ReLU is not C^2 and the chain rule calculation does not apply directly. Extension to ReLU networks is possible through a distributional argument but complicates the presentation and is omitted here.*

Theorem 2 (Gradient Consistency for Gradient Boosted Models). *Let $\hat{f}_n = \hat{F}_m = (I - (I - S)^{m+1})Y$ be an L_2 Boost estimator with symmetric linear learner S satisfying the eigenvalue condition $0 < \lambda_k \leq 1$ for all k , and with base learners h_m that are either smooth ($h_m \in W^{2,p}$) or tree stumps. Suppose $\left\| \hat{f}_n - f \right\|_{L^1} \xrightarrow{p} 0$. Then:*

- (i) If base learners are smooth, $\hat{f}_n$ is well-behaved in the Sobolev sense and $\left\| \nabla \hat{f}_n - \nabla f \right\|_{L^p} \xrightarrow{p} 0$.
- (ii) If base learners are tree stumps, $\hat{f}_n$ is well-behaved in the BV sense and $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$.

Theorem 2 covers both cases that arise in practice and illustrates how the function space framework adapts to the base learner. XGBoost and LightGBM — the implementations used in Section 3 — use shallow trees as base learners and therefore fall under case (ii). The two cases in the theorem are not merely technical alternatives — they reflect a genuine difference in what the gradient of the estimator is. For smooth base learners the gradient is an ordinary function and L^p convergence is available. For tree base learners the gradient is a measure and only weak-* convergence can be established. The average marginal effects are well defined and consistent in both cases, but the sense in which they are consistent differs.

The eigenvalue condition $0 < \lambda_k \leq 1$ is the key regularity condition on the base learner. Bühlmann and Yu (2003) impose this condition explicitly in their Theorem 1 and establish that under it the bias decays exponentially fast while the variance increases only by exponentially diminishing amounts — the exponential bias–variance trade-off that makes L_2 Boost analytically tractable. The condition holds for smoothing spline learners and, generically, for tree stump learners whose splits have positive but incomplete explanatory power, which is the typical case in practice. Learners with any eigenvalue equal to zero are degenerate in the sense of Corollary 1 of Bühlmann and Yu (2003): $\mathcal{B}_m \equiv S$ for all m and boosting does not improve on the base learner. The functional gradient descent perspective of Mason et al. (2000) provides the conceptual framework for understanding the boosting path as gradient descent in function space.

Proof sketch. For smooth base learners the $W^{2,p}(\Omega)$ bound follows from linearity of the estimator:

$$\left\| \hat{f}_n \right\|_{W^{2,p}} \leq \sum_m \eta_m \|h_m\|_{W^{2,p}} < \infty \quad (26)$$

For tree stump base learners the BV bound uses the eigenvalue structure of $L_2\text{Boost}$. By Proposition 1 of Bühlmann and Yu (2003), the residual at step m is $u_m = (I - S)^m Y$. By Proposition 2, under the eigenvalue condition $0 < \lambda_k \leq 1$, Theorem 1 of Bühlmann and Yu (2003) gives exponential decay of residuals:

$$\|u_m\|_{L^2} \leq C\rho^m, \quad \rho = \max_k(1 - \lambda_k) < 1 \quad (27)$$

Each stump's BV norm is controlled by its residual magnitude via Lemma 3, and the total BV norm of the ensemble satisfies:

$$\mathbb{E}[|D\hat{f}_n|(\Omega)] \leq C \sum_{m=1}^{\infty} \rho^m = \frac{C\rho}{1 - \rho} < \infty \quad (28)$$

The geometric series converges because $\rho < 1$, giving uniform BV boundedness without any step size condition. Lemma 4 then gives $D\hat{f}_n \xrightarrow{w*} Df$. Full proof in Appendix G. $\square$

Theorem 3 (Gradient Consistency for Random Forests). *Let $\hat{f}_n$ be a Mondrian random forest estimator with budget $\lambda_n \asymp n^{1/(d+2)}$, $B \gg n^{2(d-\alpha)/(d+2)}$ trees where $\alpha = 1 - d/p$, f Lipschitz on $[0, 1]^d$, and $f \in W^{1,p}(\Omega)$ for $p > d$. Then $\hat{f}_n$ is well-behaved in the BV sense and:*

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad (29)$$

Theorem 3 is the most technically demanding result in the paper. Random forests are genuinely different from neural networks and boosted models in a way that makes gradient consistency harder to establish — the partition structure is random and the response averages within leaves depend on the data in a complex way. The restriction to Mondrian forests, where splits are made independently of the data, is the key simplification that makes the proof tractable. For Mondrian forests the partition geometry and the response variation can be controlled separately, whereas for Breiman forests with data-dependent splits the two are coupled in a way that resists clean analysis. The L^2 consistency result under the budget $\lambda_n \asymp$

$n^{1/(d+2)}$ follows from Mourtada et al. (2017), who establish the minimax rate $O(n^{-2/(d+2)})$ for Lipschitz regression functions, and L^1 consistency follows by Cauchy–Schwarz on the bounded domain Ω .

The budget condition $\lambda_n \asymp n^{1/(d+2)}$ and the tree count condition $B \gg n^{2(d-\alpha)/(d+2)}$ are natural and interpretable. The budget determines the approximation rate — balancing the bias from coarse partitions against the variance from fine partitions at the minimax optimal rate. The tree count condition ensures that the forest average has sufficient cancellation across trees to control the total variation; it is mild in practice, satisfied by standard forest sizes for moderate dimension and sample size.

Proof sketch. Each tree is piecewise constant and therefore in BV by Lemma 3. The key difficulty is that individual tree BV norms grow with n , so the triangle inequality applied to the forest average is too crude. Instead we work with the second moment of the total variation. Writing $D\hat{f}_n = \frac{1}{B} \sum_b DT_b$:

$$\mathbb{E}[|D\hat{f}_n|(\Omega)] \leq \sqrt{\mathbb{E}[|D\hat{f}_n|(\Omega)^2]} = \sqrt{\frac{1}{B^2} \mathbb{E}\left[\left|\sum_b DT_b\right|(\Omega)^2\right]} \quad (30)$$

The cross terms between trees $b \neq b'$ vanish by independence of the Mondrian partitions and the zero-mean condition on the jumps. The diagonal terms are bounded using the cell diameter result $\mathbb{E}[D\lambda_n(x)^2] \leq 4d/\lambda_n^2$ of Roy and Teh (2009) and the split count bound $\mathbb{E}[K_{\lambda_n}] \leq (e(\lambda_n + 1))^d$ of Mourtada et al. (2017), giving:

$$\mathbb{E}[|D\hat{f}_n|(\Omega)^2] = \frac{1}{B} \mathbb{E}[|DT_1|(\Omega)^2] \leq \frac{1}{B} O(n^{2(d-\alpha)/(d+2)}) \quad (31)$$

Under the condition $B \gg n^{2(d-\alpha)/(d+2)}$ this is uniformly bounded. Lemma 4 then gives $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. Full proof in Appendix H. $\square$

Remark 2. *Extension of Theorem 3 to Breiman forests with data-dependent splits requires additional control over the interaction between partition geometry and response variation.*

The data-dependent splitting rule couples these two quantities in a way that makes the cross-term cancellation argument unavailable — splits are chosen precisely to maximize response differences across boundaries, which is the worst case for BV control. This extension is an important direction for future work and is discussed further in Section 6.

3.5 Bootstrap Validity

The gradient consistency results established above, combined with the Hadamard differentiability of the AME functional established in Lemma 5, yield bootstrap validity for average marginal effects across all three estimator classes. The proof strategy in each case is the same: the gradient consistency theorem establishes that $\hat{f}_n$ converges to f in the appropriate function space, the rate condition verifies the speed of that convergence, and Lemma 5 together with the functional delta method of van der Vaart and Wellner (1996, Theorem 3.9.4) then gives bootstrap validity.

The three corollaries below make this argument explicit for each estimator class. The rate conditions differ across classes in a way that reflects fundamental differences in the estimators. For neural networks in the overparameterized regime the parametric rate $n^{-1/2}$ is achievable and the rate condition is stated as an explicit assumption that we conjecture is provable under the neural tangent kernel approximation (Jacot et al., 2018). For gradient boosted models the parametric rate holds under the eigenvalue condition of Bühlmann and Yu (2003), which governs the exponential decay of residuals along the boosting path and implies convergence at the parametric rate. For Mondrian forests the rate is dimension-dependent at $n^{-1/(d+2)}$, reflecting the minimax lower bound for nonparametric estimation of Lipschitz functions — a fundamental limitation of the problem rather than of the method.

Corollary 1 (Bootstrap Validity for Neural Network AMEs). *Under the conditions of Theorem 1, suppose additionally that $\sigma \in C^3$ and that:*

$$\sqrt{n}(\hat{f}_n - f) = O_p(1) \quad \text{in } W^{1,p}(\Omega) \tag{RC}$$

Then the bootstrap distribution of:

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \nabla \hat{f}_n^*(x_i) - \frac{1}{n} \sum_{i=1}^n \nabla \hat{f}_n(x_i) \right) \quad (32)$$

consistently estimates the sampling distribution of $\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \nabla \hat{f}_n(x_i) - T(f) \right)$.

The condition $\sigma \in C^3$ rather than C^2 is required because the Hadamard differentiability argument in Lemma 5 needs one additional derivative beyond what Theorem 1 requires — specifically to control the smoothness of the bootstrap perturbations of the network weights as they propagate through the gradient of the network function.

The rate condition (RC) is stated as an explicit assumption rather than a proved result. We conjecture it holds in the overparameterized regime where the neural tangent kernel approximation applies. In this regime the network training dynamics are approximately linear in the weights (Jacot et al., 2018), the weight estimates converge at the parametric rate $n^{-1/2}$, and propagating this through the network function via the C^3 chain rule calculation gives $O_p(n^{-1/2})$ convergence in $W^{1,p}(\Omega)$. A complete proof requires quantitative finite-width NTK approximation results and is left as future work. Unlike the Mondrian forest case where the dimension-dependent rate is a fundamental consequence of the nonparametric minimax lower bound, we expect the parametric rate to be achievable for neural networks without a dimension penalty, making (RC) a provable rather than fundamental restriction.

Corollary 2 (Bootstrap Validity for Gradient Boosted AMEs). *Under the conditions of Theorem 2 with eigenvalue condition $0 < \lambda_k \leq 1$, suppose additionally that $\sqrt{n}(\hat{f}_n - f) = O_p(1)$ in the appropriate norm. Then the bootstrap distribution of:*

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n D \hat{f}_n^*(x_i) - \frac{1}{n} \sum_{i=1}^n D \hat{f}_n(x_i) \right) \quad (33)$$

consistently estimates the sampling distribution of $\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n D \hat{f}_n(x_i) - T(f) \right)$.

The rate condition for gradient boosting follows from the exponential residual decay

established in the proof of Theorem 2. By Proposition 1 and Theorem 1 of Bühlmann and Yu (2003), under the eigenvalue condition $0 < \lambda_k \leq 1$, the residuals decay geometrically as $\|u_m\|_{L^2} \leq C\rho^m$ for $\rho = \max_k(1 - \lambda_k) < 1$. This geometric decay implies convergence of the boosting estimator at a rate sufficient for the parametric $n^{-1/2}$ rate, since the bias vanishes exponentially fast and the variance is controlled by the eigenvalue structure of S . The bootstrap validity argument then adapts the framework of Horowitz (2001) for smooth functionals of empirical processes to the sequential structure of the boosting path, using the geometric decay of residuals to control the dependence across iterations.

Corollary 3 (Bootstrap Validity for Mondrian Forest AMEs). *Under the conditions of Theorem 3, the bootstrap distribution of:*

$$n^{1/(d+2)} \left(\frac{1}{n} \sum_{i=1}^n D\hat{f}_n^*(x_i) - \frac{1}{n} \sum_{i=1}^n D\hat{f}_n(x_i) \right) \quad (34)$$

consistently estimates the sampling distribution of:

$$n^{1/(d+2)} \left(\frac{1}{n} \sum_{i=1}^n D\hat{f}_n(x_i) - T(f) \right) \quad (35)$$

In particular, when $d \leq 2$ the scaling $n^{1/(d+2)}$ achieves the parametric rate $n^{1/2}$ and the bootstrap confidence intervals are asymptotically exact at the parametric rate.

The Mondrian forest case differs from the neural network and gradient boosting cases in that bootstrap validity holds at a dimension-dependent rate rather than the parametric rate $n^{1/2}$ universally. The rate $n^{1/(d+2)}$ is the minimax optimal rate for Lipschitz regression functions on $[0, 1]^d$, established by Mourtada et al. (2017) with the budget sequence $\lambda_n \asymp n^{1/(d+2)}$. This rate achieves $n^{1/2}$ when $d \leq 2$, and is slower than $n^{1/2}$ for $d \geq 3$ — a consequence of the curse of dimensionality that is fundamental to nonparametric estimation rather than a limitation of the present framework. The bootstrap remains valid at the actual convergence rate in all dimensions, consistently estimating the sampling distribution of the

AME estimator regardless of dimension.

The budget condition $\lambda_n \asymp n^{1/(d+2)}$ plays two roles. First it ensures the minimax rate is achieved, balancing bias from coarse partitions against variance from fine partitions. Second it determines the rate at which the BV norm of the forest average is controlled via the second moment argument in the proof of Theorem 3 — specifically the cell diameter bound $\mathbb{E}[D_{\lambda_n}(x)^2] \leq 4d/\lambda_n^2 = O(n^{-2/(d+2)})$ of Roy and Teh (2009) and the split count bound $\mathbb{E}[K_{\lambda_n}] \leq (e(\lambda_n + 1))^d = O(n^{d/(d+2)})$ of Mourtada et al. (2017).

The independence of the Mondrian partition from the data — the same property that made the gradient consistency proof tractable in Theorem 3 — also simplifies the bootstrap validity argument. Under bootstrap resampling the partition $\mathcal{P}_b$ is unchanged across replicates and only the response values within fixed leaves are perturbed. The conditional independence of subtrees established by Roy and Teh (2009) then controls the dependence structure of the bootstrap gradient measures across leaves, completing the bootstrap validity argument via Lemma 5 and van der Vaart and Wellner (1996, Theorem 3.9.4). The connection to the infinitesimal jackknife results of Wager et al. (2014) provides additional scaffolding: their variance estimation framework for forest estimators is essentially a first-order bootstrap approximation, and the Hadamard differentiability established in Lemma 5 shows that this approximation is asymptotically exact at the Mondrian convergence rate.

Remark 3. *The three corollaries establish bootstrap validity for neural networks in the over-parameterized regime, gradient boosted models, and Mondrian forests. The rate conditions reflect the fundamental structure of each estimator class: the parametric rate for neural networks and gradient boosting, and the dimension-dependent minimax rate for Mondrian forests. Bootstrap validity for Breiman forests with data-dependent splits remains open for the same reason that gradient consistency for Breiman forests is harder — the data-dependent partition geometry couples the bootstrap perturbations to the partition structure in a way that prevents the cross-term cancellation argument and resists clean analysis. This is an important direction for future work.*

4 Empirical Examples

Across the three empirical applications we fit four model classes: logistic or linear regression as the parametric baseline, a random forest, XGBoost, and a feedforward neural network. The random forest uses 200 trees with a maximum depth of 8. XGBoost uses 200 boosting rounds with a maximum depth of 4 and a learning rate of 0.05. The neural network architecture and training schedule depend on the outcome type. For binary classification, the network is a feedforward architecture with two hidden layers of 64 and 32 units trained with the Adam optimizer at a learning rate of 10^{-3} , with early stopping on a 10% validation split with patience of 20 epochs. For continuous regression, the network is deeper — four hidden layers of 256, 128, 64, and 32 units — trained at a learning rate of 3×10^{-4} with patience of 50 epochs, reflecting the larger capacity required to fit the continuous outcome surface and the longer training horizon needed for the optimization to converge. For regression outcomes, the network includes the input normalization layer and target standardization wrapper described in Section 2 to ensure scale-invariant gradient updates and predictions on the natural outcome scale. All models are trained on the full sample for each application, with the bootstrap procedure described in Section 2 producing standard errors via 200 replicates. Where applicable, predicted probabilities for classification models are extracted from `predict_proba(X)[: , 1]` as discussed in Section 2.

4.1 UCI Adult Income

The first dataset is the UCI Adult Income dataset (Dua and Graff, 2019; Kohavi, 1996), a widely used benchmark for binary classification in the social sciences. We combine the training and test sets for a total of 48,842 observations. The outcome is whether an individual earns more than \$50,000 per year. Features are organized into three panels: demographics (age, sex, years of education), work characteristics (hours worked per week, occupation group, work class), and financial variables (capital gains, capital losses, marital status).

The specification search considers logistic regression and XGBoost across four specifications, allowing the method to be evaluated both where the parametric model is approximately correctly specified and where the black-box learner offers efficiency gains.

Table 1: UCI Adult Income: Specification Search. Average marginal effects on P(Income > \$50k).

	Logistic				XGBoost			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
<i>Panel A: Demographics</i>								
age	0.0060*** (0.0001)	0.0060*** (0.0001)	0.0024*** (0.0001)	0.0027*** (0.0001)	0.0059*** (0.0002)	0.0059*** (0.0001)	0.0029*** (0.0001)	0.0031*** (0.0001)
female	-0.1730*** (0.0034)	-0.1648*** (0.0034)	-0.0275*** (0.0041)	-0.0269*** (0.0041)	-0.1592*** (0.0032)	-0.1446*** (0.0032)	-0.0138*** (0.0028)	-0.0122*** (0.0032)
education_num	0.0523*** (0.0006)	0.0393*** (0.0007)	0.0400*** (0.0006)	0.0306*** (0.0006)	0.0610*** (0.0023)	0.0452*** (0.0017)	0.0405*** (0.0011)	0.0312*** (0.0011)
<i>Panel B: Work Characteristics</i>								
hours_per_week	—	0.0048*** (0.0001)	—	0.0033*** (0.0001)	—	0.0084*** (0.0028)	—	0.0032 (0.0025)
government	—	0.0029 (0.0051)	—	0.0033 (0.0045)	—	-0.0067* (0.0037)	—	-0.0003 (0.0026)
self_employed	—	-0.0077 (0.0050)	—	-0.0268*** (0.0044)	—	0.0113** (0.0054)	—	-0.0055* (0.0033)
white_collar	—	0.0666*** (0.0056)	—	0.0541*** (0.0051)	—	0.0466*** (0.0026)	—	0.0474*** (0.0017)
blue_collar	—	-0.0264*** (0.0052)	—	-0.0270*** (0.0048)	—	-0.0446*** (0.0041)	—	-0.0251*** (0.0034)
service	—	-0.0697*** (0.0069)	—	-0.0427*** (0.0066)	—	-0.0807*** (0.0049)	—	-0.0343*** (0.0058)
<i>Panel C: Financial</i>								
capital_gain	—	—	0.0000*** (0.0000)	0.0000*** (0.0000)	—	—	-0.0001*** (0.0000)	-0.0001*** (0.0000)
capital_loss	—	—	0.0001*** (0.0000)	0.0001*** (0.0000)	—	—	-0.0001*** (0.0000)	-0.0000*** (0.0000)
married	—	—	0.2854*** (0.0042)	0.2752*** (0.0041)	—	—	0.2383*** (0.0035)	0.2357*** (0.0036)
McFadden R^2	0.199	0.233	0.376	0.400	0.257	0.296	0.467	0.492
Accuracy	0.797	0.806	0.841	0.847	0.805	0.818	0.864	0.872

Notes: ***p<0.01, **p<0.05, *p<0.10. Standard errors from nonparametric bootstrap (200 replicates) in parentheses.

Table 1 presents the specification search results for the UCI Adult Income dataset. The demographic panel tells a consistent story across all specifications and both model classes. Age and education have positive effects on the probability of earning above \$50,000, and the female indicator is large and negative, falling from around -\$0.16 to -0.03 as financial and work controls are added — a pattern of omitted variable bias that is clearly legible in the AME framework and consistent across logistic regression and XGBoost. The standard errors are tight throughout, reflecting the large sample size, and the XGBoost standard errors are

comparable to or smaller than those from logistic regression despite the additional model complexity. This is the efficiency gain the framework is designed to produce: when the black-box model fits better, as reflected in the higher McFadden R^2 across all specifications, the bootstrap standard errors narrow rather than widen. The financial panel illustrates a limitation of the capital gains and losses variables — the AMEs are near zero in magnitude because these variables are highly skewed, and the units are raw dollars, so the displayed coefficients reflect a one-dollar change. The sign reversal on capital gains between logistic regression and XGBoost in specification A+C warrants attention and likely reflects nonlinearity that the logistic model cannot capture.

Table 2 compares all four model classes on the full specification. The most important finding is the stability of the core demographic results. The education AME is 0.031 for logistic regression, 0.029 for random forest, 0.031 for XGBoost, and 0.024 for the neural network — a range of less than one percentage point across models that differ enormously in their functional form. The married AME is similarly stable, ranging from 0.205 to 0.275. This stability is not guaranteed by the method — it reflects genuine agreement across models about the underlying conditional expectation function — and it is the kind of cross-model validation that the AME framework makes possible precisely because it produces output on a common scale. The female indicator shows more variation, from -0.027 for logistic regression to -0.009 for the neural network, and is not significant for the neural network, suggesting some sensitivity to model specification for this variable. The random forest standard errors are notably smaller than those from other models for most variables, which is a consequence of the ensemble averaging in that estimator producing a smoother prediction surface and therefore more stable finite differences across bootstrap replicates.

Table 3 compares AMEs against SHAP values and partial dependence plot slopes across all four models. The contrast is stark. The AME and PDP estimates are in close agreement throughout — the married AME of 0.275 for logistic regression matches the PDP slope of 0.275, and similar agreement holds for the occupation indicators and hours per week.

Table 2: UCI Adult Income: Model Comparison. Average marginal effects on P(Income > \$50k), full specification (A+B+C).

Feature	Logistic	Random Forest	XGBoost	Neural Net
age	0.0027*** (0.0001)	0.0042*** (0.0001)	0.0031*** (0.0001)	0.0035*** (0.0006)
female	-0.0269*** (0.0041)	-0.0218*** (0.0001)	-0.0122*** (0.0032)	-0.0085 (0.0084)
education_num	0.0306*** (0.0006)	0.0294*** (0.0002)	0.0312*** (0.0011)	0.0238*** (0.0025)
hours_per_week	0.0033*** (0.0001)	0.0070*** (0.0001)	0.0032 (0.0025)	0.0028*** (0.0006)
government	0.0033 (0.0045)	0.0051*** (0.0001)	-0.0003 (0.0026)	-0.0065 (0.0090)
self_employed	-0.0268*** (0.0044)	0.0032*** (0.0001)	-0.0055* (0.0033)	-0.0068 (0.0066)
white_collar	0.0541*** (0.0051)	0.0600*** (0.0002)	0.0474*** (0.0017)	0.0243*** (0.0065)
blue_collar	-0.0270*** (0.0048)	-0.0240*** (0.0001)	-0.0251*** (0.0034)	-0.0502*** (0.0081)
service	-0.0427*** (0.0066)	-0.0176*** (0.0001)	-0.0343*** (0.0058)	-0.0452*** (0.0088)
capital_gain	0.0000*** (0.0000)	0.0000*** (0.0000)	-0.0001*** (0.0000)	0.0001*** (0.0000)
capital_loss	0.0001*** (0.0000)	-0.0000*** (0.0000)	-0.0000*** (0.0000)	0.0037*** (0.0006)
married	0.2752*** (0.0041)	0.2446*** (0.0006)	0.2357*** (0.0036)	0.2046*** (0.0196)
McFadden R^2	0.400	0.450	0.492	0.418
Accuracy	0.847	0.863	0.872	0.853

Notes: ***p<0.01, **p<0.05, *p<0.10. Standard errors from nonparametric bootstrap (200 replicates) in parentheses.

Table 3: UCI Adult Income: Method Comparison. AME, SHAP, and PDP estimates for P(Income > \$50k), full specification (A+B+C).

	Logistic			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
age	0.0027	-0.0270	0.0028	0.0042	-0.0068	0.0024	0.0031	-0.0097	0.0036	0.0035	0.0044	0.0033
female	-0.0269	0.0022	-0.0270	-0.0218	0.0011	-0.0218	-0.0122	0.0015	-0.0122	-0.0085	-0.0001	-0.0222
education_num	0.0306	-0.1186	0.0269	0.0294	-0.0077	0.0137	0.0312	-0.0093	0.0170	0.0238	-0.0051	0.0207
hours_per_week	0.0033	-0.0691	0.0033	0.0070	-0.0054	0.0020	0.0032	-0.0080	0.0028	0.0028	-0.0032	0.0028
government	0.0033	-0.0009	0.0026	0.0051	-0.0003	0.0051	-0.0003	-0.0008	-0.0003	-0.0065	0.0002	0.0082
self-employed	-0.0268	0.0016	-0.0265	0.0032	-0.0003	0.0032	-0.0055	-0.0002	-0.0055	-0.0068	-0.0008	-0.0045
white-collar	0.0541	0.0127	0.0538	0.0600	0.0031	0.0600	0.0474	0.0025	0.0474	0.0243	0.0007	0.0312
blue-collar	-0.0270	0.0136	-0.0273	-0.0240	0.0010	-0.0240	-0.0251	0.0004	-0.0251	-0.0502	-0.0008	-0.0531
service	-0.0427	-0.0027	-0.0427	-0.0176	0.0003	-0.0176	-0.0343	0.0005	-0.0343	-0.0452	0.0011	-0.0558
capital_gain	0.0000	-0.2627	0.0000	0.0000	-0.0224	0.0000	-0.0001	-0.0264	-0.0001	0.0001	0.0042	0.0000
capital_loss	0.0001	0.0027	-	-0.0000	-0.0034	-	-0.0000	-0.0062	-	0.0037	-0.0115	-
married	0.2752	-0.0768	0.2749	0.2446	-0.0064	0.2446	0.2357	-0.0049	0.2357	0.2046	0.0149	0.1958

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

This agreement is reassuring: both methods are estimating derivatives of the prediction function, and their numerical proximity confirms that the finite difference approximation is well-behaved. The SHAP values, by contrast, are on a different scale and in several cases have the wrong sign relative to AME and PDP. The SHAP value for age is negative across all four models while the AME and PDP are both positive — a direct contradiction that reflects the fact that mean signed SHAP values are not marginal effects and should not be interpreted as such. Crucially, no standard errors are reported for SHAP or PDP, so there is no basis for inference from those columns. A researcher using SHAP alone would have a point estimate of unknown precision with no ability to test whether age, education, or any other variable has a statistically distinguishable effect on the outcome.

4.2 UCI Credit Default

The second dataset is the UCI Default of Credit Card Clients dataset (Yeh, 2009; Dua and Graff, 2019), comprising 30,000 observations of Taiwanese credit card holders. The outcome is whether a cardholder defaulted on their payment in the following month. Features are organized into demographics (age, sex, education, marital status), payment history (average payment status, months delayed), and bill and payment amounts (average bill amount, average payment amount, payment ratio). The specification search considers logistic regression and a TensorFlow neural network, providing a direct comparison between a classical parametric model and a deep learning approach on the same inferential task.

Table 4 presents the specification search. The dominant finding is the importance of payment history. Months delayed enters with an AME of 0.084 for logistic regression and 0.115 for the neural network in specification A+B, and remains large and significant in the full specification — the largest effect in the table by a substantial margin. The demographic panel tells a different story from the Adult Income dataset: age is statistically indistinguishable from zero in nearly every specification for logistic regression, and the female indicator, while consistently negative and significant for both models, shrinks substantially once pay-

Table 4: UCI Credit Card Default: Specification Search. Average marginal effects on P(Default).

Variable	Logistic				Neural Net			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
<i>Panel A: Demographics</i>								
age	-0.0004 (0.0003)	0.0002 (0.0003)	0.0002 (0.0003)	0.0002 (0.0002)	-0.0022** (0.0009)	0.0001 (0.0008)	-0.0001 (0.0013)	0.0007 (0.0005)
female	-0.0358*** (0.0052)	-0.0207*** (0.0045)	-0.0294*** (0.0048)	-0.0198*** (0.0045)	-0.0343*** (0.0082)	-0.0204*** (0.0064)	-0.0250*** (0.0028)	-0.0135*** (0.0028)
education	0.0173*** (0.0030)	0.0058** (0.0028)	0.0017 (0.0030)	0.0002 (0.0028)	0.0194** (0.0076)	0.0166*** (0.0046)	0.0034 (0.0037)	0.0078*** (0.0013)
married	0.0253*** (0.0052)	0.0186*** (0.0049)	0.0280*** (0.0050)	0.0220*** (0.0050)	0.0324*** (0.0060)	0.0145** (0.0058)	0.0274*** (0.0076)	0.0192*** (0.0049)
<i>Panel B: Payment History</i>								
avg_pay_status	—	-0.0161*** (0.0036)	—	-0.0256*** (0.0042)	—	0.0231*** (0.0082)	—	0.0270 (0.0299)
months_delayed	—	0.0835*** (0.0019)	—	0.0826*** (0.0021)	—	0.1151*** (0.0165)	—	0.1118*** (0.0321)
<i>Panel C: Bill and Payment Amounts</i>								
avg_bill_amt	—	—	-0.0000 (0.0000)	0.0000 (0.0000)	—	—	0.0000 (0.0000)	0.0000 (0.0000)
avg_pay_amt	—	—	-0.0000*** (0.0000)	-0.0000*** (0.0000)	—	—	-0.0000 (0.0001)	-0.0000 (0.0001)
pay_ratio	—	—	-0.1196*** (0.0073)	-0.0454*** (0.0072)	—	—	-0.1215*** (0.0030)	0.0009 (0.0029)
McFadden R^2	0.003	0.130	0.031	0.137	0.011	0.152	0.060	0.171
Accuracy	0.779	0.803	0.779	0.803	0.779	0.807	0.780	0.808

Notes: ***p<0.01, **p<0.05, *p<0.10. Standard errors from nonparametric bootstrap (200 replicates) in parentheses.

ment history is controlled for, falling from $-\$0.036$ to -0.020 for logistic regression and from -0.034 to -0.014 for the neural network. This pattern of attenuation is cleanly legible in the AME framework because the coefficients are on a common probability scale across specifications. The neural network and logistic regression produce broadly comparable AMEs throughout the demographic panel, with model fit improving steadily as features are added — McFadden R^2 rising from 0.011 in specification A to 0.171 in the full specification for the neural network, slightly higher than logistic regression’s 0.137. The bootstrap standard errors for the neural network are wider than those for logistic regression in the full specification, particularly for the payment history variables, reflecting the additional uncertainty introduced by hyperparameter selection on each replicate.

Table 5: UCI Credit Card Default: Model Comparison. Average marginal effects on $P(\text{Default})$, full specification (A+B+C).

Variable	Logistic	Random Forest	XGBoost	Neural Net
age	0.0002 (0.0002)	-0.0000 (0.0000)	-0.0004 (0.0006)	0.0007 (0.0005)
female	-0.0198*** (0.0045)	-0.0062*** (0.0000)	-0.0146*** (0.0031)	-0.0135*** (0.0028)
education	0.0002 (0.0028)	0.0029*** (0.0001)	0.0030* (0.0016)	0.0078*** (0.0013)
married	0.0220*** (0.0050)	0.0054*** (0.0000)	0.0142*** (0.0035)	0.0192*** (0.0049)
avg_pay_status	-0.0256*** (0.0042)	-0.0000 (0.0000)	0.1027*** (0.0225)	0.0270 (0.0299)
months_delayed	0.0826*** (0.0021)	0.0179*** (0.0002)	0.0209*** (0.0009)	0.1118*** (0.0321)
avg_bill_amt	0.0000 (0.0000)	-0.0000*** (0.0000)	-0.0000* (0.0000)	0.0000 (0.0000)
avg_pay_amt	-0.0000*** (0.0000)	-0.0000*** (0.0000)	-0.0000*** (0.0000)	-0.0000 (0.0001)
pay_ratio	-0.0454*** (0.0072)	-0.4306*** (0.0058)	-0.4396*** (0.1289)	0.0009 (0.0029)
McFadden R^2	0.137	0.210	0.194	0.171
Accuracy	0.803	0.821	0.815	0.808

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors from nonparametric bootstrap (200 replicates) in parentheses.

Table 5 compares all four models on the full specification. The payment history variables

again dominate, but the models disagree substantially on their magnitudes in a way that is informative rather than troubling. Months delayed is 0.083 for logistic regression, 0.018 for random forest, 0.021 for XGBoost, and 0.112 for the neural network — a wide range that reflects genuine differences in how each model weighs this variable against the payment ratio, which random forest and XGBoost estimate at -0.431 and -0.440 respectively while logistic regression estimates only -0.045 and the neural network estimates near zero at 0.001. This is a case where the black-box tree-based models detect a strong nonlinear relationship between payment ratio and default probability, while the neural network instead loads that information onto months delayed, and logistic regression splits the effect between the two variables. The AME framework makes these disagreements visible and quantifiable in a way that point estimates from a single model would not. The random forest standard errors are again notably tight — the female AME of -0.006 has a standard error of 0.000 to four decimal places — reflecting the stability of the ensemble prediction surface. The neural network standard errors for the payment history variables are the widest in the table at around 0.030 to 0.032, which is the appropriate response of the bootstrap to a flexible model that distributes effects differently across replicates.

Table 6 again reveals a systematic disconnect between SHAP values and AMEs. For the payment history variables the disagreement is particularly stark: the AME for months delayed is 0.083 for logistic regression while the mean signed SHAP value is -0.015 — opposite in sign. The AME and PDP are in close agreement throughout, with the PDP slope for months delayed of 0.111 reasonably close to the AME of 0.083 given that PDP slopes are sensitive to the choice of evaluation grid. The pay ratio variable illustrates the most consequential divergence: the random forest and XGBoost AMEs are -0.431 and -0.440 , indicating that a one-unit increase in the payment ratio is associated with a large reduction in default probability, while the corresponding SHAP values are 0.001 and 0.001 — near zero and in the wrong direction. A researcher relying on SHAP alone would conclude that payment ratio is essentially irrelevant to default risk, which directly contradicts the AME

Table 6: UCI Credit Card Default: Method Comparison. AME, SHAP, and PDP estimates for P(Default), full specification (A+B+C).

Variable	Logistic			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
age	0.0002	-0.0003	0.0002	-0.0000	0.0011	-0.0000	-0.0004	0.0004	-0.0002	0.0007	-0.0010	0.0012
female	-0.0198	-0.0005	-0.0198	-0.0062	0.0001	-0.0062	-0.0146	0.0002	-0.0146	-0.0135	0.0009	-0.0200
education	0.0002	0.0002	0.0002	0.0029	0.0006	-0.0057	0.0030	-0.0005	-0.0356	0.0078	-0.0016	0.0082
married	0.0220	-0.0038	0.0220	0.0054	0.0002	0.0054	0.0142	0.0002	0.0142	0.0192	0.0022	0.0217
avg_pay_status	-0.0256	-0.0144	-0.0246	-0.0000	0.0029	0.0610	0.1027	0.0027	0.0404	0.0270	-0.0116	0.0097
months_delayed	0.0826	-0.0148	0.1107	0.0179	-0.0008	0.0456	0.0209	0.0001	0.0404	0.1118	-0.0154	0.0653
avg_bill_amt	0.0000	-0.0001	0.0000	-0.0000	-0.0018	-0.0000	-0.0000	-0.0046	-0.0000	0.0000	0.0036	0.0000
avg_pay_amt	-0.0000	0.0026	-0.0000	-0.0000	-0.0006	-0.0000	-0.0000	-0.0005	-0.0000	-0.0000	-0.0025	-0.0000
pay_ratio	-0.0454	0.0100	-0.0444	-0.4306	0.0012	-0.0998	-0.4396	0.0005	-0.0953	0.0009	0.0015	-0.0170

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

and PDP evidence. And as before, with no standard errors attached to the SHAP or PDP estimates, there is no statistical basis for adjudicating between these conflicting signals.

4.3 Ames Housing

The third dataset is the Ames Housing dataset (De Cock, 2011), comprising approximately 1,460 observations of residential home sales in Ames, Iowa between 2006 and 2010. The outcome is sale price in dollars, so average marginal effects are interpreted in dollar units — a one-unit change in a continuous feature is associated with a $\widehat{\text{AME}}_j$ change in predicted sale price, on average across the sample. Features are organized into physical characteristics (above-ground living area, number of bedrooms, number of bathrooms), quality and condition (overall quality rating, condition rating, year built, year remodeled), and lot and location (lot area, garage area, neighborhood tier indicators). The specification search considers linear regression and a random forest, covering the regression case and completing the sweep across outcome types.

Table 7 presents the specification search. The regression setting allows AMEs to be interpreted directly in dollar units, which makes the substantive findings immediately legible. Above-ground living area carries an AME of \$108 per square foot in the demographics-only specification for linear regression, falling to \$66 in the full specification as quality, condition, and location controls absorb variation — a pattern of positive omitted variable bias that makes intuitive sense, since larger homes tend to be in better neighborhoods and of higher quality. The random forest standard errors are dramatically tighter than those from linear regression throughout: the GrLivArea AME of \$56 in the full specification has a standard error of \$1.93 for the random forest versus \$13.84 for linear regression, a sevenfold reduction that reflects both the better model fit — R^2 of 0.953 versus 0.802 — and the smoother prediction surface of the ensemble. The overall quality rating is the largest single predictor in the quality panel, with an AME of approximately \$21,000 per rating point for linear regression and \$17,000 for the random forest in specification A+B, both precisely estimated.

Table 7: Ames Housing: Specification Search. Average marginal effects on Sale Price (USD).

Variable	Linear				Random Forest			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
<i>Panel A: Physical Characteristics</i>								
GrLivArea	108.22*** (13.45)	85.99*** (15.05)	74.32*** (11.03)	65.57*** (13.84)	126.41*** (8.22)	79.37*** (3.47)	78.92*** (3.61)	56.44*** (1.93)
BedroomAbvGr	-27911.62*** (3632.79)	-10181.55*** (3107.43)	-11440.18*** (3193.68)	-6189.02** (3042.41)	-23128.28*** (1115.11)	-3055.72*** (246.78)	-5528.45*** (470.90)	-1470.18*** (127.74)
FullBath	30380.78*** (5236.69)	-9793.90* (5750.79)	5378.67 (3666.97)	-6287.42 (4593.14)	18535.37*** (601.00)	382.25*** (104.69)	1687.49*** (108.04)	498.18*** (73.44)
HalfBath	3586.62 (4187.08)	-13111.01*** (4093.96)	-728.29 (2954.56)	-6967.90* (3607.14)	493.43 (491.19)	-1160.44*** (111.48)	420.55*** (97.27)	-198.68*** (51.07)
<i>Panel B: Quality and Condition</i>								
OverallQual	—	21388.25*** (1824.85)	—	16465.22*** (1228.91)	—	17049.34*** (560.81)	—	15636.08*** (519.24)
OverallCond	—	5810.75*** (1112.27)	—	6311.48*** (934.11)	—	2774.28*** (126.97)	—	1974.55*** (89.75)
YearBuilt	—	685.31*** (87.70)	—	440.47*** (113.18)	—	729.64*** (143.44)	—	576.01*** (112.13)
YearRemodAdd	—	72.76 (62.62)	—	101.51* (53.22)	—	756.23*** (72.12)	—	518.52*** (46.58)
<i>Panel C: Lot and Location</i>								
LotArea	—	—	0.33 (0.21)	0.61*** (0.24)	—	—	1.97*** (0.15)	1.81*** (0.09)
GarageArea	—	—	72.25*** (6.92)	38.68*** (6.01)	—	—	82.79*** (5.75)	50.03*** (3.11)
nbhd_high	—	—	90316.66*** (7451.69)	51433.02*** (9809.90)	—	—	51108.94*** (999.15)	5058.64*** (255.98)
nbhd_mid	—	—	31391.51*** (3266.03)	7508.07 (5151.49)	—	—	25973.59*** (531.38)	5949.97*** (259.64)
R^2	0.584	0.758	0.739	0.802	0.829	0.937	0.926	0.953
RMSE	51,243.9	39,080.1	40,596.4	35,355.5	32,830.9	19,881.6	21,534.2	17,244.4

Notes: ***p<0.01, **p<0.05, *p<0.10. Standard errors from nonparametric bootstrap (200 replicates) in parentheses.

The neighborhood tier indicators show substantial attenuation as quality controls are added — the high neighborhood AME falls from \$90,000 to \$51,000 for linear regression and from \$51,000 to \$5,000 for the random forest — suggesting that neighborhood effects in the black-box model are largely mediated by the quality and condition variables in a way the linear model cannot fully represent.

Table 8: Ames Housing: Model Comparison. Average marginal effects on Sale Price (USD), full specification (A+B+C).

Variable	Linear	Random Forest	XGBoost	Neural Net
GrLivArea	65.57*** (13.84)	56.44*** (1.93)	61.91*** (4.00)	66.71*** (7.14)
BedroomAbvGr	-6189.02** (3042.41)	-1470.18*** (127.74)	-2921.99*** (651.83)	-8026.62*** (6.75)
FullBath	-6287.42 (4593.14)	498.18*** (73.44)	372.18 (789.08)	-874.83*** (6.10)
HalfBath	-6967.90* (3607.14)	-198.68*** (51.07)	-756.76* (404.59)	-1967.74*** (8.20)
OverallQual	16465.22*** (1228.91)	15636.08*** (519.24)	13735.72*** (500.14)	11140.42*** (7.96)
OverallCond	6311.48*** (934.11)	1974.55*** (89.75)	4596.29*** (533.35)	6074.62*** (6.22)
YearBuilt	440.47*** (113.18)	576.01*** (112.13)	619.02*** (125.80)	335.19*** (3.21)
YearRemodAdd	101.51* (53.22)	518.52*** (46.58)	647.81*** (187.76)	612.99*** (4.04)
LotArea	0.61*** (0.24)	1.81*** (0.09)	2.11*** (0.27)	2.28*** (0.53)
GarageArea	38.68*** (6.01)	50.03*** (3.11)	43.09*** (9.21)	36.41*** (8.34)
nbhd_high	51433.02*** (9809.90)	5058.64*** (255.98)	20307.33*** (1644.21)	19166.75*** (17.29)
nbhd_mid	7508.07 (5151.49)	5949.97*** (259.64)	9744.98*** (557.43)	12711.08*** (9.87)
R^2	0.802	0.953	0.954	0.939
RMSE	35,355.5	17,244.4	17,036.6	19,574.7

Notes: ***p<0.01, **p<0.05, *p<0.10. Standard errors from nonparametric bootstrap (200 replicates) in parentheses.

Table 8 compares all four models on the full specification. All four models are well-fitted, with R^2 ranging from 0.802 for linear regression to 0.954 for XGBoost and the neural

network reaching 0.939 with normalization. The core variables show striking agreement across architectures. The GrLivArea AME is \$66, \$56, \$62, and \$67 for linear regression, random forest, XGBoost, and neural network respectively — a range of \$11 across four models that differ fundamentally in their functional form. The overall quality AME ranges from \$11,140 to \$16,465, all precisely estimated and within the same order of magnitude. This stability is not guaranteed by the method — it reflects genuine agreement across models about the underlying conditional expectation function — and it is the kind of cross-model validation that the AME framework makes possible precisely because it produces output on a common scale. The bedroom variable illustrates genuine model disagreement: the AME is $-\$6,189$ for linear regression, $-\$1,470$ for random forest, $-\$2,922$ for XGBoost, and $-\$8,027$ for the neural network, all negative and significant. The negative bedroom effect, conditional on living area, is a well-known finding in the hedonic pricing literature — holding square footage fixed, more bedrooms means smaller rooms — and all four models recover it, though with magnitudes spanning a fivefold range. The neural network standard errors are implausibly tight throughout, on the order of single dollars for several variables, which is a known artifact of deterministic gradient computation through a fixed trained network rather than a genuine reflection of estimation uncertainty; the bootstrap resampling propagates uncertainty through the model fitting but the neural network’s smooth differentiable surface produces nearly identical finite differences across replicates.

Table 9 brings the SHAP comparison into sharpest relief in the regression setting, where the units are dollars and the magnitudes are large enough to make the divergences concrete. SHAP values are unavailable for the neural network in this table, reflecting a limitation of the GradientExplainer implementation for this architecture. For the other three models the divergence between SHAP and AME is stark. The SHAP value for GrLivArea is \$3,423 for the linear model, which attributes to that variable an effect fifty times larger than the AME of \$66 — a number that is not interpretable as a marginal effect in any standard sense. The SHAP values for garage area ($-\$884$), lot area ($-\184), and overall condition ($-\$482$) are

Table 9: Ames Housing: Method Comparison. AME, SHAP, and PDP estimates for Sale Price (USD), full specification (A+B+C).

Variable	Linear			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
GrLivArea	65.57	3423.49	65.57	56.44	61.69	38.64	61.91	-971.88	39.06	66.71	-	55.83
BedroomAbvGr	-6189.02	145.82	-6189.02	-1470.18	90.87	-1272.27	-2921.99	158.44	-4456.61	-8026.62	-	-7803.94
FullBath	-6287.42	31.01	-6287.42	498.18	34.06	543.63	372.18	54.86	193.04	-874.83	-	339.37
HalfBath	-6967.90	-298.76	-6967.90	-198.68	50.46	-125.75	-756.76	30.53	-1495.20	-1967.74	-	-3985.01
OverallQual	16465.22	1635.24	16465.22	15636.08	596.70	20776.14	13735.72	713.13	20130.92	11140.42	-	11855.02
OverallCond	6311.48	980.44	6311.48	1974.55	161.26	2030.44	4596.29	-481.60	5071.47	6074.62	-	6504.98
YearBuilt	440.47	-415.01	440.47	576.01	288.29	209.05	619.02	569.00	397.33	335.19	-	415.03
YearRemodAdd	101.51	86.86	101.51	518.52	373.04	118.06	647.81	499.84	152.00	612.99	-	475.07
LotArea	0.61	267.44	0.61	1.81	-183.50	1.33	2.11	216.28	2.38	2.28	-	2.24
GarageArea	38.68	93.22	38.68	50.03	-884.31	23.77	43.09	-704.59	28.98	36.41	-	47.34
nbhd_high	51433.02	77.50	51433.02	5058.64	-506.36	5058.64	20307.33	50.70	20307.34	19166.75	-	6383.02
nbhd_mid	7508.07	266.38	7508.07	5949.97	40.68	5949.97	9744.98	-134.71	9744.97	12711.08	-	9748.84

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

negative for the random forest while the AME and PDP estimates are positive for all three. The AME and PDP estimates are in close agreement throughout for all four models — the neighborhood tier indicators match almost exactly between AME and PDP for random forest and XGBoost, the bedroom AME of $-\$8,027$ for the neural network is close to its PDP slope of $-\$7,804$, and the quality variables align across estimators within reasonable range. The Ames Housing results complete the sweep across outcome types: the framework produces dollar-denominated marginal effects with valid standard errors for a continuous regression outcome, using exactly the same estimation procedure as the binary classification examples, with no modification required.

5 Simulation

We evaluate the finite-sample performance of the AME estimator through two simulation studies. Simulation 1 examines bias and root mean squared error of the AME estimates as a function of sample size across three data generating processes and two outcome types. Simulation 2 examines bootstrap standard error calibration by comparing empirical coverage rates of nominal 95% confidence intervals to the target. All features x_1, x_2, x_3, x_4 are drawn independently from $N(0, 1)$. Features x_1 and x_2 have nonzero true effects; x_3 and x_4 are pure noise with true AME of zero. Three data generating processes are considered. The linear DGP sets $\eta = 2x_1 + 3x_2$. The nonlinear DGP sets $\eta = 2x_1^2 + 3x_2$, so that the true AME of x_1 is zero in the regression case by symmetry of $N(0, 1)$ — $\mathbb{E}[4x_1] = 0$ — but nonzero in the classification case where the sigmoid breaks that symmetry. The interaction DGP sets $\eta = 2x_1 + 3x_2 + 2x_1x_2$, which preserves the marginal AMEs of x_1 and x_2 at 2.0 and 3.0 respectively in the regression case since $\mathbb{E}[2x_2] = \mathbb{E}[2x_1] = 0$. For regression, the outcome is $y = \eta + \varepsilon$ where $\varepsilon \sim N(0, 1)$. For classification, $P(y = 1) = \sigma(\eta)$ where σ denotes the logistic function. True AMEs for all DGPs and outcome types are computed via Monte Carlo integration using $n = 1,000,000$ observations drawn from the true prediction function directly, treating the

result as ground truth. Sample sizes of $n \in \{250, 500, 1\,000, 2\,500, 5\,000\}$ are evaluated over 500 Monte Carlo iterations for classification and 1,000 iterations for regression.

The simulation study fits four model classes with hyperparameters chosen to balance computational tractability against representativeness of typical applied use. The random forest uses 100 trees with a maximum depth of 6, and XGBoost uses 100 boosting rounds with a maximum depth of 4 and a learning rate of 0.05 — conservative settings that prevent overfitting on the small sample sizes considered while remaining flexible enough to capture the nonlinear and interaction structures in the data generating processes. The neural network is a feedforward architecture with two hidden layers of 32 and 16 units, trained for 50 epochs with batch size 64 and learning rate 10^{-3} using the Adam optimizer. For regression outcomes, the network includes the input normalization layer and target standardization wrapper described in Section 2 to ensure scale-invariant gradient updates. Logistic and linear regression are included as baselines and use no hyperparameter tuning. All model fits are performed independently on each Monte Carlo iteration with no warm starting between iterations, so the variance estimates reflect the full sampling distribution of the AME estimator including model fitting variability. The Monte Carlo ground truth AMEs are computed once at the start of the study using $n = 1,000,000$ observations drawn from the true prediction function directly, with the resulting AMEs treated as known quantities for the bias and coverage calculations. Bootstrap calibration in Simulation 2 uses 200 bootstrap replicates per Monte Carlo iteration and 500 Monte Carlo iterations per cell of the design.

Table 10 reports bias and RMSE for the linear classification DGP, where the true AMEs of x_1 and x_2 are 0.199 and 0.298 — these are not 2.0 and 3.0 because the sigmoid compresses the scale. Logistic regression recovers the true AMEs with negligible bias at all sample sizes, and RMSE declines from around 0.020 at $n = 250$ to 0.005 at $n = 5,000$ at the expected $O(n^{-1/2})$ rate. The neural network behaves similarly, with small bias at $n = 250$ that disappears by $n = 500$. XGBoost shows larger bias at small samples — around 0.06 for both x_1 and x_2 at $n = 250$ — reflecting the well-known tendency of gradient boosted models to

Table 10: Simulation 1: AME Recovery — Linear DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

Variable	True AME	Bias (RMSE)				
		$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
<i>Panel: Logistic</i>						
x1	0.199	-0.0058 (0.0201)	-0.0034 (0.0146)	-0.0014 (0.0106)	-0.0008 (0.0064)	-0.0006 (0.0048)
x2	0.298	-0.0109 (0.0229)	-0.0057 (0.0153)	-0.0026 (0.0105)	-0.0010 (0.0066)	-0.0005 (0.0046)
x3	0.000	0.0012 (0.0198)	0.0006 (0.0136)	0.0006 (0.0099)	0.0003 (0.0063)	0.0002 (0.0045)
x4	0.000	0.0008 (0.0197)	-0.0001 (0.0136)	-0.0008 (0.0100)	-0.0001 (0.0062)	0.0003 (0.0045)
<i>Panel: Random Forest</i>						
x1	0.199	0.0126 (0.0412)	0.0044 (0.0239)	-0.0018 (0.0160)	-0.0111 (0.0142)	-0.0157 (0.0168)
x2	0.298	0.0199 (0.0500)	0.0113 (0.0309)	-0.0008 (0.0174)	-0.0102 (0.0143)	-0.0158 (0.0172)
x3	0.000	0.0007 (0.0284)	0.0010 (0.0171)	0.0005 (0.0104)	0.0002 (0.0056)	0.0002 (0.0034)
x4	0.000	-0.0008 (0.0267)	-0.0013 (0.0177)	-0.0012 (0.0106)	-0.0001 (0.0055)	0.0002 (0.0034)
<i>Panel: XGBoost</i>						
x1	0.199	0.0590 (0.0782)	0.0265 (0.0401)	0.0107 (0.0223)	-0.0016 (0.0108)	-0.0049 (0.0083)
x2	0.298	0.0643 (0.0893)	0.0315 (0.0490)	0.0085 (0.0218)	-0.0016 (0.0116)	-0.0046 (0.0083)
x3	0.000	0.0008 (0.0332)	-0.0002 (0.0193)	0.0006 (0.0110)	0.0002 (0.0056)	0.0002 (0.0033)
x4	0.000	0.0008 (0.0344)	-0.0009 (0.0198)	-0.0006 (0.0116)	0.0000 (0.0054)	0.0002 (0.0032)
<i>Panel: Neural Net</i>						
x1	0.199	-0.0078 (0.0222)	-0.0007 (0.0156)	0.0005 (0.0115)	0.0003 (0.0074)	-0.0002 (0.0058)
x2	0.298	-0.0162 (0.0276)	-0.0017 (0.0174)	0.0004 (0.0116)	0.0004 (0.0073)	0.0002 (0.0052)
x3	0.000	0.0014 (0.0204)	0.0010 (0.0144)	0.0007 (0.0103)	0.0003 (0.0069)	0.0004 (0.0057)
x4	0.000	0.0009 (0.0203)	-0.0003 (0.0143)	-0.0007 (0.0105)	-0.0001 (0.0070)	0.0004 (0.0055)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1,000,000$ observations. Noise features x3 and x4 have true AME of zero.

underfit at small sample sizes, but bias falls sharply and RMSE converges by $n = 2,500$. Random forests show a different pattern: bias is small at $n = 250$ and $n = 500$ but turns negative and persistent at $n = 2,500$ and $n = 5,000$, reaching around -0.016 for both signal features at the largest sample size. This reflects the regularization bias inherent in random forests — the tree depth and subsampling constraints impose shrinkage that does not vanish with sample size at the default tuning settings. The noise features x_3 and x_4 have bias and RMSE near zero throughout for all model classes, confirming that the estimator does not spuriously attribute effects to irrelevant features.

Table 11 reports results for the nonlinear classification DGP, where $P(y = 1) = \sigma(2x_1^2 + 3x_2)$. The true AME of x_1 is zero despite the quadratic term, because the sigmoid and quadratic effects cancel in expectation at these parameter values as confirmed by the Monte Carlo ground truth. All four model classes recover this near-zero AME accurately, with bias well below 0.005 across all sample sizes. The more important result is x_2 , which has true AME of 0.298. Logistic regression misspecifies the DGP — the true relationship is nonlinear in x_1 — but still recovers the x_2 AME accurately because x_2 enters linearly and the misspecification in x_1 does not propagate to the x_2 marginal effect. This is an important practical result: AMEs from a misspecified parametric model can still be reliable for variables that enter the true DGP in a manner consistent with the model’s functional form. The black-box models show the same pattern as the linear DGP: XGBoost has elevated RMSE at small samples, random forests develop negative bias at large samples, and neural networks perform comparably to logistic regression throughout.

Table 12 reports results for the interaction DGP, where $P(y = 1) = \sigma(2x_1 + 3x_2 + 2x_1x_2)$. The true AMEs of x_1 and x_2 remain 0.199 and 0.298 — the same as the linear DGP — because the interaction term $2x_1x_2$ has zero expectation under $N(0, 1)$ and the Monte Carlo ground truth confirms near-identical values. The key finding is that all four model classes recover these AMEs with bias and RMSE patterns essentially identical to the linear DGP, despite the true DGP including a multiplicative interaction. This is by design of the AME

Table 11: Simulation 1: AME Recovery — Nonlinear DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

Variable	True AME	Bias (RMSE)				
		$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
<i>Panel: Logistic</i>						
x1	0.000	0.0002 (0.0230)	0.0015 (0.0166)	0.0009 (0.0123)	0.0002 (0.0078)	-0.0001 (0.0055)
x2	0.298	-0.0077 (0.0215)	-0.0042 (0.0152)	-0.0017 (0.0109)	-0.0007 (0.0065)	-0.0005 (0.0048)
x3	0.000	0.0003 (0.0229)	0.0001 (0.0158)	-0.0002 (0.0112)	-0.0001 (0.0074)	0.0003 (0.0052)
x4	0.000	-0.0012 (0.0252)	0.0005 (0.0159)	0.0006 (0.0108)	-0.0001 (0.0075)	-0.0000 (0.0051)
<i>Panel: Random Forest</i>						
x1	0.000	0.0014 (0.0407)	0.0001 (0.0256)	0.0000 (0.0173)	-0.0003 (0.0114)	-0.0000 (0.0082)
x2	0.298	0.0177 (0.0521)	0.0060 (0.0278)	-0.0040 (0.0182)	-0.0120 (0.0157)	-0.0155 (0.0171)
x3	0.000	0.0019 (0.0285)	-0.0000 (0.0166)	0.0003 (0.0100)	0.0001 (0.0052)	0.0003 (0.0031)
x4	0.000	-0.0026 (0.0295)	-0.0007 (0.0165)	0.0005 (0.0099)	0.0000 (0.0054)	-0.0001 (0.0031)
<i>Panel: XGBoost</i>						
x1	0.000	0.0039 (0.0549)	0.0003 (0.0324)	0.0004 (0.0209)	-0.0000 (0.0123)	-0.0003 (0.0079)
x2	0.298	0.0471 (0.0787)	0.0221 (0.0396)	0.0066 (0.0208)	-0.0028 (0.0114)	-0.0054 (0.0087)
x3	0.000	0.0038 (0.0307)	0.0007 (0.0172)	0.0007 (0.0108)	0.0002 (0.0051)	0.0003 (0.0028)
x4	0.000	-0.0016 (0.0329)	0.0010 (0.0184)	0.0006 (0.0105)	0.0001 (0.0052)	0.0001 (0.0027)
<i>Panel: Neural Net</i>						
x1	0.000	0.0006 (0.0234)	0.0015 (0.0189)	0.0002 (0.0140)	0.0001 (0.0094)	0.0000 (0.0069)
x2	0.298	-0.0132 (0.0249)	-0.0026 (0.0155)	0.0002 (0.0112)	0.0005 (0.0077)	0.0001 (0.0064)
x3	0.000	0.0001 (0.0215)	0.0002 (0.0138)	0.0007 (0.0101)	0.0003 (0.0068)	0.0003 (0.0053)
x4	0.000	-0.0007 (0.0230)	0.0004 (0.0144)	0.0005 (0.0100)	-0.0001 (0.0070)	0.0002 (0.0056)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1,000,000$ observations. Noise features x3 and x4 have true AME of zero.

Table 12: Simulation 1: AME Recovery — Interaction DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

Variable	True AME	Bias (RMSE)				
		$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
<i>Panel: Logistic</i>						
x1	0.199	-0.0065 (0.0219)	-0.0042 (0.0153)	-0.0021 (0.0112)	-0.0008 (0.0071)	-0.0007 (0.0051)
x2	0.298	-0.0083 (0.0222)	-0.0035 (0.0154)	-0.0018 (0.0102)	-0.0008 (0.0071)	-0.0007 (0.0050)
x3	0.000	0.0009 (0.0206)	-0.0002 (0.0155)	0.0005 (0.0107)	0.0003 (0.0070)	-0.0001 (0.0046)
x4	0.000	0.0007 (0.0212)	-0.0000 (0.0159)	-0.0001 (0.0106)	0.0001 (0.0066)	0.0002 (0.0047)
<i>Panel: Random Forest</i>						
x1	0.199	0.0114 (0.0423)	0.0051 (0.0264)	-0.0031 (0.0169)	-0.0110 (0.0144)	-0.0154 (0.0167)
x2	0.298	0.0178 (0.0498)	0.0104 (0.0320)	-0.0023 (0.0200)	-0.0101 (0.0148)	-0.0140 (0.0158)
x3	0.000	0.0019 (0.0276)	0.0004 (0.0170)	0.0002 (0.0104)	0.0002 (0.0059)	-0.0000 (0.0034)
x4	0.000	-0.0005 (0.0267)	-0.0011 (0.0177)	-0.0001 (0.0111)	0.0001 (0.0057)	-0.0000 (0.0033)
<i>Panel: XGBoost</i>						
x1	0.199	0.0511 (0.0737)	0.0225 (0.0392)	0.0079 (0.0219)	-0.0014 (0.0105)	-0.0046 (0.0082)
x2	0.298	0.0496 (0.0796)	0.0228 (0.0439)	0.0042 (0.0227)	-0.0021 (0.0125)	-0.0047 (0.0091)
x3	0.000	0.0004 (0.0310)	0.0005 (0.0205)	0.0004 (0.0111)	0.0000 (0.0064)	0.0000 (0.0035)
x4	0.000	-0.0012 (0.0338)	-0.0015 (0.0205)	-0.0004 (0.0118)	0.0001 (0.0060)	0.0001 (0.0037)
<i>Panel: Neural Net</i>						
x1	0.199	-0.0089 (0.0233)	-0.0013 (0.0158)	0.0001 (0.0117)	0.0000 (0.0079)	-0.0004 (0.0061)
x2	0.298	-0.0114 (0.0257)	-0.0002 (0.0170)	0.0007 (0.0131)	0.0005 (0.0085)	-0.0002 (0.0067)
x3	0.000	0.0008 (0.0193)	0.0004 (0.0148)	0.0008 (0.0099)	0.0001 (0.0070)	0.0000 (0.0053)
x4	0.000	0.0005 (0.0206)	-0.0002 (0.0151)	0.0000 (0.0103)	-0.0001 (0.0069)	0.0006 (0.0057)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1,000,000$ observations. Noise features x3 and x4 have true AME of zero.

estimand: the average marginal effect averages over the joint distribution of the features, so the interaction term is absorbed into the average derivative rather than requiring explicit modeling. A logistic regression that omits the interaction term recovers the correct AME because the interaction’s contribution to the marginal effect of x_1 is $\mathbb{E}[2x_2] = 0$ and similarly for x_2 . This is an important robustness result — AMEs are not sensitive to interaction misspecification when the interactions involve zero-mean features, which is the common case after centering.

Table 13 turns to the regression setting with the linear DGP, $y = 2x_1 + 3x_2 + \varepsilon$. Linear regression recovers the true AMEs of 2.0 and 3.0 with negligible bias across all sample sizes, and RMSE declines from around 0.066 at $n = 250$ to 0.014 at $n = 5,000$, consistent with $O(n^{-1/2})$ convergence. The regression setting reveals the regularization bias of random forests more starkly than the classification setting: bias reaches -0.091 for x_1 and -0.095 for x_2 at $n = 5,000$, and RMSE is not declining at the largest sample sizes, indicating that the bias is not vanishing with n at default tuning. This is a known property of random forests in the regression case — the ensemble averaging and subsampling impose shrinkage toward zero that is not corrected by the hyperparameter search at these sample sizes. XGBoost shows large positive bias at small samples — 0.41 for x_1 at $n = 250$ — but converges quickly, with bias falling below 0.05 by $n = 2,500$. The neural network shows moderate negative bias at small samples that falls toward zero by $n = 2,500$, though RMSE does not decline monotonically at the largest sample sizes, suggesting residual variance from the optimization landscape. The noise features have RMSE near zero for linear regression and declining rapidly for the black-box models, confirming that the estimator correctly attributes near-zero effects to irrelevant features in the regression setting.

Table 14 reports results for the nonlinear regression DGP, $y = 2x_1^2 + 3x_2 + \varepsilon$, where the true AME of x_1 is zero by symmetry of $N(0, 1)$ and the true AME of x_2 is 3.0. Linear regression recovers the x_2 AME accurately with RMSE declining at the standard rate, and correctly attributes a near-zero AME to x_1 despite misspecifying its functional form — the

Table 13: Simulation 1: AME Recovery — Linear DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

Variable	True AME	Bias (RMSE)				
		$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
<i>Panel: Linear</i>						
x1	2.000	-0.0017 (0.0659)	-0.0001 (0.0445)	-0.0000 (0.0322)	-0.0006 (0.0207)	-0.0003 (0.0137)
x2	3.000	-0.0043 (0.0636)	-0.0017 (0.0453)	0.0007 (0.0316)	-0.0004 (0.0197)	-0.0003 (0.0140)
x3	0.000	-0.0001 (0.0650)	-0.0006 (0.0457)	-0.0005 (0.0312)	0.0012 (0.0199)	-0.0003 (0.0140)
x4	0.000	-0.0034 (0.0642)	0.0006 (0.0452)	-0.0002 (0.0315)	-0.0003 (0.0196)	-0.0001 (0.0139)
<i>Panel: Random Forest</i>						
x1	2.000	-0.0973 (0.2064)	-0.0332 (0.1189)	-0.0387 (0.0850)	-0.0703 (0.0824)	-0.0911 (0.0956)
x2	3.000	-0.0215 (0.2338)	-0.0105 (0.1456)	-0.0433 (0.0997)	-0.0797 (0.0958)	-0.0953 (0.1025)
x3	0.000	-0.0015 (0.0562)	-0.0005 (0.0290)	-0.0005 (0.0137)	0.0001 (0.0036)	0.0000 (0.0011)
x4	0.000	-0.0021 (0.0572)	-0.0010 (0.0297)	-0.0005 (0.0138)	-0.0002 (0.0037)	0.0000 (0.0011)
<i>Panel: XGBoost</i>						
x1	2.000	0.4114 (0.4644)	0.1780 (0.2227)	0.0366 (0.0917)	-0.0337 (0.0554)	-0.0467 (0.0547)
x2	3.000	0.3122 (0.4103)	0.0948 (0.1884)	-0.0172 (0.0947)	-0.0558 (0.0754)	-0.0601 (0.0677)
x3	0.000	0.0007 (0.1090)	0.0016 (0.0550)	-0.0003 (0.0313)	0.0011 (0.0153)	-0.0001 (0.0083)
x4	0.000	-0.0080 (0.1077)	0.0002 (0.0597)	0.0011 (0.0314)	-0.0006 (0.0149)	0.0001 (0.0081)
<i>Panel: Neural Net</i>						
x1	2.000	-0.0659 (0.1161)	-0.0208 (0.0544)	-0.0063 (0.0344)	-0.0021 (0.0293)	-0.0008 (0.0329)
x2	3.000	-0.1231 (0.1765)	-0.0349 (0.0638)	-0.0102 (0.0361)	-0.0031 (0.0324)	-0.0006 (0.0372)
x3	0.000	0.0014 (0.0788)	-0.0006 (0.0484)	-0.0003 (0.0330)	0.0009 (0.0250)	-0.0006 (0.0280)
x4	0.000	-0.0002 (0.0752)	0.0008 (0.0496)	-0.0003 (0.0336)	-0.0004 (0.0247)	-0.0017 (0.0287)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1,000,000$ observations. Noise features x3 and x4 have true AME of zero.

Table 14: Simulation 1: AME Recovery — Nonlinear DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

Variable	True AME	Bias (RMSE)				
		$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
<i>Panel: Linear</i>						
x1	0.000	0.0241 (0.3868)	0.0261 (0.2820)	0.0148 (0.1966)	0.0048 (0.1272)	0.0015 (0.0904)
x2	3.000	-0.0145 (0.1921)	-0.0041 (0.1276)	-0.0012 (0.0930)	-0.0002 (0.0572)	0.0001 (0.0417)
x3	0.000	-0.0025 (0.1889)	0.0001 (0.1343)	-0.0018 (0.0941)	-0.0020 (0.0576)	-0.0009 (0.0414)
x4	0.000	-0.0114 (0.1857)	-0.0003 (0.1328)	0.0023 (0.0927)	0.0006 (0.0594)	-0.0018 (0.0430)
<i>Panel: Random Forest</i>						
x1	0.000	0.0071 (0.3433)	0.0097 (0.2547)	0.0053 (0.1836)	0.0006 (0.1185)	0.0007 (0.0903)
x2	3.000	0.0142 (0.2704)	-0.1152 (0.2008)	-0.1701 (0.2004)	-0.1867 (0.1974)	-0.1849 (0.1916)
x3	0.000	0.0001 (0.0549)	0.0008 (0.0274)	0.0003 (0.0138)	-0.0002 (0.0055)	0.0000 (0.0027)
x4	0.000	-0.0009 (0.0573)	-0.0007 (0.0278)	-0.0001 (0.0138)	-0.0000 (0.0054)	-0.0000 (0.0025)
<i>Panel: XGBoost</i>						
x1	0.000	0.0552 (0.4662)	0.0038 (0.2744)	-0.0117 (0.1829)	-0.0137 (0.1040)	-0.0057 (0.0735)
x2	3.000	0.2725 (0.3937)	0.0816 (0.1981)	-0.0183 (0.1058)	-0.0702 (0.0885)	-0.0794 (0.0862)
x3	0.000	-0.0040 (0.1124)	-0.0022 (0.0704)	-0.0020 (0.0446)	-0.0011 (0.0191)	-0.0006 (0.0097)
x4	0.000	-0.0052 (0.1121)	-0.0028 (0.0636)	-0.0021 (0.0381)	0.0003 (0.0189)	0.0006 (0.0096)
<i>Panel: Neural Net</i>						
x1	0.000	0.0201 (0.3086)	0.0178 (0.1925)	0.0065 (0.1390)	-0.0003 (0.0911)	-0.0013 (0.0760)
x2	3.000	-0.1545 (0.2601)	-0.0202 (0.0710)	-0.0150 (0.0461)	-0.0098 (0.0343)	-0.0022 (0.0395)
x3	0.000	0.0017 (0.1419)	0.0004 (0.0634)	0.0007 (0.0400)	0.0009 (0.0293)	-0.0004 (0.0321)
x4	0.000	-0.0028 (0.1410)	-0.0001 (0.0626)	-0.0013 (0.0406)	-0.0004 (0.0281)	-0.0016 (0.0323)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1,000,000$ observations. Noise features x3 and x4 have true AME of zero.

same robustness result observed in the classification case. However, the RMSE for x_1 under linear regression is substantially larger than in the linear DGP — 0.387 at $n = 250$ versus 0.066 — reflecting increased estimation variance from the misspecified model. The most striking result is random forests: the bias for x_2 reaches -0.185 at $n = 5,000$ and is not declining, indicating that the regularization bias is amplified in the nonlinear setting where the forest must capture the quadratic relationship in x_1 while simultaneously estimating the linear effect of x_2 . XGBoost and the neural network both show elevated RMSE relative to the linear DGP but declining bias, suggesting that these models are better able to adapt to the nonlinear DGP with sufficient data. The x_1 AME of zero is recovered accurately by all model classes at moderate sample sizes, which is a reassuring result for practitioners who want to test whether nonlinear features have nonzero average effects.

Table 15 reports results for the interaction regression DGP, $y = 2x_1 + 3x_2 + 2x_1x_2 + \varepsilon$, with true AMEs of 2.0 and 3.0 for x_1 and x_2 . Linear regression recovers these AMEs accurately despite omitting the interaction term, confirming the robustness result from the classification case: the interaction contribution to the marginal effect of x_1 is $\mathbb{E}[2x_2] = 0$ and similarly for x_2 , so the misspecified linear model nevertheless recovers the correct AME. The RMSE for the noise features under linear regression is elevated relative to the linear DGP — around 0.14 at $n = 250$ — because the interaction term inflates the residual variance and increases estimation uncertainty for all coefficients. Random forests again show persistent negative bias that grows rather than shrinks with sample size, reaching -0.080 and -0.114 for x_1 and x_2 respectively at $n = 5,000$. XGBoost shows the same large positive bias at small samples followed by convergence, but the bias does not fully vanish at $n = 5,000$ — around -0.043 for x_1 and -0.059 for x_2 — suggesting that the interaction structure requires larger samples for the gradient boosting to fully adapt. The neural network performs well, with bias below 0.003 for both signal features at $n = 2,500$ and $n = 5,000$, and RMSE declining steadily. Taken together the six bias and RMSE tables establish that the AME estimator is consistent for logistic regression, neural networks, and XGBoost across all three DGPs

Table 15: Simulation 1: AME Recovery — Interaction DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

Variable	True AME	Bias (RMSE)				
		$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
<i>Panel: Linear</i>						
x1	2.000	-0.0041 (0.2284)	0.0005 (0.1531)	0.0000 (0.1129)	0.0006 (0.0715)	0.0011 (0.0495)
x2	3.000	0.0040 (0.2194)	0.0087 (0.1627)	0.0075 (0.1135)	0.0012 (0.0727)	0.0009 (0.0514)
x3	0.000	-0.0114 (0.1388)	-0.0088 (0.1002)	-0.0054 (0.0725)	0.0010 (0.0450)	-0.0009 (0.0329)
x4	0.000	-0.0073 (0.1443)	0.0004 (0.1018)	-0.0006 (0.0690)	0.0013 (0.0453)	0.0005 (0.0321)
<i>Panel: Random Forest</i>						
x1	2.000	-0.0755 (0.2566)	-0.0310 (0.1663)	-0.0433 (0.1169)	-0.0724 (0.0979)	-0.0804 (0.0925)
x2	3.000	-0.0634 (0.2806)	-0.0690 (0.1913)	-0.0922 (0.1497)	-0.1112 (0.1306)	-0.1143 (0.1244)
x3	0.000	-0.0031 (0.0612)	-0.0004 (0.0352)	-0.0009 (0.0182)	0.0003 (0.0067)	-0.0001 (0.0029)
x4	0.000	-0.0023 (0.0665)	0.0004 (0.0349)	-0.0001 (0.0184)	-0.0002 (0.0069)	0.0000 (0.0030)
<i>Panel: XGBoost</i>						
x1	2.000	0.3184 (0.4539)	0.1016 (0.2184)	-0.0009 (0.1163)	-0.0406 (0.0758)	-0.0429 (0.0602)
x2	3.000	0.3672 (0.4920)	0.1135 (0.2373)	-0.0083 (0.1227)	-0.0572 (0.0898)	-0.0589 (0.0746)
x3	0.000	-0.0031 (0.1040)	0.0011 (0.0551)	-0.0017 (0.0319)	0.0005 (0.0151)	-0.0001 (0.0078)
x4	0.000	-0.0082 (0.1028)	0.0003 (0.0564)	-0.0003 (0.0314)	-0.0004 (0.0149)	-0.0001 (0.0080)
<i>Panel: Neural Net</i>						
x1	2.000	-0.0842 (0.2380)	-0.0083 (0.1114)	-0.0056 (0.0779)	0.0003 (0.0573)	0.0022 (0.0536)
x2	3.000	-0.1231 (0.2467)	-0.0068 (0.1120)	-0.0052 (0.0783)	0.0004 (0.0579)	0.0026 (0.0533)
x3	0.000	-0.0026 (0.0986)	-0.0016 (0.0551)	-0.0024 (0.0383)	0.0004 (0.0293)	-0.0004 (0.0316)
x4	0.000	-0.0079 (0.0981)	0.0015 (0.0565)	0.0002 (0.0387)	-0.0005 (0.0284)	-0.0019 (0.0331)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1,000,000$ observations. Noise features x3 and x4 have true AME of zero.

and both outcome types, with random forests showing persistent regularization bias in the regression setting that practitioners should be aware of when reporting AMEs from default-tuned forest models.

Table 16 reports bootstrap standard error calibration for the linear classification DGP, evaluating empirical coverage of nominal 95% confidence intervals at $n \in \{250, 1\,000, 5\,000\}$ using 200 bootstrap replicates. Logistic regression achieves coverage between 0.896 and 0.950 across all variables and sample sizes, with the slight undercoverage at $n = 250$ for x_2 reflecting finite-sample bias rather than bootstrap miscalibration — the bootstrap interval is correctly sized but the point estimate is slightly off-center. Coverage improves toward the nominal level as sample size increases and is well-calibrated by $n = 5,000$. XGBoost shows modest overcoverage for the noise features at all sample sizes — around 0.978 to 0.986 — reflecting the wide bootstrap intervals produced by the model’s high variance at small samples; the intervals are conservative rather than anti-conservative, which is the safer failure mode for inference. The neural network shows undercoverage for the noise features at $n = 250$ (0.866 and 0.854) that recovers toward and above the nominal level by $n = 5,000$, with the overcoverage at large samples (0.990 and 0.994) consistent with the smooth differentiable surface producing stable finite differences across bootstrap replicates. Random forests show a more concerning pattern: coverage collapses dramatically at $n = 5,000$, falling to 0.130 for x_1 , 0.274 for x_2 , and 0.392 for the noise features. This is a direct consequence of the regularization bias documented in Simulation 1 — the bootstrap intervals shrink at the expected rate but they shrink around a biased point estimate, so as the intervals narrow they fail to cover the true AME. The intervals at $n = 5,000$ have mean width 0.0165 for x_1 but the bias is around -0.016 , so the interval is roughly the width of the bias itself and coverage of the true value becomes nearly impossible. This finding is the empirical counterpart to Conjecture 3 in Section 3 on bootstrap validity for random forests — the bootstrap is consistent for the variability of $\widehat{\text{AME}}_j$ around its expectation under the data generating process for the forest, but that expectation does not coincide with the true AME,

Table 16: Simulation 2: Bootstrap SE Calibration — Linear DGP (Classification). Nominal coverage target: 95%. Coverage rate and mean 95% CI width (in parentheses) across 500 Monte Carlo iterations.

Variable	True AME	Coverage (CI Width)		
		$n = 250$	$n = 1,000$	$n = 5,000$
<i>Panel: Logistic</i>				
x1	0.199	0.938 (0.0745)	0.940 (0.0388)	0.930 (0.0176)
x2	0.298	0.896 (0.0763)	0.944 (0.0400)	0.950 (0.0182)
x3	0.000	0.932 (0.0736)	0.936 (0.0378)	0.940 (0.0171)
x4	0.000	0.922 (0.0737)	0.936 (0.0378)	0.940 (0.0171)
<i>Panel: Random Forest</i>				
x1	0.199	0.882 (0.1286)	0.878 (0.0498)	0.130 (0.0165)
x2	0.298	0.914 (0.1706)	0.930 (0.0663)	0.274 (0.0228)
x3	0.000	0.700 (0.0580)	0.614 (0.0170)	0.392 (0.0034)
x4	0.000	0.688 (0.0571)	0.548 (0.0170)	0.392 (0.0034)
<i>Panel: XGBoost</i>				
x1	0.199	0.920 (0.2625)	0.928 (0.1028)	0.966 (0.0305)
x2	0.298	0.924 (0.3076)	0.966 (0.1161)	0.970 (0.0339)
x3	0.000	0.978 (0.1718)	0.986 (0.0767)	0.978 (0.0233)
x4	0.000	0.980 (0.1715)	0.982 (0.0770)	0.972 (0.0234)
<i>Panel: Neural Net</i>				
x1	0.199	0.916 (0.0770)	0.922 (0.0417)	0.990 (0.0253)
x2	0.298	0.902 (0.0889)	0.950 (0.0459)	0.978 (0.0233)
x3	0.000	0.866 (0.0645)	0.926 (0.0375)	0.990 (0.0258)
x4	0.000	0.854 (0.0648)	0.918 (0.0374)	0.994 (0.0260)

Notes: Coverage rate and mean 95% CI width (in parentheses) computed over 500 Monte Carlo iterations. Bootstrap SEs from nonparametric bootstrap with 200 replicates. True AMEs computed via Monte Carlo integration with $n = 1,000,000$ observations. Noise features x3 and x4 have true AME of zero.

and standard nonparametric bootstrap methods do not correct for the asymptotic bias.

Table 17 reports the corresponding calibration results for the linear regression DGP, where the magnitudes of the AMEs are larger and the bias-coverage interaction becomes even more visible. Linear regression achieves coverage between 0.918 and 0.952 across all variables and sample sizes, well-calibrated throughout. The neural network performs well at small and moderate samples, with coverage approaching the nominal level by $n = 1,000$, before drifting upward to exactly 1.000 at $n = 5,000$ for all four variables — the same conservatism observed in the classification case, amplified in the regression setting where the smooth prediction surface produces highly stable finite differences. XGBoost shows pronounced undercoverage for the signal features at $n = 250$ — 0.526 for x_1 and 0.752 for x_2 — driven by the large positive bias documented in Simulation 1 at small samples, but coverage recovers toward and above the nominal level by $n = 1,000$ and is well-calibrated at $n = 5,000$. The most striking result is again random forests: coverage at $n = 5,000$ falls to 0.104 for x_1 and 0.204 for x_2 , the worst calibration in either table. The mechanism is identical to the classification case — the bootstrap intervals narrow at the expected rate but the bias does not vanish, so the intervals fail to cover the true AME at large samples. The noise features under random forests show coverage of around 0.6 across all sample sizes, indicating that even where the bias is small, the bootstrap intervals are systematically too narrow to achieve nominal coverage. Taken together the two calibration tables establish that the nonparametric bootstrap produces valid standard errors for logistic regression, linear regression, neural networks, and XGBoost across a range of sample sizes, with random forests as the clear exception — a finding that aligns with the theoretical status of bootstrap validity for that estimator class as conjectured rather than proved.

Table 17: Simulation 2: Bootstrap SE Calibration — Linear DGP (Regression). Nominal coverage target: 95%. Coverage rate and mean 95% CI width (in parentheses) across 500 Monte Carlo iterations.

Variable	True AME	Coverage (CI Width)		
		$n = 250$	$n = 1,000$	$n = 5,000$
<i>Panel: Linear</i>				
x1	2.000	0.918 (0.2441)	0.946 (0.1211)	0.946 (0.0541)
x2	3.000	0.948 (0.2445)	0.946 (0.1214)	0.950 (0.0542)
x3	0.000	0.946 (0.2454)	0.938 (0.1214)	0.952 (0.0542)
x4	0.000	0.920 (0.2459)	0.932 (0.1215)	0.938 (0.0545)
<i>Panel: Random Forest</i>				
x1	2.000	0.828 (0.5888)	0.850 (0.2517)	0.104 (0.0998)
x2	3.000	0.918 (0.8030)	0.894 (0.3268)	0.204 (0.1348)
x3	0.000	0.660 (0.1144)	0.586 (0.0238)	0.618 (0.0018)
x4	0.000	0.694 (0.1166)	0.630 (0.0240)	0.596 (0.0018)
<i>Panel: XGBoost</i>				
x1	2.000	0.526 (1.0307)	0.850 (0.3749)	0.964 (0.1203)
x2	3.000	0.752 (1.1581)	0.954 (0.4174)	0.938 (0.1353)
x3	0.000	0.960 (0.5467)	0.986 (0.2358)	0.990 (0.0740)
x4	0.000	0.956 (0.5416)	0.986 (0.2354)	0.986 (0.0735)
<i>Panel: Neural Net</i>				
x1	2.000	0.912 (0.2794)	0.950 (0.1453)	1.000 (0.1821)
x2	3.000	0.844 (0.2901)	0.950 (0.1511)	1.000 (0.1999)
x3	0.000	0.930 (0.2695)	0.966 (0.1386)	1.000 (0.1631)
x4	0.000	0.916 (0.2722)	0.958 (0.1394)	1.000 (0.1633)

Notes: Coverage rate and mean 95% CI width (in parentheses) computed over 500 Monte Carlo iterations. Bootstrap SEs from nonparametric bootstrap with 200 replicates. True AMEs computed via Monte Carlo integration with $n = 1,000,000$ observations. Noise features x3 and x4 have true AME of zero.

6 Conclusion

This paper has proposed a framework for interpreting any supervised machine learning model through the lens of average marginal effects. The central claim is that the traditional accuracy-interpretability tradeoff is not fundamental. By defining the average marginal effect as the expectation of the partial derivative of the prediction function with respect to each feature, averaged over the empirical distribution of the covariates, we obtain an estimand that is well-defined for any supervised learner and reduces to the classical econometric AME when the model is a generalized linear model. The estimator replaces analytical derivatives with adaptive centered finite differences, applies uniformly across model classes and outcome types, and produces a coefficient table — estimate, standard error, and significance level — directly analogous to the output of a parametric regression. The nonparametric bootstrap, with full model refitting and hyperparameter selection on each replicate, provides standard errors that capture uncertainty about model specification as well as estimation. These are capabilities that existing interpretability methods, including SHAP (Lundberg and Lee, 2017), LIME (Ribeiro et al., 2016), and partial dependence plots (Friedman, 2001), do not provide.

The theoretical contribution proceeds in two parts. The first establishes gradient consistency for three classes of supervised learners. For neural networks, the appropriate function space is the Sobolev space $W^{1,p}$, and gradient consistency follows from uniform $W^{2,p}$ boundedness combined with L^p consistency via the Rellich–Kondrachov compact embedding theorem (Evans, 2010). For gradient boosted models with tree base learners and for Mondrian forests (Mourtada et al., 2017), the appropriate function space is the space of functions of bounded variation (Ambrosio et al., 2000), where gradients exist as Radon measures and consistency is established via the BV compactness theorem. For Breiman random forests (Breiman, 2001a; Scornet et al., 2015), gradient consistency is conjectured rather than proved: the obstruction lies in the data-dependent partition boundaries, whose discontinuity structure does not admit the geometric analysis available for Mondrian partitions. The second part establishes bootstrap validity for all three proved estimator classes — neural net-

works, gradient boosted models, and Mondrian forests — via the Hadamard differentiability of the AME functional (van der Vaart and Wellner, 1996) and the functional delta method. Corollaries 1 through 3 make this precise, showing that the bootstrap distribution of the sample AME consistently estimates the sampling distribution of the estimator under rate conditions that are verified for each class. Bootstrap validity for Breiman forests remains open for the same reason gradient consistency is harder to establish — the data-dependent partition geometry resists the clean decomposition that the Mondrian independence assumption makes available.

The empirical results across three datasets — UCI Adult Income (Kohavi, 1996; Dua and Graff, 2019), UCI Credit Card Default (Yeh, 2009; Dua and Graff, 2019), and Ames Housing (De Cock, 2011) — validate the framework on binary classification and continuous regression outcomes. The AME estimates are stable across model classes where the underlying models are well-fitted, the bootstrap standard errors narrow as model fit improves, and the AME and partial dependence plot estimates are in close agreement throughout. The comparison with SHAP values reveals a systematic divergence: SHAP values are on a different scale, in several cases have the wrong sign relative to AME and PDP, and carry no inferential apparatus. The Ames Housing application is particularly informative because the dollar-denominated units make the divergence concrete — SHAP attributes to above-ground living area an effect sixty times larger than the AME for the linear model, a number that is not interpretable as a marginal effect in any standard sense. An additional finding is that Breiman random forests produce less stable AME estimates than the other estimator classes, consistent with the theoretical result that gradient consistency for Breiman forests is harder to establish. Practitioners requiring random forest AMEs are advised to consider Mondrian forests, for which both gradient consistency and bootstrap validity are proved.

Several important problems remain open. The most pressing is gradient consistency and bootstrap validity for Breiman forests. The obstruction identified in the proof of Theorem 3 — that data-dependent splits couple partition geometry to response variation in a way

that prevents the key decomposition — suggests that new analytical tools will be required, possibly involving direct control over the data-dependent splitting criterion or a coupling argument relating Breiman partitions to Mondrian partitions in an appropriate sense. Establishing these results would close the gap between the theoretical guarantees and the most widely used random forest implementation.

A second open problem is the extension of gradient consistency and bootstrap validity to additional model classes. The theoretical results cover neural networks, gradient boosted models, and Mondrian forests. The BV and Sobolev architectures developed here provide a template: identify the natural function space for the estimator class, establish derivative existence and consistency in that space, verify the rate condition, and invoke the Hadamard differentiability of the AME functional. Applying this template to support vector machines with smooth kernels, Bayesian additive regression trees, generalized additive models, and kernel methods is an important direction. Each case requires identifying the appropriate function space and verifying the well-behaved conditions, but the framework makes the proof obligations explicit and the path tractable. Splines and ridge regression are particularly natural candidates since they live in Sobolev spaces by construction and the gradient consistency results are likely near-immediate from the existing theory.

A third open problem is the extension to causal inference. The average marginal effect as defined here is a predictive quantity — the average derivative of the conditional expectation function with respect to x_j , under the observational distribution of the covariates. It is not in general a causal effect. Extending the framework to settings where causal identification is possible, for example through instrumental variables or regression discontinuity designs, would substantially broaden its applicability to policy-relevant questions. The honest forest framework of Wager and Athey (2018) and the broader literature on causal machine learning suggest a natural path forward, but the connection to the AME estimand studied here has not been formalized.

A fourth direction concerns the discrete and mixed-type feature setting. The current

framework handles binary variables through a finite difference contrast and continuous variables through an adaptive centered finite difference, but ordered categorical variables, count variables, and interactions among features of mixed type are not treated systematically. The marginal effects at the mean versus average marginal effects distinction (Greene, 2012; Wooldridge, 2010) also warrants further study in the nonparametric setting, where the mean of the covariates may lie in a sparse region of the feature space and the marginal effect at the mean may therefore be a poor summary of the average effect.

Finally, the computational cost of the full bootstrap procedure — refitting the model with hyperparameter selection on each of B replicates — limits practical applicability for very large datasets or very expensive models. Approximations based on the infinitesimal jackknife (Wager et al., 2014), subsampling, or warm-started hyperparameter search reduce this cost substantially, but the statistical properties of these approximations relative to the full bootstrap have not been characterized. The Hadamard differentiability established in Lemma 5 provides a natural starting point for analyzing such approximations: any procedure that consistently estimates the derivative T'_f of the AME functional will yield asymptotically valid inference, and this opens a principled path toward computationally efficient alternatives to the full bootstrap. Developing and characterizing such procedures is an important practical problem that we leave to future work.

A Proof of Lemma 1 (Derivative Existence, Sobolev)

A.1 What Needs to Be Shown

That $\hat{f}_n \in W^{2,p}(\Omega)$ for $p > d$ implies $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$ with the norm bound:

$$\left\| \nabla \hat{f}_n \right\|_{L^p} \leq \left\| \hat{f}_n \right\|_{W^{1,p}} \leq \left\| \hat{f}_n \right\|_{W^{2,p}} \quad (36)$$

Remark 4. *The condition $p > d$ is required by the proof. It is the condition under which*

Theorem 6.5 of Evans and Gariepy (2015) guarantees pointwise differentiability of $W_{loc}^{1,p}$ functions. This is slightly stronger than the condition $p > 1$ stated in the main text and is an explicit assumption of the lemma.

A.2 Step 1: Existence of the Gradient as an L^p Function

Since $\hat{f}_n \in W^{2,p}(\Omega)$ and $p > d$, we have in particular that $\hat{f}_n \in W^{1,p}(\Omega)$. Extending $\hat{f}_n$ to a $W_{loc}^{1,p}(\mathbb{R}^d)$ function by a standard extension operator (Evans and Gariepy, 2015, Section 4.4), we may apply Theorem 6.5 of Evans and Gariepy (2015):

Theorem 6.5 (Evans and Gariepy, 2015). Assume $f \in W_{loc}^{1,p}(\mathbb{R}^n)$ for some $n < p \leq \infty$. Then f is differentiable $\mathcal{L}^n$ -a.e., and its classical derivative equals its weak derivative $\mathcal{L}^n$ -a.e.

Applying this theorem to $\hat{f}_n$ with $n = d$ and $p > d$, we conclude that $\hat{f}_n$ is classically differentiable at $\mathcal{L}^d$ -almost every $x \in \Omega$, and:

$$\nabla \hat{f}_n(x) = D\hat{f}_n(x) \quad \mathcal{L}^d\text{-a.e.} \quad (37)$$

where $D\hat{f}_n$ denotes the weak gradient of $\hat{f}_n$. Since $\hat{f}_n \in W^{1,p}(\Omega)$, the weak gradient $D\hat{f}_n \in L^p(\Omega)^d$ by definition of the Sobolev space. Therefore $\nabla \hat{f}_n \in L^p(\Omega)^d$.

Applying the same argument to each first-order weak derivative $\partial_j \hat{f}_n \in W^{1,p}(\Omega)$ — which belongs to $W^{1,p}(\Omega)$ because $\hat{f}_n \in W^{2,p}(\Omega)$ — we conclude that each $\partial_j \hat{f}_n$ is classically differentiable $\mathcal{L}^d$ -a.e. with classical second derivatives in $L^p(\Omega)$. Therefore $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$.

A.3 Step 2: The Norm Bound

We establish the chain of inequalities:

$$\left\| \nabla \hat{f}_n \right\|_{L^p} \leq \left\| \hat{f}_n \right\|_{W^{1,p}} \leq \left\| \hat{f}_n \right\|_{W^{2,p}} \quad (38)$$

First Inequality

By definition of the $W^{1,p}(\Omega)$ norm:

$$\left\| \hat{f}_n \right\|_{W^{1,p}} = \left\| \hat{f}_n \right\|_{L^p} + \sum_{j=1}^d \left\| \partial_j \hat{f}_n \right\|_{L^p} = \left\| \hat{f}_n \right\|_{L^p} + \left\| \nabla \hat{f}_n \right\|_{L^p} \quad (39)$$

Since $\left\| \hat{f}_n \right\|_{L^p} \geq 0$:

$$\left\| \hat{f}_n \right\|_{W^{1,p}} = \left\| \hat{f}_n \right\|_{L^p} + \left\| \nabla \hat{f}_n \right\|_{L^p} \geq \left\| \nabla \hat{f}_n \right\|_{L^p} \quad (40)$$

Second Inequality

By definition of the $W^{2,p}(\Omega)$ norm using multi-index notation:

$$\left\| \hat{f}_n \right\|_{W^{2,p}} = \sum_{|\alpha| \leq 2} \left\| \partial^\alpha \hat{f}_n \right\|_{L^p} \quad (41)$$

The multi-indices with $|\alpha| \leq 1$ form a strict subset of those with $|\alpha| \leq 2$, and all terms are nonnegative, so:

$$\left\| \hat{f}_n \right\|_{W^{2,p}} = \sum_{|\alpha| \leq 2} \left\| \partial^\alpha \hat{f}_n \right\|_{L^p} \geq \sum_{|\alpha| \leq 1} \left\| \partial^\alpha \hat{f}_n \right\|_{L^p} = \left\| \hat{f}_n \right\|_{W^{1,p}} \quad (42)$$

A.4 Conclusion

For $p > d$, membership of $\hat{f}_n$ in $W^{2,p}(\Omega)$ implies via Theorem 6.5 of Evans and Gariepy (2015) that $\nabla \hat{f}_n$ exists $\mathcal{L}^d$ -almost everywhere as a classical gradient, coincides with the weak gradient, and belongs to $W^{1,p}(\Omega)^d \subset L^p(\Omega)^d$. The gradient operator $\nabla : W^{2,p}(\Omega) \rightarrow W^{1,p}(\Omega)^d$ is bounded with:

$$\left\| \nabla \hat{f}_n \right\|_{L^p} \leq \left\| \hat{f}_n \right\|_{W^{1,p}} \leq \left\| \hat{f}_n \right\|_{W^{2,p}} \quad (43)$$

□

B Proof of Lemma 2 (Derivative Consistency, Sobolev)

B.1 What Needs to Be Shown

That well-behavedness in the Sobolev sense implies $\left\| \nabla \hat{f}_n - \nabla f \right\|_{L^p} \xrightarrow{p} 0$.

B.2 Proof Strategy

The proof uses a subsequence argument. We show that every subsequence of $\{\hat{f}_n\}$ has a further subsequence along which $\left\| \nabla \hat{f}_n - \nabla f \right\|_{L^p} \rightarrow 0$ almost surely. By the standard metric space characterization of convergence in probability — a sequence converges in probability if and only if every subsequence has an almost surely convergent further subsequence — this implies $\left\| \nabla \hat{f}_n - \nabla f \right\|_{L^p} \xrightarrow{p} 0$.

Suppose for contradiction that $\left\| \nabla \hat{f}_n - \nabla f \right\|_{L^p}$ does not converge to zero in probability. Then there exist $\epsilon > 0$ and $\delta > 0$ and a subsequence $\{n_k\}$ such that:

$$\mathbb{P}\left(\left\| \nabla \hat{f}_{n_k} - \nabla f \right\|_{L^p} > \epsilon\right) > \delta \quad \forall k \quad (44)$$

We derive a contradiction by extracting a further subsequence along which the gradient converges to zero almost surely.

B.3 Step 1: Extract a Strongly Convergent Subsequence

Take any subsequence $\{\hat{f}_{n_k}\}$. By the uniform $W^{2,p}(\Omega)$ boundedness condition:

$$\sup_k \left\| \hat{f}_{n_k} \right\|_{W^{2,p}} < \infty \quad \text{almost surely} \quad (45)$$

The Rellich–Kondrachov compact embedding theorem states that for bounded Lipschitz Ω , the embedding $W^{2,p}(\Omega) \hookrightarrow W^{1,p}(\Omega)$ is compact (Evans, 2010, Theorem 5, Section 5.7). Compactness of the embedding means that every sequence bounded in $W^{2,p}(\Omega)$ has a subse-

quence converging strongly in $W^{1,p}(\Omega)$. Therefore there exists a further subsequence $\{\hat{f}_{n_{k_j}}\}$ and a function $g \in W^{1,p}(\Omega)$ such that:

$$\left\| \hat{f}_{n_{k_j}} - g \right\|_{W^{1,p}} \rightarrow 0 \quad \text{almost surely as } j \rightarrow \infty \quad (46)$$

B.4 Step 2: Identify the Limit

Strong $W^{1,p}(\Omega)$ convergence implies strong L^p convergence, since by definition of the $W^{1,p}$ norm:

$$\left\| \hat{f}_{n_{k_j}} - g \right\|_{L^p} \leq \left\| \hat{f}_{n_{k_j}} - g \right\|_{W^{1,p}} \rightarrow 0 \quad \text{almost surely} \quad (47)$$

Therefore $\hat{f}_{n_{k_j}} \rightarrow g$ in L^p almost surely. By the L^p consistency assumption, $\left\| \hat{f}_n - f \right\|_{L^p} \xrightarrow{p} 0$, so in particular along the subsequence $\hat{f}_{n_{k_j}} \rightarrow f$ in L^p in probability, and hence almost surely along a further subsequence if necessary. By uniqueness of limits in $L^p(\Omega)$:

$$g = f \quad \mathcal{L}^d\text{-a.e. on } \Omega \quad (48)$$

Therefore the strongly convergent subsequence satisfies:

$$\left\| \hat{f}_{n_{k_j}} - f \right\|_{W^{1,p}} \rightarrow 0 \quad \text{almost surely} \quad (49)$$

B.5 Step 3: Extract Gradient Convergence

Since $\hat{f}_{n_{k_j}} \rightarrow f$ strongly in $W^{1,p}(\Omega)$, expanding the $W^{1,p}$ norm:

$$\left\| \hat{f}_{n_{k_j}} - f \right\|_{W^{1,p}} = \left\| \hat{f}_{n_{k_j}} - f \right\|_{L^p} + \left\| \nabla \hat{f}_{n_{k_j}} - \nabla f \right\|_{L^p} \rightarrow 0 \quad \text{almost surely} \quad (50)$$

Since both terms are nonnegative and their sum goes to zero, each term goes to zero individually. In particular:

$$\left\| \nabla \hat{f}_{n_{k_j}} - \nabla f \right\|_{L^p} \rightarrow 0 \quad \text{almost surely as } j \rightarrow \infty \quad (51)$$

B.6 Conclusion

We have shown that from any subsequence $\{\hat{f}_{n_k}\}$ we can extract a further subsequence $\{\hat{f}_{n_{k_j}}\}$ along which $\left\| \nabla \hat{f}_{n_{k_j}} - \nabla f \right\|_{L^p} \rightarrow 0$ almost surely. This contradicts the assumption that $\mathbb{P}(\left\| \nabla \hat{f}_{n_k} - \nabla f \right\|_{L^p} > \epsilon) > \delta$ for all k . Therefore the assumption was false and:

$$\left\| \nabla \hat{f}_n - \nabla f \right\|_{L^p} \xrightarrow{p} 0 \quad (52)$$

which is the conclusion of Lemma 2. □

Remark 5. *The key step is the application of Rellich–Kondrachov in Step 1. The theorem requires Ω to be bounded and Lipschitz — both of which are assumed throughout — and $p > 1$, which is assumed in the lemma statement. The compact embedding $W^{2,p} \hookrightarrow W^{1,p}$ is strictly stronger than the continuous embedding $W^{2,p} \hookrightarrow W^{1,p}$: the latter follows from the norm bound in Lemma 1, while the former provides the compactness needed to extract convergent subsequences. It is this compactness, not merely the norm bound, that allows the interchange of limit and gradient in the proof.*

C Proof of Lemma 3 (Derivative Existence, BV)

C.1 What Needs to Be Shown

Two things: that BV membership implies $D\hat{f}_n$ exists as a Radon measure with finite total variation, and the explicit formula for piecewise constant functions.

C.2 Step 1: Existence of $D\hat{f}_n$ as a Radon Measure

We establish existence in three steps, using the Riesz representation theorem and the Radon-Nikodym theorem.

Substep 1a: The Distributional Gradient as a Linear Functional

For any $\hat{f}_n \in BV(\Omega)$, the distributional gradient is defined as the linear functional $\Lambda : C_c^1(\Omega)^d \rightarrow \mathbb{R}$ given by:

$$\Lambda(\phi) = - \int_{\Omega} \hat{f}_n \operatorname{div} \phi \, dx \quad \forall \phi \in C_c^1(\Omega)^d \quad (53)$$

By the definition of $BV(\Omega)$, this functional has finite total variation:

$$|\Lambda|(\Omega) = \sup \{ \Lambda(\phi) : \phi \in C_c^1(\Omega)^d, \|\phi\|_{L^\infty} \leq 1 \} < \infty \quad (54)$$

This is precisely the condition that $\hat{f}_n \in BV(\Omega)$ imposes — the distributional gradient is a bounded linear functional on $C_c^1(\Omega)^d$ with respect to the supremum norm.

Substep 1b: Riesz Representation

Since Λ is a bounded linear functional on $C_c^1(\Omega)^d$ with finite total variation, the Riesz representation theorem (Evans and Gariepy, 2015, Theorem 1.38) guarantees the existence of a vector-valued Radon measure $D\hat{f}_n \in \mathcal{M}(\Omega)^d$ such that:

$$\Lambda(\phi) = \int_{\Omega} \phi \cdot d(D\hat{f}_n) \quad \forall \phi \in C_c^1(\Omega)^d \quad (55)$$

That is:

$$- \int_{\Omega} \hat{f}_n \operatorname{div} \phi \, dx = \int_{\Omega} \phi \cdot d(D\hat{f}_n) \quad \forall \phi \in C_c^1(\Omega)^d \quad (56)$$

This is the distributional gradient equation, and $D\hat{f}_n$ is the distributional gradient of $\hat{f}_n$ represented as a Radon measure.

Substep 1c: Finiteness of Total Variation

By the Radon-Nikodym theorem for Radon measures (Evans and Gariepy, 2015, Theorem 1.29), the derivative $D_{\mathcal{L}^d}(D\hat{f}_n)$ exists and is finite $\mathcal{L}^d$ -almost everywhere on Ω . Since Ω is bounded and the total variation of Λ is finite by the BV assumption, we conclude:

$$|D\hat{f}_n|(\Omega) = |\Lambda|(\Omega) < \infty \quad (57)$$

Therefore $D\hat{f}_n \in \mathcal{M}(\Omega)^d$ with $|D\hat{f}_n|(\Omega) < \infty$, as claimed.

C.3 Step 2: Explicit Formula for Piecewise Constant Functions

Let $\hat{f}_n = \sum_j \bar{y}_j \mathbf{1}_{R_j}$ where $\{R_j\}$ is a finite partition of Ω into sets with Lipschitz boundaries, and $\bar{y}_j$ denotes the response mean on leaf R_j . We derive the explicit formula for $D\hat{f}_n$ by applying the Gauss–Green theorem to each leaf.

Substep 2a: Applying the Definition

For any test function $\phi \in C_c^1(\Omega)^d$, the distributional gradient equation from Substep 1b gives:

$$\int_{\Omega} \phi \cdot d(D\hat{f}_n) = - \int_{\Omega} \hat{f}_n \operatorname{div} \phi \, dx = - \sum_j \bar{y}_j \int_{R_j} \operatorname{div} \phi \, dx \quad (58)$$

where the second equality uses the piecewise constant structure of $\hat{f}_n$ and linearity of the integral.

Substep 2b: Gauss–Green on Each Leaf

Since each R_j has Lipschitz boundary, the Gauss–Green theorem for BV functions (Evans and Gariepy, 2015, Theorem 5.16) applies to each leaf:

$$\int_{R_j} \operatorname{div} \phi \, dx = \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} \quad (59)$$

where ν_j denotes the outward unit normal to ∂R_j and $\mathcal{H}^{d-1}$ is the $(d-1)$ -dimensional Hausdorff measure on the boundary.

Substep 2c: Cancellation on Interior Boundaries

Substituting the Gauss–Green formula into the expression from Substep 2a:

$$\int_{\Omega} \phi \cdot d(D\hat{f}_n) = - \sum_j \bar{y}_j \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} \quad (60)$$

Every interior boundary $\partial R_j \cap \partial R_k$ between adjacent leaves j and k appears twice in this sum — once with outward normal ν_j for leaf R_j and once with outward normal $\nu_k = -\nu_j$ for leaf R_k . Collecting these paired contributions and writing $\nu_{jk} = \nu_j$ for the unit normal pointing from R_j into R_k :

$$\begin{aligned} - \sum_j \bar{y}_j \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} &= - \sum_{j \sim k} \bar{y}_j \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \\ &\quad + \sum_{j \sim k} \bar{y}_k \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \\ &= \sum_{j \sim k} (\bar{y}_k - \bar{y}_j) \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \end{aligned} \quad (61)$$

Boundary terms on $\partial\Omega$ vanish because $\phi \in C_c^1(\Omega)^d$ has compact support in Ω .

Substep 2d: Identifying the Measure

Since the equation:

$$\int_{\Omega} \phi \cdot d(D\hat{f}_n) = \sum_{j \sim k} (\bar{y}_k - \bar{y}_j) \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} d\mathcal{H}^{d-1} \quad (62)$$

holds for all $\phi \in C_c^1(\Omega)^d$, the uniqueness part of the Riesz representation theorem (Evans and Gariepy, 2015, Theorem 1.38) identifies $D\hat{f}_n$ uniquely as the measure:

$$D\hat{f}_n = \sum_{j \sim k} (\bar{y}_k - \bar{y}_j) \cdot \nu_{jk} \cdot \mathcal{H}^{d-1}|_{\partial R_j \cap \partial R_k} \quad (63)$$

where $\mathcal{H}^{d-1}|_{\partial R_j \cap \partial R_k}$ denotes the $(d-1)$ -dimensional Hausdorff measure restricted to the shared boundary between leaves j and k . This is the formula stated in the lemma. $\square$

D Proof of Lemma 4 (Derivative Consistency, BV)

D.1 What Needs to Be Shown

That well-behavedness in the BV sense implies $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$, and the strict convergence upgrade under the additional condition $\mathbb{E}[|D\hat{f}_n|(\Omega)] \rightarrow |Df|(\Omega)$.

D.2 Proof Strategy

The proof mirrors the subsequence argument of Lemma 2, replacing the Rellich–Kondrachov compact embedding with the BV compactness theorem and replacing strong L^p convergence with weak-* convergence of measures. We show that every subsequence of $\{D\hat{f}_n\}$ has a further subsequence converging weak-* to Df . By the metric space characterization of convergence in probability, this implies $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$ in probability.

Suppose for contradiction that $D\hat{f}_n$ does not converge weak-* to Df in probability. Then there exist $\epsilon > 0$, $\delta > 0$, a test function $\phi \in C_0(\Omega)^d$ with $\|\phi\|_{\infty} \leq 1$, and a subsequence $\{n_k\}$

such that:

$$\mathbb{P}\left(\left|\int_{\Omega} \phi \cdot d(D\hat{f}_{n_k}) - \int_{\Omega} \phi \cdot d(Df)\right| > \epsilon\right) > \delta \quad \forall k \quad (64)$$

We derive a contradiction by extracting a further subsequence along which weak-* convergence holds almost surely.

D.3 Part I: Weak-* Convergence

Step 1: Bound the Full BV Norm

To apply the BV compactness theorem we require a uniform bound on the full BV norm:

$$\left\|\hat{f}_n\right\|_{BV} = \left\|\hat{f}_n\right\|_{L^1} + |D\hat{f}_n|(\Omega) \quad (65)$$

For the total variation term, the well-behaved condition gives:

$$\sup_n \mathbb{E}\left[|D\hat{f}_n|(\Omega)\right] < \infty \quad (66)$$

so $|D\hat{f}_n|(\Omega)$ is bounded in expectation and hence bounded almost surely along subsequences by Markov's inequality.

For the L^1 term, the consistency assumption $\left\|\hat{f}_n - f\right\|_{L^1} \xrightarrow{p} 0$ implies that $\left\|\hat{f}_n\right\|_{L^1}$ is eventually bounded in probability. Since $f \in W^{1,p}(\Omega) \subset L^1(\Omega)$, we have $\|f\|_{L^1} < \infty$, and:

$$\left\|\hat{f}_n\right\|_{L^1} \leq \left\|\hat{f}_n - f\right\|_{L^1} + \|f\|_{L^1} \quad (67)$$

which is bounded in probability. Combining the two terms:

$$\sup_n \left\|\hat{f}_n\right\|_{BV} < \infty \quad \text{almost surely} \quad (68)$$

Step 2: Extract a Weakly-* Convergent Subsequence

Take any subsequence $\{\hat{f}_{n_k}\}$. By Step 1 it is bounded in $BV(\Omega)$ almost surely. The BV compactness theorem (Evans and Gariepy, 2015, Theorem 5.5) states that every sequence bounded in $BV(\Omega)$ for bounded Lipschitz Ω has a subsequence $\{\hat{f}_{n_{k_j}}\}$ and $g \in BV(\Omega)$ such that:

1. $\hat{f}_{n_{k_j}} \rightarrow g$ strongly in $L^1(\Omega)$
2. $D\hat{f}_{n_{k_j}} \xrightarrow{w*} Dg$ in $\mathcal{M}(\Omega)^d$

both holding almost surely. This is the BV analog of the Rellich–Kondrachov compact embedding: the BV compactness theorem gives simultaneously the L^1 convergence of the functions and the weak-* convergence of their gradient measures, in a single extraction.

Step 3: Identify the Limit

From item (1) of Step 2, $\hat{f}_{n_{k_j}} \rightarrow g$ strongly in $L^1(\Omega)$ almost surely. By the L^1 consistency assumption $\|\hat{f}_n - f\|_{L^1} \xrightarrow{p} 0$, along the subsequence $\hat{f}_{n_{k_j}} \rightarrow f$ in L^1 in probability, hence almost surely along a further subsequence. By uniqueness of L^1 limits:

$$g = f \quad \mathcal{L}^d\text{-a.e. on } \Omega \quad (69)$$

Therefore $Dg = Df$ as Radon measures, and from item (2) of Step 2:

$$D\hat{f}_{n_{k_j}} \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad \text{almost surely} \quad (70)$$

In particular, for the test function ϕ from the contradiction assumption:

$$\int_{\Omega} \phi \cdot d(D\hat{f}_{n_{k_j}}) \rightarrow \int_{\Omega} \phi \cdot d(Df) \quad \text{almost surely} \quad (71)$$

Step 4: Derive the Contradiction

The almost sure convergence in Step 3 implies:

$$\mathbb{P}\left(\left|\int_{\Omega} \phi \cdot d(D\hat{f}_{n_{k_j}}) - \int_{\Omega} \phi \cdot d(Df)\right| > \epsilon\right) \rightarrow 0 \quad (72)$$

This contradicts the assumption that the probability exceeds $\delta > 0$ for all k . Therefore the assumption was false.

Conclusion of Part I

Since every subsequence $\{D\hat{f}_{n_k}\}$ has a further subsequence converging weak-* to Df almost surely, the full sequence satisfies:

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad \text{in probability} \quad (73)$$

D.4 Part II: Strict Convergence Upgrade

We now prove the additional claim: if $\mathbb{E}[|D\hat{f}_n|(\Omega)] \rightarrow |Df|(\Omega)$ then $D\hat{f}_n \rightarrow Df$ strictly in $\mathcal{M}(\Omega)^d$, meaning weak-* convergence holds and $|D\hat{f}_n|(\Omega) \rightarrow |Df|(\Omega)$.

Step 1: Lower Semicontinuity Gives One Direction

The total variation functional $\mu \mapsto |\mu|(\Omega)$ is lower semicontinuous with respect to weak-* convergence of Radon measures (Evans and Gariepy, 2015, Theorem 5.2). Since $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$ by Part I:

$$|Df|(\Omega) \leq \liminf_{n \rightarrow \infty} |D\hat{f}_n|(\Omega) \quad (74)$$

This gives the lower bound on the total variation of the limit.

Step 2: The Additional Condition Gives the Other Direction

By the additional assumption $\mathbb{E}[|D\hat{f}_n|(\Omega)] \rightarrow |Df|(\Omega)$, we have:

$$\limsup_{n \rightarrow \infty} |D\hat{f}_n|(\Omega) \leq |Df|(\Omega) \quad (75)$$

in expectation, hence in probability along subsequences.

Step 3: Pin the Total Variation

Step 1 gives $|Df|(\Omega) \leq \liminf_{n \rightarrow \infty} |D\hat{f}_n|(\Omega)$ by lower semicontinuity of total variation under weak-* convergence. Step 2 gives $\limsup_{n \rightarrow \infty} |D\hat{f}_n|(\Omega) \leq |Df|(\Omega)$ from the additional assumption $\mathbb{E}[|D\hat{f}_n|(\Omega)] \rightarrow |Df|(\Omega)$. Since $\liminf \leq \limsup$ always, combining gives:

$$|Df|(\Omega) \leq \liminf_{n \rightarrow \infty} |D\hat{f}_n|(\Omega) \leq \limsup_{n \rightarrow \infty} |D\hat{f}_n|(\Omega) \leq |Df|(\Omega) \quad (76)$$

All inequalities are therefore equalities and $|D\hat{f}_n|(\Omega) \rightarrow |Df|(\Omega)$ in probability. Combined with the weak-* convergence $D\hat{f}_n \xrightarrow{w*} Df$ established in Part I, this is precisely the definition of strict convergence in $\mathcal{M}(\Omega)^d$ (Ambrosio et al., 2000, Definition 3.13). $\square$

Remark 6. *The parallel between Lemma 2 and Lemma 4 is exact at the structural level. In both cases: uniform boundedness in a strong space gives compactness via a compact embedding or compactness theorem; the consistency condition identifies the limit; and the subsequence argument promotes deterministic convergence to convergence in probability. The difference is entirely in the function spaces and the topology of convergence — strong L^p in the Sobolev case, weak-* of measures in the BV case — reflecting the fundamental difference in the inductive biases of smooth and piecewise constant estimators.*

Remark 7. *The strict convergence upgrade requires the additional assumption $\mathbb{E}[|D\hat{f}_n|(\Omega)] \rightarrow |Df|(\Omega)$, which is not implied by weak-* convergence alone. This is because weak-* convergence of measures does not in general imply convergence of total masses — mass can escape*

to the boundary or concentrate in ways invisible to continuous test functions. The lower semicontinuity of total variation (Evans and Gariepy, 2015, Theorem 5.2) gives the $\liminf$ inequality for free; pinning the $\limsup$ requires the additional assumption. In applications where only weak- $*$ convergence is needed — as in the bootstrap corollaries — the additional assumption is not required.

E Proof of Lemma 5 (Hadamard Differentiability of the AME Functional)

E.1 What Needs to Be Shown

That the functional $T(f) = \int_{\Omega} \nabla f(x) dP(x)$ is Hadamard differentiable at f tangentially to $C(\Omega)^d$, with derivative $T'_f(h) = \int_{\Omega} \nabla h(x) dP(x)$.

Recall that Hadamard differentiability tangentially to $C(\Omega)^d$ means that for every sequence $t_n \rightarrow 0$ and every sequence $h_n \rightarrow h$ in $C(\Omega)^d$ (van der Vaart and Wellner, 1996, Definition 3.9.2):

$$\frac{T(f + t_n h_n) - T(f)}{t_n} \rightarrow T'_f(h) \tag{77}$$

Once Hadamard differentiability is established, the functional delta method (van der Vaart and Wellner, 1996, Theorem 3.9.4) and the bootstrap validity result of Horowitz (2001, Chapter 3.6) apply directly to give Corollaries 1, 2, and 3.

E.2 Step 1: Linearity and Boundedness of the Candidate Derivative

The candidate derivative $T'_f(h) = \int_{\Omega} \nabla h \, dP$ is linear in h by linearity of the integral and the gradient operator. It is bounded on $C(\Omega)^d$ since:

$$|T'_f(h)| \leq \int_{\Omega} |\nabla h(x)| \, dP(x) \leq \|\nabla h\|_{L^\infty(\Omega)} \cdot P(\Omega) = \|\nabla h\|_{L^\infty(\Omega)} \quad (78)$$

using that P is a probability measure so $P(\Omega) = 1$. Therefore $T'_f : C(\Omega)^d \rightarrow \mathbb{R}^d$ is a bounded linear map.

E.3 Step 2: Verify the Hadamard Derivative Formula

The key observation is that T is itself a linear functional of f . Therefore for any $t_n \rightarrow 0$ and $h_n \rightarrow h$ in $C(\Omega)^d$:

$$\begin{aligned} \frac{T(f + t_n h_n) - T(f)}{t_n} &= \frac{1}{t_n} \int_{\Omega} [\nabla(f + t_n h_n)(x) - \nabla f(x)] \, dP(x) \\ &= \frac{1}{t_n} \int_{\Omega} t_n \nabla h_n(x) \, dP(x) \\ &= \int_{\Omega} \nabla h_n(x) \, dP(x) \end{aligned} \quad (79)$$

where the second equality uses linearity of the gradient operator: $\nabla(f + t_n h_n) = \nabla f + t_n \nabla h_n$.

It remains to show that $\int_{\Omega} \nabla h_n \, dP \rightarrow \int_{\Omega} \nabla h \, dP$. Since $h_n \rightarrow h$ in $C(\Omega)^d$, we have $\nabla h_n \rightarrow \nabla h$ uniformly on Ω . The density of P is bounded on Ω by assumption, so P is absolutely continuous with respect to $\mathcal{L}^d$ with bounded Radon-Nikodym derivative. Applying the dominated convergence theorem with dominating function $\|\nabla h_n\|_{L^\infty} \cdot p_{\max}$ where $p_{\max}$ is the essential supremum of the density of P :

$$\int_{\Omega} \nabla h_n \, dP \rightarrow \int_{\Omega} \nabla h \, dP = T'_f(h) \quad (80)$$

This establishes the Hadamard derivative formula.

E.4 Step 3: Tangential Differentiability

The differentiability is tangential to $C(\Omega)^d$ rather than to all of $W^{1,p}(\Omega)$ for the following reason. The derivative formula requires passing the limit $h_n \rightarrow h$ through the gradient and integral. This exchange is justified by uniform convergence of $\nabla h_n \rightarrow \nabla h$, which holds when $h_n \rightarrow h$ in $C(\Omega)^d$ but not in general when $h_n \rightarrow h$ only in $W^{1,p}(\Omega)$.

Formally, following van der Vaart and Wellner (1996, Definition 3.9.2), tangential differentiability to $C(\Omega)^d$ means the derivative formula holds for all sequences $t_n \rightarrow 0$ and $h_n \rightarrow h$ with $h_n, h \in C(\Omega)^d$. The condition $p > d$ in the lemma statement ensures via the Sobolev embedding $W^{1,p}(\Omega) \hookrightarrow C^{0,\alpha}(\Omega)$ for $\alpha = 1 - d/p$ (Evans, 2010, Section 5.6) that the estimator sequences $\{\hat{f}_n\}$ take values in $C(\Omega)^d$ with probability one, so the tangential differentiability condition is the operative one for the bootstrap corollaries.

E.5 Conclusion

The functional $T(f) = \int_{\Omega} \nabla f dP$ is Hadamard differentiable at f tangentially to $C(\Omega)^d$ with derivative $T'_f(h) = \int_{\Omega} \nabla h dP$. The proof required only the linearity of T , the uniform convergence of $h_n \rightarrow h$ in $C(\Omega)^d$, and the dominated convergence theorem. By van der Vaart and Wellner (1996, Theorem 3.9.4) and Horowitz (2001, Chapter 3.6), Hadamard differentiability tangentially to $C(\Omega)^d$ together with the rate conditions verified in Corollaries 1, 2, and 3 implies bootstrap validity for the AME estimator under each of the three estimator classes. $\square$

Remark 8. *Lemma 5 is the shortest proof in the appendix precisely because the AME functional is linear. Hadamard differentiability of a linear functional is immediate once the derivative formula is identified — the quotient $[T(f + t_n h_n) - T(f)]/t_n$ equals $T(h_n)$ exactly, with no remainder term, and convergence to $T(h)$ follows from continuity. The hard work*

is in Lemmas 2 and 4, which establish that the estimator sequences live in the right function spaces with the right regularity. Lemma 5 then harvests those results with a one-line calculation.

Remark 9. *The condition that P has density bounded away from zero and infinity on Ω is used in Step 2 to apply the dominated convergence theorem and in Step 3 to ensure the Sobolev embedding gives the required regularity. The lower bound on the density ensures the integral $\int_{\Omega} \nabla f dP$ is well defined and nondegenerate. The upper bound provides the dominating function for the dominated convergence argument. Both are mild conditions satisfied by any covariate distribution with reasonable support and no extreme mass concentrations.*

F Proof of Theorem 1 (Neural Networks)

F.1 Setup and Notation

Let $\hat{f}_n : \Omega \rightarrow \mathbb{R}$ be a feedforward neural network with L layers, weight matrices $W_{\ell}^{(n)} \in \mathbb{R}^{d_{\ell} \times d_{\ell-1}}$, bias vectors $b_{\ell}^{(n)} \in \mathbb{R}^{d_{\ell}}$, and activation function $\sigma \in C^2(\mathbb{R})$. The network computes:

$$\hat{f}_n(x) = W_L^{(n)} z_{L-1} + b_L^{(n)} \quad (81)$$

where the intermediate representations are defined recursively by:

$$z_{\ell} = \sigma\left(W_{\ell}^{(n)} z_{\ell-1} + b_{\ell}^{(n)}\right), \quad z_0 = x \quad (82)$$

with σ applied elementwise. Denote the pre-activation at layer ℓ by $a_{\ell} = W_{\ell}^{(n)} z_{\ell-1} + b_{\ell}^{(n)}$ and the diagonal matrix of activation derivatives by $\Sigma'_{\ell} = \text{diag}(\sigma'(a_{\ell}^{(1)}), \dots, \sigma'(a_{\ell}^{(d_{\ell})}))$ and similarly $\Sigma''_{\ell} = \text{diag}(\sigma''(a_{\ell}^{(1)}), \dots, \sigma''(a_{\ell}^{(d_{\ell})}))$.

Define the product of weight matrices from layer k to layer ℓ as:

$$M_{\ell:k} = W_{\ell}^{(n)} \Sigma'_{\ell-1} W_{\ell-1}^{(n)} \Sigma'_{\ell-2} \cdots W_{k+1}^{(n)} \Sigma'_k \quad \text{for } \ell > k \quad (83)$$

with the convention $M_{\ell:\ell} = \Sigma'_\ell$. This notation captures the Jacobian of the composition of layers from k to ℓ .

The proof proceeds in two steps. Step 1 establishes uniform $W^{2,p}(\Omega)$ boundedness. Step 2 applies Lemma 2 to conclude.

F.2 Step 1: Uniform $W^{2,p}$ Boundedness

Claim: $\sup_n \left\| \hat{f}_n \right\|_{W^{2,p}(\Omega)} < \infty$.

Substep 1a: Bound on $\hat{f}_n$ in L^p

Since $\left\| \hat{f}_n - f \right\|_{L^p} \xrightarrow{p} 0$ and $f \in W^{1,p}(\Omega) \subset L^p(\Omega)$, the triangle inequality gives:

$$\left\| \hat{f}_n \right\|_{L^p} \leq \left\| \hat{f}_n - f \right\|_{L^p} + \|f\|_{L^p} \quad (84)$$

The first term converges to zero in probability and the second is finite since $f \in L^p(\Omega)$. Therefore $\left\| \hat{f}_n \right\|_{L^p}$ is eventually bounded in probability and this bound can be made uniform in n by passing to a subsequence or imposing a mild boundedness condition on the network output.

Substep 1b: First Derivative Bound

Differentiating $\hat{f}_n$ with respect to the input x via the chain rule through all L layers, the gradient at $x \in \Omega$ is:

$$\nabla \hat{f}_n(x) = \left(W_L^{(n)} M_{L-1:1} \right)^\top \quad (85)$$

where $M_{L-1:1} = W_{L-1}^{(n)} \Sigma'_{L-2} W_{L-2}^{(n)} \cdots \Sigma'_1 W_1^{(n)}$ is the product of weight-activation Jacobians from layer 1 to layer $L-1$. More explicitly, for the j -th input coordinate:

$$\frac{\partial \hat{f}_n}{\partial x_j}(x) = \left(W_L^{(n)} \right)_j \cdot \prod_{\ell=L-1}^1 \left(W_\ell^{(n)} \Sigma'_{\ell-1} \right) \cdot e_j \quad (86)$$

where e_j is the j -th standard basis vector.

Taking the operator norm of this product and applying submultiplicativity:

$$\left| \frac{\partial \hat{f}_n}{\partial x_j}(x) \right| \leq \|W_L^{(n)}\| \cdot \prod_{\ell=1}^{L-1} \|W_\ell^{(n)}\| \cdot \|\Sigma'_\ell\| \leq \prod_{\ell=1}^L \|W_\ell^{(n)}\| \cdot \|\sigma'\|_\infty^{L-1} \quad (87)$$

using that $\|\Sigma'_\ell\| = \max_i |\sigma'(a_\ell^{(i)})| \leq \|\sigma'\|_\infty < \infty$ since $\sigma \in C^2$. This bound holds pointwise for every $x \in \Omega$, so integrating over Ω :

$$\|\nabla \hat{f}_n\|_{L^p} \leq \|\nabla \hat{f}_n\|_{L^\infty} \cdot |\Omega|^{1/p} \leq \prod_{\ell} \|W_\ell^{(n)}\| \cdot \|\sigma'\|_\infty^{L-1} \cdot |\Omega|^{1/p} \quad (88)$$

Substep 1c: Second Derivative Bound

Differentiating the gradient formula once more with respect to x_i using the product rule and chain rule, the (i, j) entry of the Hessian $\nabla^2 \hat{f}_n(x)$ is:

$$\frac{\partial^2 \hat{f}_n}{\partial x_i \partial x_j}(x) = \sum_{\ell=1}^{L-1} \left(W_L^{(n)} M_{L-1:\ell+1} \right)_\cdot \cdot \Sigma''_\ell \cdot (M_{\ell:1} e_j) \cdot (M_{\ell:1} e_i)^\top \cdot \left(W_L^{(n)} M_{L-1:\ell+1} \right)_\cdot^\top \quad (89)$$

where the sum runs over all layers ℓ at which the second derivative of σ is applied. The structure is: the forward pass from the input to layer ℓ contributes $M_{\ell:1} e_j$ for the j -th coordinate direction, the activation second derivative at layer ℓ contributes Σ''_ℓ , and the backward pass from layer ℓ to the output contributes $W_L^{(n)} M_{L-1:\ell+1}$.

Taking absolute values and applying submultiplicativity of operator norms:

$$\begin{aligned} \left| \frac{\partial^2 \hat{f}_n}{\partial x_i \partial x_j}(x) \right| &\leq \sum_{\ell=1}^{L-1} \left\| W_L^{(n)} M_{L-1:\ell+1} \right\|^2 \cdot \|\Sigma''_\ell\| \cdot \|M_{\ell:1}\|^2 \\ &\leq \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^L \|W_k^{(n)}\| \right)^2 \cdot \|\sigma''\|_\infty \cdot \left(\prod_{k=1}^{\ell} \|W_k^{(n)}\| \right)^2 \cdot \|\sigma'\|_\infty^{2(L-2)} \\ &\leq (L-1) \cdot \left(\prod_{\ell=1}^L \|W_\ell^{(n)}\| \right)^2 \cdot \|\sigma''\|_\infty \cdot \|\sigma'\|_\infty^{2(L-2)} \end{aligned} \quad (90)$$

using $\|\Sigma_\ell''\| \leq \|\sigma''\|_\infty < \infty$ since $\sigma \in C^2$, and collecting all weight norm factors into a single product. This bound holds pointwise for every $x \in \Omega$, so:

$$\left\| \nabla^2 \hat{f}_n \right\|_{L^p} \leq \left\| \nabla^2 \hat{f}_n \right\|_{L^\infty} \cdot |\Omega|^{1/p} \leq (L-1) \cdot \left(\prod_\ell \left\| W_\ell^{(n)} \right\| \right)^2 \cdot \|\sigma''\|_\infty \cdot \|\sigma'\|_\infty^{2(L-2)} \cdot |\Omega|^{1/p} \quad (91)$$

Combining Substeps 1a–1c

Summing the three bounds:

$$\left\| \hat{f}_n \right\|_{W^{2,p}} = \left\| \hat{f}_n \right\|_{L^p} + \left\| \nabla \hat{f}_n \right\|_{L^p} + \left\| \nabla^2 \hat{f}_n \right\|_{L^p} \leq C \left(\prod_\ell \left\| W_\ell^{(n)} \right\|, \|\sigma'\|_\infty, \|\sigma''\|_\infty, L, |\Omega| \right) \quad (92)$$

where C is a finite constant depending on the indicated quantities but not on n or x . By the assumption $\sup_n \prod_\ell \left\| W_\ell^{(n)} \right\| < \infty$ — which formalizes the effect of L^2 regularization during training (?) — and the finiteness of $\|\sigma'\|_\infty$ and $\|\sigma''\|_\infty$ from $\sigma \in C^2$, the right-hand side is uniformly bounded in n . Therefore:

$$\sup_n \left\| \hat{f}_n \right\|_{W^{2,p}} < \infty \quad (93)$$

This completes Step 1.

F.3 Step 2: L^p Consistency

The L^p consistency condition $\left\| \hat{f}_n - f \right\|_{L^p} \xrightarrow{p} 0$ is the standard universal approximation result for smooth neural networks. Hornik et al. (1990) establish that multilayer feedforward networks with smooth activation functions are dense in $W^{1,p}(\Omega)$ with respect to the $W^{1,p}$ norm, implying in particular density in $L^p(\Omega)$. Combined with standard empirical risk minimization results establishing that the training procedure converges to the best approximation in the function class, this gives $\left\| \hat{f}_n - f \right\|_{L^p} \xrightarrow{p} 0$.

F.4 Step 3: Applying Lemma 2

By Step 1, $\sup_n \|\hat{f}_n\|_{W^{2,p}} < \infty$. By Step 2, $\|\hat{f}_n - f\|_{L^p} \xrightarrow{p} 0$. These are precisely the two conditions of the well-behaved definition in the Sobolev sense (Definition 1). Lemma 2 then gives:

$$\|\nabla \hat{f}_n - \nabla f\|_{L^p} \xrightarrow{p} 0 \quad (94)$$

which is the conclusion of Theorem 1. $\square$

Remark 10. *The weight norm assumption $\sup_n \prod_\ell \|W_\ell^{(n)}\| < \infty$ is the key regularity condition on the estimator. It formalizes the effect of L^2 weight decay regularization during training, which penalizes large weights and keeps the product of operator norms bounded. Under standard conditions on the regularization parameter and optimization procedure, this bound holds with high probability over the training randomness (?). The condition is stated as an assumption of the theorem rather than derived from first principles because its verification depends on details of the training procedure that are outside the scope of the present analysis.*

Remark 11. *Theorem 1 is stated for smooth activations $\sigma \in C^2$. For ReLU networks $\sigma \notin C^2$ and the Hessian calculation in Substep 1c does not apply directly. The first derivative exists almost everywhere with $\sigma' = \mathbf{1}_{(0,\infty)}$ bounded, so Substep 1b goes through unchanged. The distributional second derivative is $\sigma'' = \delta_0$, a Dirac mass at zero, placing $\hat{f}_n$ in a mixed $W^{1,p} \cap BV$ space rather than $W^{2,p}(\Omega)$. A modified version of Lemma 2 applicable in this mixed space, or a restriction to input distributions placing zero mass on the kink set $\{x : a_\ell(x) = 0\}$, would be required to handle this case. We leave this extension for future work.*

G Proof of Theorem 2 (Gradient Boosting)

G.1 Setup and Notation

By Proposition 1 of Bühlmann and Yu (2003), the L_2 Boost estimator after m iterations can be represented as:

$$\hat{F}_m = (I - (I - S)^{m+1})Y \quad (95)$$

where $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the base learner operator. The residual at step m is:

$$u_m = Y - \hat{F}_m = (I - S)^m Y \quad (96)$$

so that each base learner fit is $\hat{f}_m = Su_m$ and the ensemble updates as $\hat{F}_m = \hat{F}_{m-1} + \hat{f}_m$.

By Proposition 2 of Bühlmann and Yu (2003), for a symmetric linear learner S with eigenvalues $\{\lambda_k; k = 1, \dots, n\}$, the eigenvalues of the boosting operator $\mathcal{B}_m = I - (I - S)^{m+1}$ are $\{1 - (1 - \lambda_k)^{m+1}; k = 1, \dots, n\}$.

Under the eigenvalue condition $0 < \lambda_k \leq 1$ for all k (Bühlmann and Yu, 2003, Theorem 1), let:

$$\rho = \max_k (1 - \lambda_k) \in [0, 1) \quad (97)$$

Since $\lambda_k > 0$ for all k , we have $\rho < 1$. The bias decays exponentially fast: by Theorem 1(a) of Bühlmann and Yu (2003):

$$\|u_m\|_{L^2}^2 = \|(I - S)^m Y\|_{L^2}^2 \leq \rho^{2m} \|Y\|_{L^2}^2 \quad (98)$$

In particular:

$$\|u_m\|_{L^2} \leq C \rho^m, \quad C = \|Y\|_{L^2} \quad (99)$$

This geometric decay is the key engine for both cases of the proof.

The two cases of the theorem are handled separately.

G.2 Case (i): Smooth Base Learners

Step 1: $W^{2,p}$ Bound for a Single Base Learner

Each $\hat{f}_m = Su_m \in W^{2,p}(\Omega)$ by assumption on the base learner class. For shallow neural net base learners this follows from Appendix F. For spline base learners it follows from standard spline approximation theory (?). We require a uniform bound:

$$\sup_m \left\| \hat{f}_m \right\|_{W^{2,p}} \leq C' \|u_m\|_{L^2} < \infty \quad (100)$$

where the bound on $\left\| \hat{f}_m \right\|_{W^{2,p}}$ in terms of $\|u_m\|_{L^2}$ follows from the boundedness of S as an operator from L^2 to $W^{2,p}$.

Step 2: Ensemble Bound via Geometric Decay

By linearity of the $W^{2,p}$ norm and the triangle inequality:

$$\left\| \hat{F}_m \right\|_{W^{2,p}} \leq \sum_{j=1}^m \left\| \hat{f}_j \right\|_{W^{2,p}} \leq C' \sum_{j=1}^m \|u_j\|_{L^2} \leq C' C \sum_{j=1}^m \rho^j \leq \frac{C' C \rho}{1 - \rho} < \infty \quad (101)$$

using the geometric decay (99) and the fact that $\sum_{j=1}^{\infty} \rho^j = \rho/(1 - \rho) < \infty$ for $\rho < 1$. The bound is uniform in m .

Step 3: Apply Lemma 2

L^p consistency is assumed. Uniform $W^{2,p}(\Omega)$ boundedness follows from Step 2. Lemma 2 gives $\left\| \nabla \hat{F}_m - \nabla f \right\|_{L^p} \xrightarrow{p} 0$. $\square$

G.3 Case (ii): Tree Stump Base Learners

Each base learner $\hat{f}_m = Su_m$ is a depth-1 tree (stump) of the form:

$$\hat{f}_m(x) = c_m^L \mathbf{1}[x_j \leq t_m] + c_m^R \mathbf{1}[x_j > t_m] \quad (102)$$

for some split variable j , threshold t_m , and leaf constants:

$$c_m^L = \frac{1}{|I_L|} \sum_{i \in I_L} u_m(x_i), \quad c_m^R = \frac{1}{|I_R|} \sum_{i \in I_R} u_m(x_i) \quad (103)$$

where $I_L = \{i : x_{ij} \leq t_m\}$ and $I_R = \{i : x_{ij} > t_m\}$ are the left and right leaf index sets.

Step 1: BV Norm of a Single Stump

By Lemma 3 the distributional gradient of $\hat{f}_m$ is:

$$D\hat{f}_m = (c_m^R - c_m^L) \cdot \nu_m \cdot \mathcal{H}^{d-1} \llcorner_{\{x_j=t_m\} \cap \Omega} \quad (104)$$

so the total variation is:

$$|D\hat{f}_m|(\Omega) = |c_m^R - c_m^L| \cdot \mathcal{H}^{d-1}(\{x_j = t_m\} \cap \Omega) \quad (105)$$

The geometric term satisfies $\mathcal{H}^{d-1}(\{x_j = t_m\} \cap \Omega) \leq \text{diam}(\Omega)^{d-1}$, a constant depending only on Ω . For the response term, by Cauchy–Schwarz applied to the leaf means:

$$|c_m^R - c_m^L| \leq \frac{1}{|I_R|} \sum_{i \in I_R} |u_m(x_i)| + \frac{1}{|I_L|} \sum_{i \in I_L} |u_m(x_i)| \leq 2 \|u_m\|_{L^2} \quad (106)$$

Therefore:

$$|D\hat{f}_m|(\Omega) \leq 2 \text{diam}(\Omega)^{d-1} \cdot \|u_m\|_{L^2} \leq 2 \text{diam}(\Omega)^{d-1} \cdot C \rho^m \quad (107)$$

using the geometric decay (99).

Step 2: Ensemble BV Bound via Geometric Series

By subadditivity of total variation:

$$|D\hat{F}_m|(\Omega) \leq \sum_{j=1}^m |D\hat{f}_j|(\Omega) \leq 2 \text{diam}(\Omega)^{d-1} \cdot C \sum_{j=1}^m \rho^j \leq \frac{2 \text{diam}(\Omega)^{d-1} \cdot C \rho}{1 - \rho} < \infty \quad (108)$$

The geometric series $\sum_{j=1}^{\infty} \rho^j = \rho/(1 - \rho)$ converges because $\rho < 1$, which follows from the eigenvalue condition $\lambda_k > 0$ for all k . Taking expectations:

$$\mathbb{E}[|D\hat{F}_m|(\Omega)] \leq \frac{2 \operatorname{diam}(\Omega)^{d-1} \cdot \mathbb{E}[C]\rho}{1 - \rho} < \infty \quad (109)$$

uniformly in m , where $\mathbb{E}[C] = \mathbb{E}[\|Y\|_{L^2}] < \infty$ by assumption.

Step 3: Apply Lemma 4

L^1 consistency is assumed. Uniform BV boundedness follows from Step 2. Lemma 4 gives $D\hat{F}_m \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. $\square$

Remark 12. *The key step in both cases is the geometric decay $\|u_m\|_{L^2} \leq C\rho^m$, which follows from the eigenvalue condition $0 < \lambda_k \leq 1$ via Proposition 1 and Theorem 1 of Bühlmann and Yu (2003). This decay is much faster than any polynomial rate and implies that the geometric series $\sum_j \rho^j$ converges without any condition on step sizes. The eigenvalue condition is the operative regularity assumption for both cases of the theorem, replacing the step size conditions common in the boosting literature.*

Remark 13. *When $\lambda_k \in \{0, 1\}$ for all k — as occurs for linear projection operators — Corollary 1 of Bühlmann and Yu (2003) shows that $\mathcal{B}_m \equiv S$ for all m , meaning boosting does not improve on the base learner. In this degenerate case $\rho = \max_k(1 - \lambda_k)$ takes the value 1 for any k with $\lambda_k = 0$, and the geometric series does not converge. The strict positivity condition $\lambda_k > 0$ excludes this case and corresponds to the requirement that the base learner has positive but incomplete explanatory power — the standard “weak learner” assumption in the boosting literature.*

H Proof of Theorem 3 (Random Forests)

H.1 Setup

Let $\hat{f}_n = \frac{1}{B} \sum_{b=1}^B T_b$ where T_b are individual Mondrian trees. Each T_b is piecewise constant on a random partition $\mathcal{P}_b$ of $\Omega = [0, 1]^d$ generated by the Mondrian process $MP(\lambda_n, [0, 1]^d)$ independently of the data, with budget $\lambda_n \asymp n^{1/(d+2)}$ (Mourtada et al., 2017). The partition $\mathcal{P}_b$ consists of $K_{\lambda_n} + 1$ hyperrectangular cells, where K_{λ_n} is the number of splits (Roy and Teh, 2009).

H.2 Step 1: Each Tree is in BV

Each T_b is piecewise constant on a finite partition and therefore in $BV(\Omega)$ by Lemma 3. The forest average $\hat{f}_n = \frac{1}{B} \sum_b T_b$ is a finite convex combination of BV functions, hence $\hat{f}_n \in BV(\Omega)$ with:

$$|D\hat{f}_n|(\Omega) \leq \frac{1}{B} \sum_{b=1}^B |DT_b|(\Omega) \quad (110)$$

by convexity of the total variation functional (Ambrosio et al., 2000, Proposition 3.6).

H.3 Step 2: Uniform BV Boundedness via the Second Moment Argument

Claim: $\sup_n \mathbb{E}[|D\hat{f}_n|(\Omega)] < \infty$.

The triangle inequality applied to the forest average is too crude because individual tree BV norms grow with n . Instead we work with the second moment and apply Cauchy–Schwarz:

$$\mathbb{E}[|D\hat{f}_n|(\Omega)] \leq \sqrt{\mathbb{E}[|D\hat{f}_n|(\Omega)^2]} \quad (111)$$

It therefore suffices to show $\mathbb{E}[|D\hat{f}_n|(\Omega)^2] < \infty$ uniformly in n .

Substep 2a: Expand the Second Moment

Writing $D\hat{f}_n = \frac{1}{B} \sum_b DT_b$ and expanding:

$$\mathbb{E}\left[|D\hat{f}_n|(\Omega)^2\right] = \frac{1}{B^2} \mathbb{E}\left[\left|\sum_b DT_b\right|(\Omega)^2\right] = \frac{1}{B^2} \sum_{b,b'} \mathbb{E}\left[\int \phi \cdot d(DT_b) \cdot \int \phi \cdot d(DT_{b'})\right] \quad (112)$$

for a suitable test function ϕ with $\|\phi\|_\infty \leq 1$. The double sum splits into diagonal terms $b = b'$ and cross terms $b \neq b'$.

Substep 2b: Cross Terms Vanish

For $b \neq b'$ the partitions $\mathcal{P}_b$ and $\mathcal{P}_{b'}$ are independent of each other and of the data. By independence:

$$\mathbb{E}\left[\int \phi \cdot d(DT_b) \cdot \int \phi \cdot d(DT_{b'})\right] = \mathbb{E}\left[\int \phi \cdot d(DT_b)\right] \cdot \mathbb{E}\left[\int \phi \cdot d(DT_{b'})\right] \quad (113)$$

We claim each factor is zero. By Lemma 3 and the explicit formula for DT_b :

$$\mathbb{E}\left[\int \phi \cdot d(DT_b)\right] = \mathbb{E}\left[\sum_{j \sim k} (\bar{y}_k^b - \bar{y}_j^b) \int_{\partial R_j^b \cap \partial R_k^b} \phi \cdot \nu_{jk} d\mathcal{H}^{d-1}\right] \quad (114)$$

Since $\mathcal{P}_b$ is independent of the data, we condition on the partition and take the expectation over the data first:

$$= \mathbb{E}_{\mathcal{P}_b} \left[\sum_{j \sim k} \mathbb{E}_{\text{data}} [\bar{y}_k^b - \bar{y}_j^b] \int_{\partial R_j^b \cap \partial R_k^b} \phi \cdot \nu_{jk} d\mathcal{H}^{d-1} \right] \quad (115)$$

The leaf means satisfy $\mathbb{E}_{\text{data}} [\bar{y}_j^b] \rightarrow f(\bar{x}_j^b)$ as $n \rightarrow \infty$, where $\bar{x}_j^b$ is the center of leaf R_j^b . For adjacent leaves j and k , by the Lipschitz continuity of f :

$$\mathbb{E}_{\text{data}} [\bar{y}_k^b - \bar{y}_j^b] \approx f(\bar{x}_k^b) - f(\bar{x}_j^b) = O(D_{\lambda_n}) \quad (116)$$

where D_{λ_n} is the cell diameter. As $\lambda_n \rightarrow \infty$ the cell diameter $D_{\lambda_n} \rightarrow 0$ and each term vanishes. More precisely, since $f \in W^{1,p}(\Omega)$ with $p > d$, the Sobolev embedding $W^{1,p} \hookrightarrow C^{0,\alpha}$ for $\alpha = 1 - d/p$ gives:

$$|\mathbb{E}_{\text{data}}[\bar{y}_k^b - \bar{y}_j^b]| \leq C \cdot \text{diam}(R_j^b \cup R_k^b)^\alpha \leq C \cdot D_{\lambda_n}^\alpha \rightarrow 0 \quad (117)$$

Therefore $\mathbb{E}[\int \phi \cdot d(DT_b)] \rightarrow 0$ and the cross terms vanish asymptotically. For n large enough the cross terms are negligible and the second moment is dominated by the diagonal terms.

Substep 2c: Bound the Diagonal Terms

For the diagonal terms $b = b'$:

$$\mathbb{E}[|DT_b|(\Omega)^2] = \mathbb{E}\left[\left(\sum_{j \sim k} |\bar{y}_k^b - \bar{y}_j^b| \cdot \mathcal{H}^{d-1}(\partial R_j^b \cap \partial R_k^b)\right)^2\right] \quad (118)$$

Expanding the square and separating geometric and response terms using the independence of $\mathcal{P}_b$ from the data:

$$\mathbb{E}[|DT_b|(\Omega)^2] \leq \mathbb{E}_{\mathcal{P}_b}[K_{\lambda_n}^2] \cdot \sup_{j \sim k} \mathbb{E}_{\text{data}}[|\bar{y}_k^b - \bar{y}_j^b|^2] \quad (119)$$

where K_{λ_n} is the number of adjacent pairs, which equals the number of splits.

Geometric factor. The Mondrian process generates cuts according to a recursive tree structure. By the conditional independence of subtrees (Roy and Teh, 2009), the number of splits satisfies the distributional recursion:

$$K_{\lambda_n} = 1 + K_{\lambda_n, L} + K_{\lambda_n, R} \quad (120)$$

where $K_{\lambda_n, L}$ and $K_{\lambda_n, R}$ are the split counts in the left and right subtrees, conditionally independent given the first cut. From Mourtada et al. (2017):

$$\mathbb{E}[K_{\lambda_n}] \leq (e(\lambda_n + 1))^d = O(n^{d/(d+2)}) \quad (121)$$

Taking the second moment via the recursion and using conditional independence:

$$\mathbb{E}[K_{\lambda_n}^2] = \mathbb{E}[K_{\lambda_n}] + \mathbb{E}[K_{\lambda_n}]^2 = O(n^{d/(d+2)}) + O(n^{2d/(d+2)}) = O(n^{2d/(d+2)}) \quad (122)$$

where the first equality uses the Poisson-like variance structure of the Mondrian process implied by the conditional independence recursion.

Response factor. For adjacent leaves j and k , by Cauchy–Schwarz on the sample means:

$$\mathbb{E}_{\text{data}}[|\bar{y}_k^b - \bar{y}_j^b|^2] \leq 4 \mathbb{E}_{\text{data}}[\|r^{(b)}\|_{L^2}^2] \quad (123)$$

where $r^{(b)}$ denotes the residual within the relevant cells. Using the cell diameter bound of Roy and Teh (2009):

$$\mathbb{E}[D_{\lambda_n}(x)^2] \leq \frac{4d}{\lambda_n^2} = O(n^{-2/(d+2)}) \quad (124)$$

and the Hölder continuity of $f \in W^{1,p}$ with $p > d$ giving $\alpha = 1 - d/p$:

$$\mathbb{E}_{\text{data}}[|\bar{y}_k^b - \bar{y}_j^b|^2] \leq C^2 (\mathbb{E}[D_{\lambda_n}(x)^2])^\alpha \leq C^2 \left(\frac{4d}{\lambda_n^2}\right)^\alpha = O(n^{-2\alpha/(d+2)}) \quad (125)$$

Combining.

$$\mathbb{E}[|DT_b|(\Omega)^2] \leq O(n^{2d/(d+2)}) \cdot O(n^{-2\alpha/(d+2)}) = O(n^{2(d-\alpha)/(d+2)}) \quad (126)$$

Therefore:

$$\mathbb{E}\left[|D\hat{f}_n|(\Omega)^2\right] = \frac{1}{B^2} \sum_b \mathbb{E}[|DT_b|(\Omega)^2] = \frac{1}{B} O(n^{2(d-\alpha)/(d+2)}) \quad (127)$$

For this to be bounded uniformly in n we require:

$$B \gg n^{2(d-\alpha)/(d+2)} \quad (128)$$

This is a condition on the number of trees relative to the dimension and smoothness of f .

Under this condition:

$$\mathbb{E}\left[|D\hat{f}_n|(\Omega)\right] \leq \sqrt{\mathbb{E}\left[|D\hat{f}_n|(\Omega)^2\right]} = \sqrt{\frac{1}{B} O(n^{2(d-\alpha)/(d+2)})} < \infty \quad (129)$$

uniformly in n , completing the uniform BV boundedness claim.

Remark 14. *The condition $B \gg n^{2(d-\alpha)/(d+2)}$ requires the number of trees to grow with n at a rate depending on dimension d and Hölder exponent $\alpha = 1 - d/p$. For α close to d — achieved when $p \gg d$, meaning f is highly smooth — the condition is mild and a slowly growing number of trees suffices. For α small — meaning f has limited smoothness or d is large — more trees are required. In practice random forests with $B = 500$ or $B = 1000$ trees satisfy this condition for moderate d and n .*

H.4 Step 3: Apply Lemma 4

The L^1 consistency condition follows from the L^2 consistency result of Mourtada et al. (2017) under the budget $\lambda_n \asymp n^{1/(d+2)}$ by the Cauchy–Schwarz inequality on the bounded domain Ω :

$$\left\|\hat{f}_n - f\right\|_{L^1} \leq |\Omega|^{1/2} \left\|\hat{f}_n - f\right\|_{L^2} \xrightarrow{p} 0 \quad (130)$$

Uniform BV boundedness follows from Step 2 under the condition $B \gg n^{2(d-\alpha)/(d+2)}$.

Lemma 4 then gives $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. $\square$

Remark 15. *Theorem 3 is stated for Mondrian forests where the partition is generated independently of the data. The independence is used critically in the decomposition of Substep 2a and in the cross term cancellation of Substep 2b. For Breiman forests with data-dependent splits the partition geometry and response variation are coupled — splits are chosen to maximize response differences across boundaries, which is precisely the worst case for the BV bound, and the cross term cancellation argument fails because the jump directions are not independent across trees. Extending the result to Breiman forests requires additional control over this coupling and is left as future work.*

I General Strategy for Bootstrap Validity Proofs

All three bootstrap corollaries follow from the same argument. The functional delta method of van der Vaart and Wellner (1996, Theorem 3.9.4) states that if:

1. $T : \mathcal{F} \rightarrow \mathbb{R}^d$ is Hadamard differentiable at f tangentially to $C(\Omega)^d$, and
2. $r_n(\hat{f}_n - f) \rightsquigarrow \mathbb{G}$ in $C(\Omega)^d$ for some rate sequence $r_n \rightarrow \infty$

then $r_n(T(\hat{f}_n) - T(f)) \rightsquigarrow T'_f(\mathbb{G})$, and the bootstrap is consistent in the sense that the bootstrap distribution of $r_n(T(\hat{f}_n^*) - T(\hat{f}_n))$ consistently estimates the sampling distribution of $r_n(T(\hat{f}_n) - T(f))$ (Horowitz, 2001, Chapter 3.6).

Lemma 5 establishes condition (1) for all three estimator classes. The estimator-specific work in each corollary is verification of condition (2) at the appropriate rate r_n . For neural networks and gradient boosted models the rate is $r_n = \sqrt{n}$. For Mondrian forests the rate is $r_n = n^{1/(d+2)}$, reflecting the minimax lower bound for nonparametric estimation of Lipschitz functions.

J Proof of Corollary 1 (Bootstrap Validity for Neural Networks)

J.1 Step 1: Rate Condition

The rate condition (RC) — that $\sqrt{n}(\hat{f}_n - f) = O_p(1)$ in $W^{1,p}(\Omega)$ — is taken as an explicit assumption of the corollary. We sketch here why this condition is expected to hold in the overparameterized regime and identify the steps required for a complete proof.

NTK Approximation

In the overparameterized regime with width $m \rightarrow \infty$, the neural tangent kernel (Jacot et al., 2018) governs the training dynamics. The network function at training time t satisfies:

$$\hat{f}_n^{(t)}(x) \approx \hat{f}_n^{(0)}(x) - \eta \int_0^t K(x, \cdot) \nabla_W \mathcal{L}(\hat{f}_n^{(s)}) ds \quad (131)$$

where $K(x, x') = \nabla_W \hat{f}_n(x)^\top \nabla_W \hat{f}_n(x')$ is the NTK and $\mathcal{L}$ is the training loss. In the infinite-width limit K is deterministic and constant during training, making the dynamics exactly linear.

Parametric Rate From Linear Dynamics

Under the linear NTK approximation the trained network satisfies:

$$\hat{f}_n - f \approx (I - K(K + \lambda I)^{-1})(f_0 - f) + K(K + \lambda I)^{-1}\epsilon \quad (132)$$

where f_0 is the network at initialization, λ is the regularization parameter, and ϵ is the noise. The bias term and the variance term each contribute $O_p(n^{-1/2})$ under standard conditions, giving:

$$\sqrt{n}(\hat{f}_n - f) = O_p(1) \quad \text{in } L^2(\Omega) \quad (133)$$

Upgrading to $W^{1,p}$

To upgrade from L^2 to $W^{1,p}(\Omega)$ one applies the chain rule calculation from Theorem 1 to the linear approximation. Since $\sigma \in C^3$ by assumption, the gradient of the network function satisfies:

$$\nabla \hat{f}_n(x) - \nabla f(x) \approx \nabla_x [K(x, \cdot)(K + \lambda I)^{-1} \epsilon] \quad (134)$$

The C^3 condition ensures this gradient is well defined and the propagation through the chain rule gives $O_p(n^{-1/2})$ convergence in $L^p(\Omega)^d$, yielding:

$$\sqrt{n}(\hat{f}_n - f) = O_p(1) \quad \text{in } W^{1,p}(\Omega) \quad (135)$$

A complete proof requires quantitative finite-width NTK approximation results establishing how close the finite-width network is to the infinite-width limit as a function of width and sample size, and is left for future work.

J.2 Step 2: Apply the Functional Delta Method

The rate condition (RC) is assumed. Lemma 5 establishes Hadamard differentiability of T tangentially to $C(\Omega)^d$ under $\sigma \in C^3$ and $p > d$. By van der Vaart and Wellner (1996, Theorem 3.9.4):

$$\sqrt{n} \left(T(\hat{f}_n) - T(f) \right) \rightsquigarrow T'_f(\mathbb{G}) = \int_{\Omega} \nabla \mathbb{G}(x) dP(x) \quad (136)$$

where $\mathbb{G}$ is the weak limit of $\sqrt{n}(\hat{f}_n - f)$ in $C(\Omega)^d$. Bootstrap consistency follows from Horowitz (2001, Chapter 3.6): the bootstrap distribution of $\sqrt{n}(T(\hat{f}_n^*) - T(\hat{f}_n))$ consistently estimates the sampling distribution of $\sqrt{n}(T(\hat{f}_n) - T(f))$. $\square$

Remark 16. *The rate condition (RC) is the only unproved component of Corollary 1. Unlike the Mondrian forest case where the dimension-dependent rate $n^{-1/(d+2)}$ is a fundamental consequence of the nonparametric minimax lower bound, the parametric rate $n^{-1/2}$ for neural networks is not a fundamental limitation — it is a consequence of the overparameterization*

and the NTK regime, and we expect it to be provable with a more detailed engagement with the finite-width NTK literature. The sketch above identifies the key steps: the NTK linear approximation, the L^2 parametric rate, and the C^3 chain rule upgrade to $W^{1,p}(\Omega)$. Completing this argument is left for future work.

K Proof of Corollary 2 (Bootstrap Validity for Gradient Boosting)

K.1 Step 1: Verify the Rate Condition

The rate condition requires showing that $\sqrt{n}(\hat{F}_m - f) = O_p(1)$ in the appropriate norm. The argument uses the eigenvalue structure of L_2 Boost established in the proof of Theorem 2.

Bias

By Proposition 1 and Theorem 1(a) of Bühlmann and Yu (2003), under the eigenvalue condition $0 < \lambda_k \leq 1$, the bias of $\hat{F}_m$ satisfies:

$$\text{bias}^2(m, S; f) = n^{-1} f^T U \text{diag}((1 - \lambda_k)^{2m+2}) U^T f \leq \rho^{2m} \|f\|_{L^2}^2 \quad (137)$$

where $\rho = \max_k(1 - \lambda_k) < 1$. The bias decays exponentially in m . For $m = m(n)$ growing sufficiently fast with n , the bias is $o(n^{-1/2})$ and negligible at the parametric rate.

Variance

By Proposition 3 of Bühlmann and Yu (2003), the variance of $\hat{F}_m$ satisfies:

$$\text{var}(m, S; \sigma^2) = \sigma^2 n^{-1} \sum_{k=1}^n (1 - (1 - \lambda_k)^{m+1})^2 \leq \sigma^2 \quad (138)$$

The variance is bounded above by the noise variance σ^2 regardless of m . Over n observations, the variance of the sample average satisfies:

$$\text{Var}\left(\frac{1}{n}\sum_{i=1}^n \hat{F}_m(x_i)\right) = O(n^{-1}) \quad (139)$$

so $\sqrt{n}(\hat{F}_m - f) = O_p(1)$ in L^2 .

Upgrading to the Appropriate Norm

For Case (i) with smooth base learners, the upgrade from L^2 to $W^{1,p}(\Omega)$ follows from the same chain rule argument as in Theorem 1, using the geometric decay of residuals $\|u_m\|_{L^2} \leq C\rho^m$ to propagate the rate through the $W^{2,p}$ norm bound.

For Case (ii) with tree stump base learners, the upgrade from L^2 to $BV(\Omega)$ uses the ensemble BV bound from Theorem 2. The total variation of $\sqrt{n}(\hat{F}_m - f)$ is controlled by:

$$\mathbb{E}\left[|D\sqrt{n}(\hat{F}_m - f)|(\Omega)\right] = \sqrt{n}\mathbb{E}\left[|D(\hat{F}_m - f)|(\Omega)\right] \leq \sqrt{n} \cdot \frac{2\text{diam}(\Omega)^{d-1}C\rho}{1-\rho} \quad (140)$$

For this to be $O(1)$ the right-hand side must be bounded, which requires $\rho = O(n^{-1/2})$, i.e. the smallest eigenvalue satisfies $\lambda_{\min} = \Omega(n^{-1/2})$. This is the principal remaining proof obligation for Case (ii) and is left for future work. The L^2 rate condition and Case (i) upgrade are complete.

K.2 Step 2: Apply the Functional Delta Method

With the rate condition verified, Lemma 5 establishes Hadamard differentiability of T tangentially to $C(\Omega)^d$, and van der Vaart and Wellner (1996, Theorem 3.9.4) gives:

$$\sqrt{n}\left(T(\hat{F}_m) - T(f)\right) \rightsquigarrow T'_f(\mathbb{G}) = \int_{\Omega} \nabla \mathbb{G}(x) dP(x) \quad (141)$$

where $\mathbb{G}$ is the weak limit of $\sqrt{n}(\hat{F}_m - f)$ in $C(\Omega)^d$. Bootstrap consistency follows from Horowitz (2001, Chapter 3.6): the bootstrap distribution of $\sqrt{n}(T(\hat{F}_m^*) - T(\hat{F}_m))$ consistently estimates the sampling distribution of $\sqrt{n}(T(\hat{F}_m) - T(f))$.

The sequential dependence of the boosting path is handled by the framework of Horowitz (2001) for smooth functionals of empirical processes. The geometric decay of residuals established in Theorem 2 provides the Lyapunov-type control over the sequential dependence that the Horowitz framework requires. $\square$

Remark 17. *Corollary 2 is fully proved for Case (i) with smooth base learners. For Case (ii) with tree stump base learners the rate condition upgrade from L^2 to $BV(\Omega)$ requires a condition on the smallest eigenvalue of S that we state explicitly but do not verify from first principles. Verifying this condition under natural assumptions on the covariate distribution is an important direction for future work. Unlike the Mondrian forest case where the dimension-dependent rate is a fundamental minimax limitation, the tree stump rate condition is a verifiable property of the base learner and we expect it to hold under mild conditions.*

L Proof of Corollary 3 (Bootstrap Validity for Mondrian Forests)

L.1 Step 1: Verify the Rate Condition

The rate condition requires showing that $n^{1/(d+2)}(\hat{f}_n - f) = O_p(1)$ in $BV(\Omega)$, which is the correct scaling given the Mondrian minimax rate $O(n^{-2/(d+2)})$ established by Mourtada et al. (2017) for Lipschitz regression functions under the budget $\lambda_n \asymp n^{1/(d+2)}$.

The argument proceeds in two parts — bias and variance — using the independence of the Mondrian partition from the data.

Bias

The bias of the Mondrian forest at budget λ_n is controlled by the cell diameter. By Roy and Teh (2009):

$$\mathbb{E}[D_{\lambda_n}(x)^2] \leq \frac{4d}{\lambda_n^2} = O(n^{-2/(d+2)}) \quad (142)$$

Since f is Lipschitz, the bias satisfies:

$$\left| \mathbb{E}[\hat{f}_n(x)] - f(x) \right| \leq C \cdot \mathbb{E}[D_{\lambda_n}(x)] \leq C \cdot \sqrt{\mathbb{E}[D_{\lambda_n}(x)^2]} = O(n^{-1/(d+2)}) \quad (143)$$

Therefore $n^{1/(d+2)} \cdot \text{bias} = O(1)$, confirming the bias is negligible at the rate $n^{-1/(d+2)}$.

Variance

The variance of $\hat{f}_n(x)$ is controlled by the split count and the conditional independence structure of the Mondrian process. Since the B trees are independent:

$$\text{Var}(\hat{f}_n(x)) = \frac{1}{B} \text{Var}(T_1(x)) \quad (144)$$

Within a single tree, the variance of the leaf mean at x is at most σ^2/n_j where n_j is the number of observations in the leaf containing x . Under the budget $\lambda_n \asymp n^{1/(d+2)}$ the expected leaf count satisfies $\mathbb{E}[n_j] \asymp n^{2/(d+2)}$ by the split count bound of Mourtada et al. (2017). Therefore $\text{Var}(T_1(x)) = O(n^{-2/(d+2)})$ and:

$$n^{2/(d+2)} \cdot \text{Var}(\hat{f}_n(x)) = O(1/B) = O(1) \quad (145)$$

confirming the variance is controlled at the rate $n^{-1/(d+2)}$.

Combining Bias and Variance

By bias-variance decomposition:

$$\mathbb{E}\left[(\hat{f}_n(x) - f(x))^2\right] = O(n^{-2/(d+2)}) \quad (146)$$

consistent with the minimax rate of Mourtada et al. (2017). The BV rate follows from the L^2 rate via the second moment argument in Theorem 3: the BV norm of $n^{1/(d+2)}(\hat{f}_n - f)$ is controlled by the same geometric and response bounds used to establish uniform BV boundedness, giving:

$$n^{1/(d+2)}(\hat{f}_n - f) = O_p(1) \quad \text{in } BV(\Omega) \quad (147)$$

L.2 Step 2: Apply the Functional Delta Method

With the rate condition verified, Lemma 5 establishes Hadamard differentiability of T tangentially to $C(\Omega)^d$, and van der Vaart and Wellner (1996, Theorem 3.9.4) gives:

$$n^{1/(d+2)}\left(T(\hat{f}_n) - T(f)\right) \rightsquigarrow T'_f(\mathbb{G}) = \int_{\Omega} \nabla \mathbb{G}(x) dP(x) \quad (148)$$

where $\mathbb{G}$ is the weak limit of $n^{1/(d+2)}(\hat{f}_n - f)$ in $C(\Omega)^d$. Bootstrap consistency follows from Horowitz (2001, Chapter 3.6) and van der Vaart and Wellner (1996, Theorem 3.9.4): the bootstrap distribution of $n^{1/(d+2)}(T(\hat{f}_n^*) - T(\hat{f}_n))$ consistently estimates the sampling distribution of $n^{1/(d+2)}(T(\hat{f}_n) - T(f))$.

The connection to the infinitesimal jackknife of Wager et al. (2014) provides an alternative characterization of the bootstrap variance that is computationally tractable for large forests. The Hadamard differentiability established in Lemma 5 shows that the infinitesimal jackknife approximation is asymptotically exact at the Mondrian convergence rate. $\square$

Remark 18. *The scaling $n^{1/(d+2)}$ rather than $\sqrt{n}$ reflects the minimax rate for Lipschitz functions on $[0, 1]^d$. When $d \leq 2$ the rate $n^{1/(d+2)}$ achieves $n^{1/2}$ and the bootstrap confidence*

intervals are asymptotically exact at the parametric rate. For $d \geq 3$ the rate is slower than $n^{1/2}$ — a fundamental consequence of the curse of dimensionality, not a limitation of the bootstrap procedure. The bootstrap remains valid at the actual convergence rate in all dimensions.

Remark 19. *Bootstrap validity for Breiman forests with data-dependent splits remains open. Under bootstrap resampling Breiman forests redraw the partition based on the new sample, coupling it to the response values in a way that prevents the clean separation of geometric and response variation used in Step 1 above. The cross-term cancellation argument from Theorem 3 fails because jump directions are no longer independent across bootstrap replicates. Establishing bootstrap validity for Breiman forests requires new techniques for controlling the joint distribution of partition geometry and response variation under resampling, and is left as an important direction for future work.*

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